

Solutions to

Daron Acemoglu

Introduction to Modern Economic Growth

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Chapters 2–3

The exercises that the instructor's solutions manual of Peters and Simsek (2009) leaves unsolved: Exercises 2.1–2.6, 2.8–2.10, 2.13, 2.15, 2.24–2.26 and 3.3–3.8, 3.11, with the numerical parts computed by the Python programs in `code/`.

The exercise statements are not reproduced here: read them in Acemoglu's *Introduction to Modern Economic Growth*.

Each solution is headed by the number of its exercise in the book; notation, the part letters (a), (b), ... follow the statements there.

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Chapter 2

The Solow Growth Model

Solutions to the fourteen exercises of Chapter 2 of Daron Acemoglu, *Introduction to Modern Economic Growth* (Princeton University Press, 2009), that are left unsolved in the instructor's solutions manual of Peters and Simsek (2009): Exercises 2.1–2.6, 2.8, 2.9, 2.10, 2.13, 2.15, 2.24, 2.25 and 2.26. The remaining thirteen exercises of the chapter (2.7, 2.11, 2.12, 2.14, 2.16–2.23 and 2.27) are solved there. Assumptions, propositions and equations cited as ASSUMPTION 2, PROPOSITION 2.5 or (2.18) refer to the book. Throughout, F is the aggregate production function, $f(k) \equiv F(k, 1, A)$ the per capita production function, $k \equiv K/L$ the capital-labor ratio, $w(k) \equiv f(k) - kf'(k)$ the wage, and $R(k) \equiv f'(k)$ the rental rate of capital. The numerical parts are computed by `code/ch2.py`.

Exercise 2.1

Fix a date t and write $\bar{L}(t) > 0$ for the (finite, inelastically supplied) labor endowment and $K(t)$ for the capital stock used in production. In a competitive equilibrium the representative firm solves (2.5),

$$\max_{K \geq 0, L \geq 0} F(K, L, A(t)) - R(t)K - w(t)L,$$

taking $(R(t), w(t))$ as given, and factor markets clear in the complementary slackness form (2.4):

$$L(t) \leq \bar{L}(t), \quad w(t) \geq 0, \quad (L(t) - \bar{L}(t))w(t) = 0.$$

Suppose, to get a contradiction, that $w(t) = 0$. Let $(K(t), L(t))$ be the equilibrium factor demands, which must be finite since they cannot exceed the endowments. Consider raising employment to $L(t) + \varepsilon$ for $\varepsilon > 0$, holding capital fixed. Profits change by

$$F(K(t), L(t) + \varepsilon, A(t)) - F(K(t), L(t), A(t)) - w(t)\varepsilon = \int_{L(t)}^{L(t)+\varepsilon} F_L(K(t), \ell, A(t)) d\ell > 0,$$

because $w(t) = 0$ and $F_L > 0$ everywhere by ASSUMPTION 1. So the firm strictly prefers more labor at *every* finite L : the maximization problem has no solution, and in particular $(K(t), L(t))$ is not profit maximizing. That contradicts firm optimization, so $w(t) > 0$.

Given $w(t) > 0$, the complementary slackness condition in (2.4) forces $L(t) - \bar{L}(t) = 0$, which is exactly the simple market clearing condition (2.3), $L(t) = \bar{L}(t)$. This is why (2.3) can be used throughout in place of (2.4).

Remark. Only $F_L > 0$ is used; neither constant returns nor the Inada conditions are needed. The identical argument with $F_K > 0$ shows $R(t) > 0$, so the capital market also clears with equality, $K(t) = \bar{K}(t)$. Note also that *some* bound on labor demand is essential to the argument: it is the finiteness of the endowment that turns “the firm always wants more labor” into a contradiction.

Exercise 2.2

Fix A and write $F(K, L)$ for $F(K, L, A)$, which by ASSUMPTION 1 is twice differentiable and homogeneous of degree 1 in (K, L) , with $F_{KK} < 0$ and $F_{LL} < 0$.

The Hessian is singular. By THEOREM 2.1 (Euler's Theorem), the partial derivatives F_K and F_L are homogeneous of degree 0. Applying Euler's Theorem to each of them,

$$KF_{KK} + LF_{KL} = 0 \quad \text{and} \quad KF_{LK} + LF_{LL} = 0.$$

Hence, for $K, L > 0$,

$$F_{KL} = -\frac{K}{L}F_{KK} > 0 \quad \text{and} \quad F_{KL} = -\frac{L}{K}F_{LL} > 0,$$

the signs following from $F_{KK} < 0$ and $F_{LL} < 0$. Multiplying the two expressions,

$$F_{KL}^2 = \left(-\frac{K}{L}F_{KK}\right)\left(-\frac{L}{K}F_{LL}\right) = F_{KK}F_{LL},$$

so that the Hessian

$$\mathbf{H} = \begin{pmatrix} F_{KK} & F_{KL} \\ F_{KL} & F_{LL} \end{pmatrix} \quad \text{has} \quad \det \mathbf{H} = F_{KK}F_{LL} - F_{KL}^2 = 0.$$

Concavity. A twice differentiable function is concave if and only if its Hessian is negative semidefinite everywhere. Here the leading principal minors are $F_{KK} < 0$ and $\det \mathbf{H} = 0$, so $\mathbf{H}$ is negative semidefinite (its eigenvalues are 0 and $F_{KK} + F_{LL} < 0$). Therefore F is concave in (K, L) .

But not strictly. Strict concavity would require $\mathbf{H}$ to be negative definite, which fails since $\det \mathbf{H} = 0$; the null direction is $(K, L)'$ itself, because $\mathbf{H} \begin{pmatrix} K \\ L \end{pmatrix} = \begin{pmatrix} KF_{KK} + LF_{KL} \\ KF_{LK} + LF_{LL} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The failure is easy to see directly: along any ray through the origin, constant returns make F linear,

$$F(\lambda K + (1 - \lambda)\lambda'K, \lambda L + (1 - \lambda)\lambda'L) = [\lambda + (1 - \lambda)\lambda']F(K, L) = \lambda F(K, L) + (1 - \lambda)F(\lambda'K, \lambda'L),$$

so the concavity inequality holds with equality for any two points on a ray. This is the sense in which the production function can be “concave but not strictly concave” even though $F_{KK} < 0$ and $F_{LL} < 0$: it curves in every direction except the direction in which the two inputs are scaled together.

Remark. The per capita function $f(k) \equiv F(k, 1)$ is strictly concave, since $f''(k) = F_{KK}(k, 1) < 0$. Strict concavity of f , not of F , is what the uniqueness arguments of the chapter use.

Exercise 2.3

Write the profit function as

$$\pi(K, L) \equiv F(K, L, A) - RK - wL.$$

Since F is homogeneous of degree 1 in (K, L) , so is π : for every $\lambda > 0$,

$$\pi(\lambda K, \lambda L) = \lambda \pi(K, L).$$

For $L > 0$ write $\kappa \equiv K/L$; then constant returns give

$$\pi(K, L) = L[f(\kappa) - R\kappa - w] \equiv L\varphi(\kappa), \quad f(\kappa) \equiv F(\kappa, 1, A).$$

Everything therefore depends on the sign of the maximized value of φ , and only three cases can arise.

Case 1: $\varphi(\kappa) > 0$ for some κ (equivalently $\pi(K_0, L_0) > 0$ for some (K_0, L_0)). Then $\pi(\lambda K_0, \lambda L_0) = \lambda\pi(K_0, L_0) \rightarrow \infty$ as $\lambda \rightarrow \infty$: profits are unbounded and no maximizer exists. This is the case in which factor prices are “too low” relative to marginal products.

Case 2: $\varphi(\kappa) < 0$ for every $\kappa \geq 0$. Then $\pi(K, L) < 0$ whenever $L > 0$. On the boundary $L = 0$ constant returns give $F(K, 0, A) = K F(1, 0, A)$, and

$$F(1, 0, A) = \lim_{L \rightarrow 0} Lf(1/L) = \lim_{k \rightarrow \infty} \frac{f(k)}{k} = 0$$

by ASSUMPTION 2, so $\pi(K, 0) = -RK \leq 0$ with equality only at $K = 0$. Since $\pi(0, 0) = 0$, the unique maximizer is $K = L = 0$ and maximal profit is zero.

Case 3: $\max_{\kappa} \varphi(\kappa) = 0$, attained at some $\kappa^* > 0$. Then $\pi(K, L) = 0$ for every (K, L) with $K/L = \kappa^*$, and $\pi(K, L) < 0$ otherwise (for $L > 0$), so the solution set is the entire ray

$$\{(K, L) : K/L = \kappa^*, L \geq 0\},$$

a continuum: the scale of the firm is completely indeterminate, only its capital-labor ratio is pinned down.

These are the only possibilities, since $\max_{\kappa} \varphi$ is either positive, negative, or zero. In equilibrium it is Case 3 that must obtain: factor supplies are fixed and finite, so Case 1 is inconsistent with market clearing and Case 2 would leave the supplies unemployed. Hence equilibrium factor prices satisfy $\varphi(\kappa^*) = 0$ with $\varphi'(\kappa^*) = f'(\kappa^*) - R = 0$, which is (2.7), and then $w = f(\kappa^*) - \kappa^* f'(\kappa^*)$, which is (2.6); profits are zero, as PROPOSITION 2.1 states. The indeterminacy of scale is why the model determines only aggregates, and why the representative firm can be taken to operate at the economy's endowment point.

Exercise 2.4

It is convenient to factor the function once and for all:

$$f(k) = k(k-1)(k-2)(k-3), \quad \text{so} \quad \frac{f(k)}{k} = (k-1)(k-2)(k-3).$$

Also

$$\begin{aligned} f'(k) &= 4k^3 - 18k^2 + 22k - 6, & f''(k) &= 12k^2 - 36k + 22, \\ w(k) &= f(k) - kf'(k) = -k^2(3k^2 - 12k + 11). \end{aligned}$$

(a) *Violations.* The underlying production function is $F(K, L) = Lf(K/L)$, so $F_K = f'(k)$, $F_{KK} = f''(k)/L$, $F_L = w(k)$ and $F_{LL} = (K^2/L^3)f''(k)$. Constant returns to scale hold by construction, and f is a polynomial, hence twice (indeed infinitely) differentiable. Everything else in ASSUMPTION 1 fails somewhere:

- *Nonnegativity of output.* $f(k) < 0$ for $k \in (0, 1) \cup (2, 3)$ — for instance $f(1/2) = -0.9375$ — so F does not map into $\mathbb{R}_+$.
- *Positive marginal products.* $f'(0) = -6 < 0$, so $F_K > 0$ fails near $k = 0$ (and again on part of $(2, 3)$). Likewise $w(k) < 0$ for $k < 2 - 1/\sqrt{3}$ and for $k > 2 + 1/\sqrt{3}$, e.g. $w(3) = -18$, so $F_L > 0$ fails.
- *Diminishing marginal products.* $f''(k) = 12k^2 - 36k + 22$ is positive outside the interval between its roots $k = (9 \pm \sqrt{15})/6$, i.e. on $(0, 0.8545) \cup (2.1455, \infty)$ — for instance $f''(0) = 22 > 0$. So $F_{KK} < 0$ and $F_{LL} < 0$ both fail there.

Of ASSUMPTION 2, only $F(0, L, A) = 0$ survives ($f(0) = 0$). Both Inada conditions for capital fail: $\lim_{k \rightarrow 0} f'(k) = -6 \neq \infty$ and $f'(k) \rightarrow +\infty$ rather than 0 as $k \rightarrow \infty$ (e.g. $f'(10) = 2414$). The Inada conditions for labor fail as well, since $F_L = w(k) \rightarrow -\infty$ as $k \rightarrow \infty$.

(b) *Three steady states.* With population growth n and depreciation δ , the fundamental law of motion (2.33) is

$$\frac{\dot{k}(t)}{k(t)} = s \frac{f(k(t))}{k(t)} - (n + \delta),$$

so an interior steady state is a solution of

$$(k-1)(k-2)(k-3) = h, \quad h \equiv \frac{n+\delta}{s} > 0. \quad (*)$$

The cubic on the left has derivative $3k^2 - 12k + 11$, which vanishes at $k = 2 \mp 1/\sqrt{3}$; it therefore rises to a local maximum at $k = 2 - 1/\sqrt{3} \approx 1.42265$, falls to a local minimum at $k = 2 + 1/\sqrt{3}$, and rises thereafter. The two extreme values are

$$\left(1 - \frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{3}}\right)\left(-1 - \frac{1}{\sqrt{3}}\right) = \frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9} \approx 0.38490 \quad \text{and} \quad -\frac{2\sqrt{3}}{9}.$$

Hence (*) has exactly three positive roots whenever

$$0 < \frac{n+\delta}{s} < \frac{2\sqrt{3}}{9} \approx 0.38490,$$

one in $(1, 2 - 1/\sqrt{3})$, one in $(2 - 1/\sqrt{3}, 2)$, and one in $(3, \infty)$; it has exactly one (in $(3, \infty)$) when $h > 2\sqrt{3}/9$. For example, with $n = 0$ and $\delta/s = 1/4$ the three steady states are

$$k_1^* = 1.162435, \quad k_2^* = 1.730406, \quad k_3^* = 3.107160.$$

Figure 1 plots the cubic together with the horizontal line h .

(c) *Stability.* Since $\dot{k} = k s [(k-1)(k-2)(k-3) - h]$, the sign of $\dot{k}$ is the sign of the vertical distance between the cubic and the line, and at a steady state

$$\left. \frac{\partial \dot{k}}{\partial k} \right|_{k=k^*} = s k^* \cdot \frac{d}{dk} \left[\frac{f(k)}{k} \right]_{k=k^*},$$

so a steady state is locally stable exactly when the cubic crosses the line *downward*. The cubic crosses upward at k_1^* (rising branch), downward at k_2^* (falling branch), and upward again at k_3^* :

$$\text{at } k_1^* : +1.1045, \quad \text{at } k_2^* : -0.7820, \quad \text{at } k_3^* : +2.6774,$$

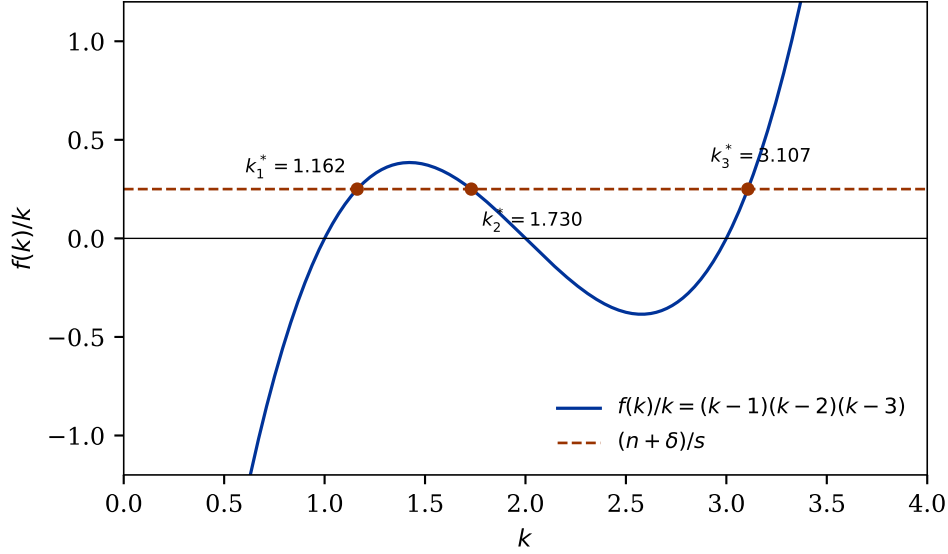


FIGURE 1. Steady states of the quartic economy: the interior steady states are the intersections of $f(k)/k = (k-1)(k-2)(k-3)$ with the horizontal line $(n+\delta)/s$, drawn here at $1/4$.

for the derivative of $f(k)/k$. Therefore k_2^* is locally asymptotically stable while k_1^* and k_3^* are unstable. In addition $k = 0$ is a rest point of the differential equation, and it is locally stable: for small $k > 0$ we have $f(k)/k < 0 < h$, so $\dot{k} < 0$ and the economy collapses to zero. The complete phase line is

$$\underbrace{0}_{\text{stable}} \longleftarrow \underbrace{k_1^*}_{\text{unstable}} \longrightarrow \underbrace{k_2^*}_{\text{stable}} \longleftarrow \underbrace{k_3^*}_{\text{unstable}} \longrightarrow \infty.$$

An economy starting with $k(0) < k_1^*$ collapses to $k = 0$; one starting in (k_1^*, k_3^*) converges to k_2^* ; one starting above k_3^* grows without bound. This is a poverty-trap configuration: history (the initial condition) decides the long-run outcome.

Finally, *no* steady state can be globally stable. Global asymptotic stability requires convergence to the same point from every initial condition; but an economy that starts at another steady state stays there forever, so the existence of more than one rest point rules global stability out immediately.

Remark. Part (c) of the exercise asks one to prove that *two* of the three steady states in (b) are locally stable. Of the three *interior* steady states exactly one — the middle one — is locally stable, as the sign pattern above shows: on the rising branches of $f(k)/k$ the economy is pushed away from the steady state. The count in the book is recovered only if the degenerate rest point $k = 0$, which the text elsewhere excludes as having “no economic content”, is included among the steady states; then two of the four rest points ($k = 0$ and k_2^*) are stable and two are unstable.

Exercise 2.5

We must show that under ASSUMPTIONS 1 and 2 the continuous-time Solow model with population growth at the rate n has a unique steady-state equilibrium, with $k^* \in (0, \infty)$ solving (2.34), $y^* = f(k^*)$ and $c^* = (1-s)f(k^*)$.

By (2.33) the equilibrium path satisfies $\dot{k}(t)/k(t) = sf(k(t))/k(t) - (n + \delta)$, so a steady state with $k > 0$ is precisely a solution of

$$\psi(k) \equiv \frac{f(k)}{k} = \frac{n + \delta}{s}. \quad (*)$$

ψ is continuous and strictly decreasing. Continuity is immediate from ASSUMPTION 1. For monotonicity, differentiate:

$$\psi'(k) = \frac{f'(k)k - f(k)}{k^2} = -\frac{w(k)}{k^2}, \quad w(k) = f(k) - kf'(k).$$

Since f is strictly concave with $f(0) = 0$ (ASSUMPTION 2), strict concavity gives $f(k) > f(0) + kf'(k) = kf'(k)$, that is $w(k) > 0$ for every $k > 0$; equivalently, $w(k) = F_L > 0$ is the wage, which is positive by ASSUMPTION 1. Hence $\psi'(k) < 0$ for all $k > 0$.

The limits. Both $f(k) \rightarrow 0$ and $k \rightarrow 0$ as $k \rightarrow 0$, so l'Hôpital's rule and ASSUMPTION 2 give

$$\lim_{k \rightarrow 0} \psi(k) = \lim_{k \rightarrow 0} f'(k) = \lim_{K \rightarrow 0} F_K = \infty, \quad \lim_{k \rightarrow \infty} \psi(k) = \lim_{k \rightarrow \infty} f'(k) = \lim_{K \rightarrow \infty} F_K = 0.$$

(The second limit also follows from $0 < f(k)/k$ and concavity.)

Existence and uniqueness. ψ is continuous on $(0, \infty)$ with limits ∞ at 0 and 0 at ∞ , so by the Intermediate Value Theorem it attains the value $(n + \delta)/s > 0$ at some $k^* \in (0, \infty)$; since ψ is strictly decreasing, that value is attained exactly once. Hence there is a unique steady-state capital-labor ratio, given by (2.34).

Finally, at k^* the level of output per capita is $y^* = f(k^*)$ by definition of f , and since a constant fraction s of income is saved, consumption per capita is $c^* = (1 - s)f(k^*)$, with the corresponding investment $sf(k^*) = (n + \delta)k^*$ exactly replacing depreciated capital and equipping the new workers.

Remark. The proof is that of PROPOSITION 2.2 with δ replaced by $n + \delta$: going from discrete to continuous time, and adding population growth, changes nothing essential. Population growth acts exactly like faster depreciation, since both erode the capital-labor ratio at a constant proportional rate.

Exercise 2.6

Let $f(k) = A\tilde{f}(k)$ with $\tilde{f}$ satisfying ASSUMPTIONS 1 and 2, and let k^* be the unique steady state of Exercise 2.5, i.e. the solution of

$$G(k^*; A, s, \delta, n) \equiv sA \frac{\tilde{f}(k^*)}{k^*} - (n + \delta) = 0.$$

Writing $\tilde{w}(k) \equiv \tilde{f}(k) - k\tilde{f}'(k) > 0$ for the wage per unit of A , the computation in Exercise 2.5 gives

$$\frac{\partial G}{\partial k^*} = sA \cdot \frac{d}{dk} \left[\frac{\tilde{f}(k)}{k} \right]_{k=k^*} = -\frac{sA \tilde{w}(k^*)}{(k^*)^2} < 0,$$

so the Implicit Function Theorem applies at every parameter configuration and yields $\partial k^*/\partial x = -(\partial G/\partial x)/(\partial G/\partial k^*)$ for each parameter x . Since $\partial G/\partial A = s\tilde{f}(k^*)/k^* > 0$, $\partial G/\partial s = A\tilde{f}(k^*)/k^* > 0$, $\partial G/\partial \delta = -1$ and $\partial G/\partial n = -1$,

$$\frac{\partial k^*}{\partial A} = \frac{k^* \tilde{f}(k^*)}{A \tilde{w}(k^*)} > 0, \quad \frac{\partial k^*}{\partial s} = \frac{k^* \tilde{f}(k^*)}{s \tilde{w}(k^*)} > 0, \quad \frac{\partial k^*}{\partial \delta} = \frac{\partial k^*}{\partial n} = -\frac{(k^*)^2}{sA \tilde{w}(k^*)} < 0.$$

This proves the four statements about k^* .

For output, $y^* = A\tilde{f}(k^*)$, so for $x \in \{s, \delta, n\}$ the chain rule gives

$$\frac{\partial y^*}{\partial x} = A\tilde{f}'(k^*)\frac{\partial k^*}{\partial x},$$

which has the same sign as $\partial k^*/\partial x$ because $\tilde{f}' > 0$: $\partial y^*/\partial s > 0$ and $\partial y^*/\partial \delta, \partial y^*/\partial n < 0$. For A there is in addition a direct effect, and both effects push the same way:

$$\frac{\partial y^*}{\partial A} = \tilde{f}(k^*) + A\tilde{f}'(k^*)\frac{\partial k^*}{\partial A} > 0.$$

The economics is transparent. A higher saving rate or a better technology raises investment at every capital-labor ratio and so shifts the $sf(k)/k$ schedule up, moving its intersection with the horizontal line $n + \delta$ to the right. A higher depreciation rate or a higher rate of population growth raises the amount of investment needed merely to hold k constant — capital must be replaced, and new workers must be equipped — and so moves the intersection to the left. The only new result relative to PROPOSITION 2.3 is the effect of n , and it enters exactly as δ does: countries with faster population growth are, other things equal, poorer in per capita terms.

Exercise 2.8

Throughout, F is twice differentiable and exhibits constant returns to scale, $f(k) = F(k, 1, A)$ is concave (so $f'' \leq 0$) with $f(0) = 0$, and the Inada conditions of ASSUMPTION 2 hold. What is *not* assumed is $F_{KK} < 0$, $F_{LL} < 0$, or even $F_K > 0$ and $F_L > 0$. The first thing to notice is that the missing sign restrictions on the first derivatives come for free.

Step 1: the marginal product of capital is nonnegative. Concavity makes f' nonincreasing. If $f'(k_0) < 0$ for some k_0 , then $f'(k) \leq f'(k_0) < 0$ for every $k \geq k_0$, contradicting the Inada condition $\lim_{k \rightarrow \infty} f'(k) = 0$. Hence $f'(k) \geq 0$, that is $F_K \geq 0$, for all $k > 0$. (One cannot get more: a technology that is affine from some k_0 on has $f' \equiv 0$ beyond k_0 and still satisfies every hypothesis. As the rest of the argument shows, $f' \geq 0$ is all that is needed.)

Step 2: the wage is strictly positive. Write $w(k) = f(k) - kf'(k)$. Since $f(0) = 0$,

$$w(k) = \int_0^k [f'(x) - f'(k)] dx,$$

and the integrand is nonnegative because f' is nonincreasing. It is moreover strictly positive on a set of positive measure: $f'(x) \rightarrow \infty$ as $x \rightarrow 0$ while $f'(k) < \infty$, so $f'(x) > f'(k)$ for all small x . Hence $w(k) > 0$ for every $k > 0$; that is, $F_L > 0$, and the “not necessarily strictly” concave technology still pays a positive wage.

PROPOSITION 2.2: existence and uniqueness of the steady state. Exactly as in Exercise 2.5, $\psi(k) \equiv f(k)/k$ satisfies $\psi'(k) = -w(k)/k^2 < 0$ by Step 2, so ψ is *strictly* decreasing even though f is only weakly concave. Its limits are ∞ at 0 and 0 at ∞ by ASSUMPTION 2 and l'Hôpital's rule. The Intermediate Value Theorem therefore gives a unique $k^* \in (0, \infty)$ with $f(k^*)/k^* = \delta/s$, and then $y^* = f(k^*)$ and $c^* = (1-s)f(k^*)$. Nothing in the argument used strict concavity; it used only that $w > 0$, which Step 2 delivers.

PROPOSITION 2.5: global stability. Let $g(k) \equiv sf(k) + (1-\delta)k$, so that the discrete time law of motion is $k(t+1) = g(k(t))$ and $k^* = g(k^*)$. By Step 1, $g'(k) = sf'(k) + 1 - \delta > 0$. By Step 2,

$\delta = sf(k^*)/k^* > sf'(k^*)$, whence

$$g'(k^*) = sf'(k^*) + 1 - \delta \in (0, 1),$$

and local stability follows from COROLLARY 2.1. For global stability, take $k(t) \in (0, k^*)$. Since g is strictly increasing, $k(t+1) = g(k(t)) < g(k^*) = k^*$; and since ψ is strictly decreasing,

$$\frac{k(t+1) - k(t)}{k(t)} = s \frac{f(k(t))}{k(t)} - \delta > s \frac{f(k^*)}{k^*} - \delta = 0.$$

So $k(t+1) \in (k(t), k^*)$: the sequence is increasing and bounded above by k^* , hence convergent, and its limit is a fixed point of g in $(0, k^*]$, which can only be k^* by uniqueness. The argument starting from $k(0) > k^*$ is identical with the inequalities reversed. Monotone global convergence therefore survives the weakening of ASSUMPTION 1.

PROPOSITION 2.6 *must be weakened to weak monotonicity*. The proposition asserts that $\{w(t)\}$ is an increasing and $\{R(t)\}$ a decreasing sequence when $k(0) < k^*$. With $f'' \leq 0$ instead of $f'' < 0$ we get

$$R'(k) = f''(k) \leq 0 \quad \text{and} \quad w'(k) = -kf''(k) \geq 0,$$

so the correct statement is that $\{w(t)\}$ is *nondecreasing* and $\{R(t)\}$ is *nonincreasing* (with the inequalities reversed when $k(0) > k^*$). Both can be locally constant: on any interval where f is affine — the production function is then locally an AK technology — the capital-labor ratio keeps rising while factor prices do not move at all. Strict monotonicity is restored as soon as $f'' < 0$ on the relevant range, which is what ASSUMPTION 1 guarantees.

Exercise 2.9

Let ASSUMPTIONS 1 and 2 hold and suppose $k(0) < k^*$. By PROPOSITION 2.5 the capital-labor ratio increases monotonically to k^* : $k(t+1) > k(t)$ for every t , with $k(t) \rightarrow k^*$.

In equilibrium factor prices are the marginal products, and with constant returns they depend on the capital-labor ratio alone:

$$R(t) = F_K(K(t), L(t), A) = f'(k(t)),$$

$$w(t) = F_L(K(t), L(t), A) = f(k(t)) - k(t)f'(k(t)) = w(k(t)).$$

Differentiating the two expressions in k ,

$$\frac{dR}{dk} = f''(k) < 0, \quad \frac{dw}{dk} = f'(k) - f'(k) - kf''(k) = -kf''(k) > 0,$$

both strict by ASSUMPTION 1, which has $f''(k) = F_{KK}(k, 1, A) < 0$.

So $R(\cdot)$ is a strictly decreasing and $w(\cdot)$ a strictly increasing function of k . Composing with the strictly increasing sequence $\{k(t)\}$ gives

$$w(t+1) = w(k(t+1)) > w(k(t)) = w(t), \quad R(t+1) = f'(k(t+1)) < f'(k(t)) = R(t),$$

that is, $\{w(t)\}_{t=0}^\infty$ is an increasing and $\{R(t)\}_{t=0}^\infty$ a decreasing sequence, and they converge to $w(k^*)$ and $f'(k^*)$. If instead $k(0) > k^*$, PROPOSITION 2.5 gives a strictly decreasing $\{k(t)\}$ and the two conclusions are reversed.

The economics is the usual scarcity argument: capital deepening makes capital abundant relative to labor, so the marginal product of capital — and with it the rental rate and the interest rate $r = R - \delta$ — falls, while the marginal product of labor, and hence the wage, rises. Along the transition the entire increase in output per worker that is not saved accrues to labor.

Exercise 2.10

Part 1. The linear equation $\dot{x}(t) = ax(t)$ with $x(0)$ given has the unique solution $x(t) = x(0) \exp(at)$. If $a < 0$ then $\exp(at) \rightarrow 0$, so $x(t) \rightarrow 0$ for every initial condition: the steady state $x^* = 0$ is globally asymptotically stable. (If $a > 0$ the solution diverges unless $x(0) = 0$, and if $a = 0$ every point is a steady state.)

Part 2. Let g be differentiable in a neighborhood of x^* , with $g(x^*) = 0$ and $g'(x^*) < 0$. This is the scalar case ($n = 1$) of THEOREM 2.5: the Jacobian is the number $g'(x^*)$, its unique eigenvalue is $g'(x^*) < 0$, which has negative real part, so x^* is locally asymptotically stable. The direct argument is just as short. Since g is continuous with $g(x^*) = 0$ and $g'(x^*) < 0$, there is $\varepsilon > 0$ with

$$g(x) < 0 \text{ for } x \in (x^*, x^* + \varepsilon) \quad \text{and} \quad g(x) > 0 \text{ for } x \in (x^* - \varepsilon, x^*),$$

so on that neighborhood $\dot{x}$ always points toward x^* ; Part 3 applied to the interval $(x^* - \varepsilon, x^* + \varepsilon)$ then gives convergence.

Part 3. Let g be continuously differentiable with $g(x^*) = 0$, $g(x) < 0$ for $x > x^*$ and $g(x) > 0$ for $x < x^*$. Take any $x(0) < x^*$ (the case $x(0) > x^*$ is symmetric, and $x(0) = x^*$ is trivial).

Because g is continuously differentiable it is locally Lipschitz, so by the Picard–Lindelöf theorem the solution through any point is unique. The constant function $x \equiv x^*$ solves the equation, so the trajectory starting below x^* can never reach x^* in finite time, and *a fortiori* never crosses it: $x(t) < x^*$ for all t . Consequently $g(x(t)) > 0$ and $\dot{x}(t) > 0$ for all t , so $x(\cdot)$ is strictly increasing and bounded above by x^* . A monotone bounded function converges; let $L \equiv \lim_{t \rightarrow \infty} x(t) \leq x^*$.

Suppose $L < x^*$. By continuity $\dot{x}(t) = g(x(t)) \rightarrow g(L) > 0$, so there are T and $\eta > 0$ with $\dot{x}(t) \geq \eta$ for all $t \geq T$, whence $x(t) \geq x(T) + \eta(t - T) \rightarrow \infty$, contradicting $x(t) \leq x^*$. Therefore $L = x^*$, and $x(t) \rightarrow x^*$ from every initial condition: x^* is globally asymptotically stable.

Remark. Part 3 is what makes the continuous-time Solow model so much better behaved than its discrete-time counterpart. In discrete time a one-dimensional map can jump over its fixed point, so a sign condition on g alone does not preclude two-cycles; EXERCISE 2.21 constructs exactly such a cycle. In continuous time trajectories are continuous and cannot cross a rest point, which is why the analogue of Part 3 fails in discrete time.

Exercise 2.13

The law of motion. Labor income per worker is the wage $w(k) = f(k) - kf'(k)$, of which a fraction s is saved; capital income $f'(k)k$ is entirely consumed. Aggregate saving is therefore $sw(k(t))L(t)$, and with population growth at the rate n and depreciation δ the capital-labor ratio obeys

$$\dot{k}(t) = sw(k(t)) - (n + \delta)k(t), \quad \text{so} \quad \frac{\dot{k}(t)}{k(t)} = s\phi(k(t)) - (n + \delta), \quad \phi(k) \equiv \frac{w(k)}{k}.$$

Interior steady states are the solutions of

$$\phi(k) = \frac{w(k)}{k} = \frac{n + \delta}{s}. \quad (*)$$

The contrast with the baseline model is exactly the replacement of $f(k)/k$ by $w(k)/k$. In the baseline model $f(k)/k$ is strictly decreasing, which is what delivered uniqueness in Exercise 2.5. Nothing of the sort holds for $w(k)/k$.

When is ϕ increasing? Differentiating,

$$\phi'(k) = \frac{k w'(k) - w(k)}{k^2} = \frac{-k^2 f''(k) - w(k)}{k^2},$$

using $w'(k) = -k f''(k)$. Now recall that for a constant returns technology the elasticity of substitution (2.37) can be written as

$$\sigma(k) = \frac{f'(k) w(k)}{-k f(k) f''(k)},$$

so that $-k^2 f''(k) = \alpha_K(k) w(k)/\sigma(k)$ with $\alpha_K(k) \equiv k f'(k)/f(k)$ the share of capital in national income. Substituting,

$$\phi'(k) = \frac{w(k)}{k^2} \left[\frac{\alpha_K(k)}{\sigma(k)} - 1 \right], \quad \text{hence} \quad \phi'(k) > 0 \iff \sigma(k) < \alpha_K(k).$$

So $w(k)/k$ rises precisely where capital and labor are hard to substitute relative to the capital share. Intuitively, when σ is small an increase in capital raises the wage more than proportionately, because labor becomes the scarce complementary factor; saving out of wages then feeds back into more capital, and the economy can have several rest points.

An explicit example. Let

$$f(k) = A[\gamma k^\rho + (1 - \gamma)]^{1/\rho} + Bk^\alpha, \quad \rho = \frac{\sigma - 1}{\sigma} < 0, \quad \alpha \in (0, 1), \quad B > 0 :$$

a CES technology with a low elasticity of substitution plus a small Cobb–Douglas term. Both terms are increasing, strictly concave and homogeneous of degree one in (K, L) when written as $Lf(K/L)$, so ASSUMPTION 1 holds. The CES term alone would violate ASSUMPTION 2, since $f'_{\text{CES}}(0) = A\gamma^{1/\rho} < \infty$ and $w_{\text{CES}}(\infty) = A(1 - \gamma)^{1/\rho} < \infty$; the Cobb–Douglas term repairs both, because Bk^α contributes $B\alpha k^{\alpha-1} \rightarrow \infty$ to $f'(k)$ as $k \rightarrow 0$ and $(1 - \alpha)Bk^\alpha \rightarrow \infty$ to $w(k)$ as $k \rightarrow \infty$. So f satisfies ASSUMPTIONS 1 and 2 in full, and

$$w(k) = A(1 - \gamma)[\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho} + (1 - \alpha)Bk^\alpha.$$

Take $A = 1$, $\gamma = 1/2$, $\sigma = 1/4$ (so $\rho = -3$), $B = 0.02$ and $\alpha = 1/2$. Then $\phi(k) = w(k)/k$ falls from $+\infty$ to a local minimum $\phi = 0.03005$ at $k = 0.1509$, rises to a local maximum $\phi = 0.60360$ at $k = 1.4340$, and falls back to 0; the turning points are exactly where $\sigma(k)$ crosses $\alpha_K(k)$:

k	0.05	0.30	1.00	3.00
$\sigma(k)$	1.8871	0.3581	0.2537	0.3264
$\alpha_K(k)$	0.9667	0.9603	0.5000	0.0483
$\phi'(k)$	-0.438	+0.286	+0.495	-0.115

Consequently $(*)$ has *three* solutions for every $(n + \delta)/s$ strictly between 0.03005 and 0.60360. With $(n + \delta)/s = 0.045$ they are

$$k_1^* = 0.04972, \quad k_2^* = 0.27684, \quad k_3^* = 29.19752,$$

and since $\dot{k}$ has the sign of $\phi(k) - (n + \delta)/s$, the outer two are locally asymptotically stable ($\phi'(k_1^*) = -0.442$, $\phi'(k_3^*) = -0.002$) while the middle one is unstable ($\phi'(k_2^*) = +0.240$). Figure 2 plots ϕ and the horizontal line. The economy converges to the poor or to the rich steady state according to whether it starts below or above k_2^* : with saving out of wages alone, the Solow model produces a poverty trap.

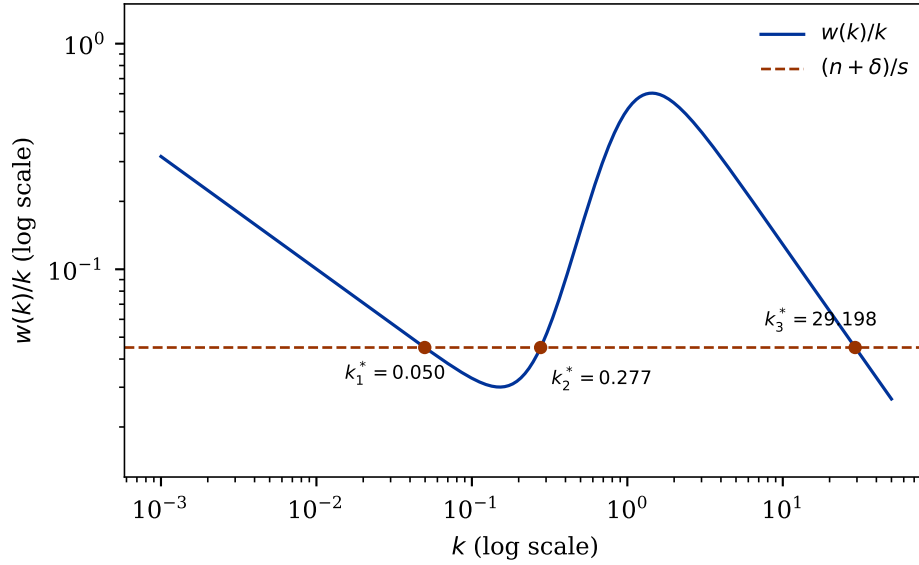


FIGURE 2. Saving out of labor income only: steady states are the intersections of $w(k)/k$ with the horizontal line $(n + \delta)/s$, drawn here at 0.045, for the technology of the text.

Remark. The comparison with EXERCISE 2.12 is instructive. When only capital income is saved, the law of motion is $\dot{k}/k = s_K f'(k) - (n + \delta)$, and f' is strictly decreasing by ASSUMPTION 1: the steady state is unique and globally stable, whatever the elasticity of substitution. It is the wage — the factor price that *rises* with capital — that can generate the self-reinforcing accumulation behind multiplicity.

Exercise 2.15

With constant returns, output per worker and the wage are both functions of the capital-labor ratio alone:

$$y = f(k), \quad w = F_L = f(k) - k f'(k),$$

the second equality being the competitive wage. Along the family of economies indexed by k ,

$$\frac{dy}{dk} = f'(k), \quad \frac{dw}{dk} = f'(k) - f'(k) - k f''(k) = -k f''(k) > 0,$$

so w is a strictly increasing function of k and can be inverted; by the chain rule,

$$\frac{\partial y}{\partial w} = \frac{dy/dk}{dw/dk} = \frac{f'(k)}{-k f''(k)}.$$

Therefore

$$\varepsilon_{y,w} = \frac{\partial y}{\partial w} \cdot \frac{w}{y} = \frac{f'(k)}{-kf''(k)} \cdot \frac{f(k) - kf'(k)}{f(k)} = \frac{f'(k)[f(k) - kf'(k)]}{-kf(k)f''(k)}. \quad (*)$$

It remains to check that the right-hand side of (*) is the elasticity of substitution (2.37). With constant returns, $F_K = f'(k)$ and $F_L = f(k) - kf'(k)$, so

$$\frac{\partial \log(F_K/F_L)}{\partial \log k} = k \frac{d}{dk} \left[\log f'(k) - \log(f(k) - kf'(k)) \right] = k \left[\frac{f''(k)}{f'(k)} + \frac{kf''(k)}{f(k) - kf'(k)} \right].$$

Putting the bracket over a common denominator,

$$\frac{\partial \log(F_K/F_L)}{\partial \log k} = kf''(k) \frac{[f(k) - kf'(k)] + kf'(k)}{f'(k)[f(k) - kf'(k)]} = \frac{kf(k)f''(k)}{f'(k)[f(k) - kf'(k)]},$$

and hence, by the definition (2.37),

$$\sigma = - \left[\frac{\partial \log(F_K/F_L)}{\partial \log(K/L)} \right]^{-1} = \frac{f'(k)[f(k) - kf'(k)]}{-kf(k)f''(k)},$$

which is exactly (*). Therefore $\varepsilon_{y,w} = \sigma$.

The result is the basis of the empirical strategy of Arrow et al. (1961) taken up in EXERCISE 2.16: regressing the logarithm of output per capita on the logarithm of the wage across economies (or across time) estimates the elasticity of substitution directly, and a slope of 1 — which is what those authors found to be a good approximation — says that the technology is Cobb–Douglas. It also gives a clean reading of σ : it measures how much richer, in output per worker, an economy must be in order to sustain a given proportional increase in its wage.

Exercise 2.24

Normalize the technology terms to $A_H = A_K = A_L = 1$ (they only rescale the arguments and the level, and none of the limits below depends on them) and write the per capita form of (2.38) as

$$f(k) = [\gamma k^\rho + (1 - \gamma)]^{1/\rho}, \quad \rho \equiv \frac{\sigma - 1}{\sigma} \in (-\infty, 1), \quad \gamma \in (0, 1),$$

so that $\sigma > 1 \iff \rho \in (0, 1)$, $\sigma < 1 \iff \rho < 0$ and $\sigma = 1$ is the Cobb–Douglas limit $f(k) = k^\gamma$. Differentiating,

$$f'(k) = \gamma k^{\rho-1} [\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho}, \quad w(k) = f(k) - kf'(k) = (1 - \gamma) [\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho}.$$

Recall that ASSUMPTION 2 requires four limits — $F_K \rightarrow \infty$ as $K \rightarrow 0$, $F_K \rightarrow 0$ as $K \rightarrow \infty$, $F_L \rightarrow \infty$ as $L \rightarrow 0$, $F_L \rightarrow 0$ as $L \rightarrow \infty$ — together with $F(0, L, A) = 0$. In per capita terms $F_K = f'(k)$ and $F_L = w(k)$, and $L \rightarrow 0$ with K fixed means $k \rightarrow \infty$, while $L \rightarrow \infty$ means $k \rightarrow 0$.

Case $\sigma > 1$ (i.e. $0 < \rho < 1$). As $k \rightarrow 0$ the bracket tends to $1 - \gamma > 0$, so

$$f(0) = (1 - \gamma)^{1/\rho} > 0,$$

and capital is not essential: $F(0, L, A) > 0$, the last requirement of ASSUMPTION 2, fails. As $k \rightarrow \infty$ the bracket behaves like γk^ρ , so

$$f'(k) \longrightarrow \gamma k^{\rho-1} (\gamma k^\rho)^{(1-\rho)/\rho} = \gamma^{1/\rho} > 0,$$

so $\lim_{K \rightarrow \infty} F_K = 0$ fails too: the marginal product of capital is bounded below by a positive number, and the technology is asymptotically linear in capital — an AK technology in the limit, which is exactly why EXERCISE 2.23(a) finds sustained growth in this case. Finally, as $k \rightarrow 0$ (that is, $L \rightarrow \infty$ for given K), the bracket tends to $1 - \gamma$ and

$$w(k) \rightarrow (1 - \gamma)(1 - \gamma)^{(1-\rho)/\rho} = (1 - \gamma)^{1/\rho} > 0,$$

so $\lim_{L \rightarrow \infty} F_L = 0$ fails as well: labor remains productive at the margin however abundant it is, because output does not require capital. (The two remaining limits, $f'(0) = \infty$ and $w(\infty) = \infty$, do hold, the latter because $w(k) \sim (1 - \gamma)\gamma^{(1-\rho)/\rho}k^{1-\rho}$ with $1 - \rho > 0$.)

Case $\sigma < 1$ (i.e. $\rho < 0$). Now $k^\rho \rightarrow \infty$ as $k \rightarrow 0$, so the bracket diverges and, since $1/\rho < 0$, $f(0) = 0$: capital is essential, as ASSUMPTION 2 wants. But

$$f'(k) \rightarrow \gamma k^{\rho-1} (\gamma k^\rho)^{(1-\rho)/\rho} = \gamma^{1/\rho} < \infty \quad \text{as } k \rightarrow 0,$$

so the Inada condition $\lim_{K \rightarrow 0} F_K = \infty$ fails. At the other end, as $k \rightarrow \infty$ the bracket tends to $1 - \gamma$ and

$$w(k) \rightarrow (1 - \gamma)^{1/\rho} < \infty,$$

so $\lim_{L \rightarrow 0} F_L = \infty$ fails as well. (The two remaining limits, $f'(\infty) = 0$ and $w(0) = 0$, do hold.)

Case $\sigma = 1$. Then $f(k) = k^\gamma$ with $\gamma \in (0, 1)$: $f(0) = 0$, $f'(k) = \gamma k^{\gamma-1} \rightarrow \infty$ as $k \rightarrow 0$ and $\rightarrow 0$ as $k \rightarrow \infty$, and $w(k) = (1 - \gamma)k^\gamma \rightarrow 0$ as $k \rightarrow 0$ and $\rightarrow \infty$ as $k \rightarrow \infty$. All the requirements of ASSUMPTION 2 hold.

Hence ASSUMPTION 2 holds for the CES technology if and only if $\sigma = 1$. The failure is economically meaningful rather than technical: with $\sigma > 1$ the two factors are good enough substitutes that capital alone can produce, and accumulating it never drives its marginal product to zero; with $\sigma < 1$ they are such poor substitutes that the first units of capital are not infinitely productive, and an economy with a fixed labor force cannot make the wage arbitrarily large.

Remark. ASSUMPTION 1, by contrast, holds for every finite $\sigma > 0$. Logarithmic differentiation of f' gives

$$\frac{f''(k)}{f'(k)} = \frac{\rho - 1}{k} + \frac{(1 - \rho)\gamma k^{\rho-1}}{\gamma k^\rho + (1 - \gamma)} = -\frac{(1 - \rho)(1 - \gamma)}{k[\gamma k^\rho + (1 - \gamma)]} < 0 \quad \text{for all } \rho < 1,$$

so $f'' < 0$, and $F_{KK}, F_{LL} < 0$ follow. The text on page 55 states instead that “the CES production function with $\sigma > 1$ violates Assumption 1”; what the computation above shows, and what this exercise asks one to prove, is that what fails is ASSUMPTION 2.

Exercise 2.25

With Harrod-neutral technological progress at the rate g and population growth at the rate n , the effective capital-labor ratio $k(t) = K(t)/(A(t)L(t))$ obeys (2.51), and the BGP value k^* solves (2.52),

$$H(k^*; s, \delta, n, g) \equiv s \frac{f(k^*)}{k^*} - (\delta + g + n) = 0.$$

The level of technology does not enter. $A(0)$ appears nowhere in this equation — it has been normalized away by the definition of k — so

$$\frac{\partial k^*(A(0), s, \delta, n, g)}{\partial A(0)} = 0.$$

The other comparative statics. Exactly as in Exercise 2.5,

$$\frac{\partial H}{\partial k^*} = s \cdot \frac{d}{dk} \left[\frac{f(k)}{k} \right]_{k=k^*} = -\frac{s w(k^*)}{(k^*)^2} < 0,$$

where $w(k) = f(k) - k f'(k) > 0$. The Implicit Function Theorem applies and gives, with $\partial H / \partial s = f(k^*) / k^* > 0$ and $\partial H / \partial \delta = \partial H / \partial n = \partial H / \partial g = -1$,

$$\frac{\partial k^*}{\partial s} = \frac{k^* f(k^*)}{s w(k^*)} > 0, \quad \frac{\partial k^*}{\partial n} = \frac{\partial k^*}{\partial \delta} = \frac{\partial k^*}{\partial g} = -\frac{(k^*)^2}{s w(k^*)} < 0.$$

Output per capita. Along the BGP, income per capita is $y^*(t) = A(t)f(k^*) = A(0)e^{gt}f(k^*)$ by (2.50). Hence, for each t ,

$$\frac{\partial y^*}{\partial A(0)} = e^{gt}f(k^*) > 0,$$

which is the one comparative static that is new relative to PROPOSITION 2.8: a higher initial technology level leaves the effective capital-labor ratio untouched but raises income per capita proportionately, at every date. For the remaining parameters the effect runs through k^* alone,

$$\frac{\partial y^*}{\partial x} = A(0)e^{gt}f'(k^*)\frac{\partial k^*}{\partial x}, \quad x \in \{s, \delta, n\},$$

which is positive for s and negative for δ and n , since $f' > 0$.

Remark. The proposition lists no comparative static with respect to g , and for good reason: a higher g lowers k^* and hence the level of income per capita at any fixed date, but it raises its growth rate, so $\partial y^*(t) / \partial g$ changes sign as t grows. This is the first appearance of a theme that recurs throughout the book — level effects and growth effects need not point the same way.

Exercise 2.26

Let ASSUMPTIONS 1 and 2 hold and consider the law of motion (2.51) of the effective capital-labor ratio,

$$\dot{k}(t) = G(k(t)), \quad G(k) \equiv sf(k) - (\delta + g + n)k.$$

G is continuously differentiable on $(0, \infty)$ by ASSUMPTION 1, and by PROPOSITION 2.11 it has a unique zero $k^* \in (0, \infty)$, characterized by $f(k^*)/k^* = (\delta + g + n)/s$.

Write $G(k) = k[s\psi(k) - (\delta + g + n)]$ with $\psi(k) \equiv f(k)/k$. As established in Exercise 2.5, ψ is strictly decreasing, with $\psi(k) \rightarrow \infty$ as $k \rightarrow 0$ and $\psi(k) \rightarrow 0$ as $k \rightarrow \infty$. Therefore

$$k < k^* \implies \psi(k) > \psi(k^*) = \frac{\delta + g + n}{s} \implies G(k) > 0,$$

$$k > k^* \implies \psi(k) < \psi(k^*) \implies G(k) < 0.$$

That is exactly the hypothesis of part 3 of COROLLARY 2.2 (proved in Exercise 2.10), which therefore gives global asymptotic stability: starting from any $k(0) > 0$, $k(t) \rightarrow k^*$. Moreover the convergence is monotone, since $\dot{k}$ keeps a constant sign along the path: an economy that starts

below its BGP accumulates effective capital throughout the transition, and one that starts above decumulates it.

Local stability can also be read off the derivative at the rest point. Using $sf(k^*) = (\delta + g + n)k^*$,

$$G'(k^*) = sf'(k^*) - (\delta + g + n) = sf'(k^*) - \frac{sf(k^*)}{k^*} = -\frac{sw(k^*)}{k^*} < 0,$$

because the wage $w(k^*) = f(k^*) - k^*f'(k^*)$ is strictly positive. So the linearization has a strictly negative eigenvalue, confirming local asymptotic stability by part 2 of COROLLARY 2.2, and giving in addition the speed of convergence: near the BGP the gap $k(t) - k^*$ closes at the exponential rate $sw(k^*)/k^* = (\delta + g + n)\alpha_L(k^*)$, where $\alpha_L = w/f$ is the share of labor in national income.

Since $k(t) \rightarrow k^*$ and $y(t) = A(t)f(k(t))$, output per capita converges to the balanced growth path $A(0)e^{gt}f(k^*)$ and grows asymptotically at the rate g : with Harrod-neutral progress, transitional dynamics and long-run growth separate cleanly, and the model behaves exactly as it did without technological change once variables are measured per efficiency unit of labor.

Chapter 3

The Solow Model and the Data

The seven exercises of Chapter 3 left unsolved in the instructor's manual: Exercises 3.3–3.8 and 3.11. The manual solves 3.1, 3.2, 3.9 and 3.10. Notation follows the book: $k \equiv K/(AL)$ is the effective capital-labor ratio, $\hat{y} \equiv Y/(AL)$, $y \equiv Y/L$, $y^*(t) \equiv A(t)f(k^*)$ is the steady-state (BGP) level of output per capita, and $\varepsilon_k(k) \equiv f'(k)k/f(k)$ is the elasticity of the per capita production function (3.9). The numerical parts are computed by `code/ch3.py`.

Exercise 3.3

Write the technology in labor-augmenting form, $Y = K^\alpha(AL)^{1-\alpha}$, so that in terms of the effective capital-labor ratio $k = K/(AL)$,

$$\hat{y} = k^\alpha, \quad y = Ak^\alpha, \quad \frac{\dot{k}}{k} = sk^{\alpha-1} - (\delta + g + n), \quad k^* = \left(\frac{s}{\delta + g + n} \right)^{\frac{1}{1-\alpha}}.$$

The exact equation. Differentiating $y = Ak^\alpha$ logarithmically and substituting the law of motion, with no approximation at all,

$$\frac{\dot{y}}{y} = g + \alpha \frac{\dot{k}}{k} = g + \alpha [sk^{\alpha-1} - (\delta + g + n)] = g + \alpha(\delta + g + n) \left[\left(\frac{k}{k^*} \right)^{\alpha-1} - 1 \right],$$

the last step using $s(k^*)^{\alpha-1} = \delta + g + n$. Since $y/y^* = (k/k^*)^\alpha$, we have $\log k - \log k^* = \alpha^{-1}(\log y - \log y^*)$, and therefore

$$\boxed{\frac{\dot{y}(t)}{y(t)} = g + \alpha(\delta + g + n) \left[\exp\left(-\frac{1-\alpha}{\alpha} (\log y(t) - \log y^*(t)) \right) - 1 \right]} \quad (*)$$

which is the exact analogue of (3.10). Expanding the exponential to first order around $\log y = \log y^*$ returns $\dot{y}/y \approx g - (1-\alpha)(\delta + g + n)(\log y - \log y^*)$, which is (3.11): the book's equation is the linearization of (*).

The rate of convergence is not a constant. Write $z \equiv \log y(t) - \log y^*(t)$ for the gap and define the convergence coefficient implicitly by $\dot{y}/y = g - \beta(z)z$, so that from (*)

$$\beta(z) = \alpha(\delta + g + n) \frac{1 - e^{-cz}}{z}, \quad c \equiv \frac{1-\alpha}{\alpha} > 0.$$

The function $\varphi(u) \equiv (1 - e^{-u})/u$ has $\varphi'(u) = e^{-u}[u + 1 - e^u]/u^2 < 0$ for $u \neq 0$, because $e^u > 1 + u$; and $\varphi(u) \rightarrow 1$ as $u \rightarrow 0$. Hence

$$\beta \text{ is strictly decreasing in } z, \quad \beta(0) = \alpha(\delta + g + n)c = (1-\alpha)(\delta + g + n),$$

$$\beta(z) \rightarrow \infty \text{ as } z \rightarrow -\infty, \quad \beta(z) \rightarrow 0 \text{ as } z \rightarrow +\infty.$$

So for a country *below* its steady state ($z < 0$, the empirically relevant case) the true speed of convergence *exceeds* the constant $(1-\alpha)(\delta + g + n)$ of the linearized equation, and it does so by more the poorer the country is; for a country above its steady state convergence is slower than the

linear approximation says. With the parameter values of EXAMPLE 3.1 ($\alpha = 1/3$, $\delta + g + n = 0.08$, so $(1 - \alpha)(\delta + g + n) = 0.0533$):

y/y^*	0.10	0.25	0.50	0.80	1.25	2.00	10.0
β	1.1465	0.2885	0.1154	0.0672	0.0430	0.0289	0.0115
$\beta/\beta(0)$	21.5	5.41	2.16	1.26	0.81	0.54	0.21

A country at a tenth of its steady-state income converges twenty times faster than the log-linear coefficient suggests. Figure 3 plots β .

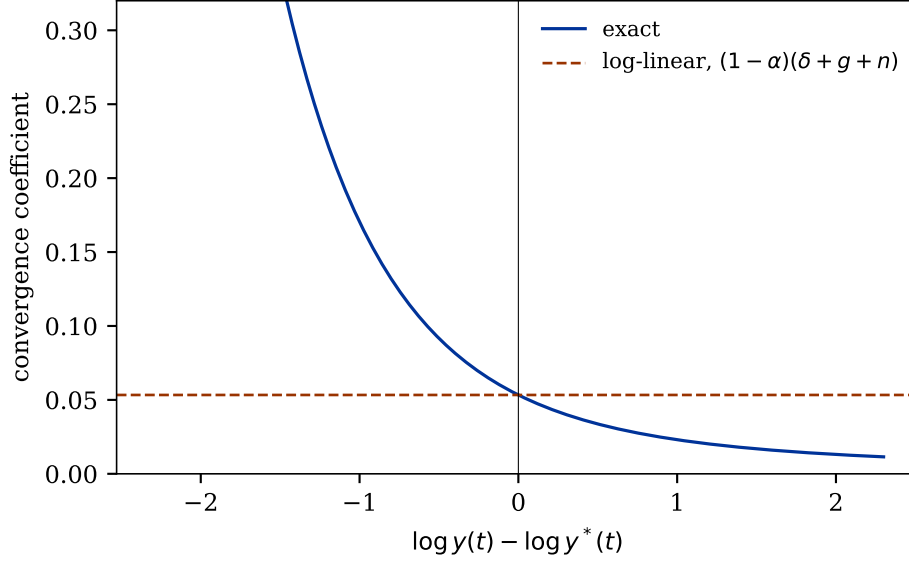


FIGURE 3. The exact convergence coefficient β as a function of the gap $\log y - \log y^*$, against the constant coefficient of the log-linearized equation (3.11).

Remark. This is a first hint of why growth regressions of the form (3.12)–(3.14), which impose a *constant* coefficient b_1 on $\log y_{i,t-1}$, are hard to interpret. Even within the Solow model with a common technology, the correct coefficient varies systematically with how far a country is from its own steady state, so the estimated b_1 is some average of very different speeds — and the average depends on the composition of the sample.

Exercise 3.4

Log-linear answer. By (3.11), each country's gap $z_i(t) \equiv \log y_i(t) - \log y^*(t)$ obeys $\dot{z}_i = -\beta z_i$ with $\beta = (1 - \alpha)(\delta + g + n)$. Since the two countries share $y^*(t)$, the *relative* gap $D(t) \equiv \log y_1(t) - \log y_2(t) = z_1(t) - z_2(t)$ obeys the same linear equation,

$$\dot{D} = -\beta D \quad \implies \quad D(t) = D(0)e^{-\beta t}.$$

With $\alpha = 1/3$, $g = 0.02$, $n = 0.01$ and $\delta = 0.05$ we get $\beta = \frac{2}{3} \times 0.08 = 0.0533$ per year. Starting from $D(0) = \log 2 = 0.6931$ and requiring $D(T) = \log 1.1 = 0.0953$,

$$T = \frac{1}{\beta} \log \frac{\log 2}{\log 1.1} = \frac{\log(7.273)}{0.0533} = \frac{1.9843}{0.0533} \approx 37.2 \text{ years.}$$

The half-life of the gap is $\log 2/\beta = 13.0$ years, which is the “little more than 10 years” quoted in EXAMPLE 3.1.

Exact answer. The Cobb–Douglas model can be solved in closed form, so the approximation can be checked. With $x \equiv k^{1-\alpha}$ the law of motion is linear, $\dot{x} = (1 - \alpha)[s - (\delta + g + n)x]$, so

$$\frac{x(t)}{x^*} = 1 + \left(\frac{x(0)}{x^*} - 1\right)e^{-\beta t}, \quad \frac{y(t)}{y^*(t)} = \left(\frac{x(t)}{x^*}\right)^{\frac{\alpha}{1-\alpha}},$$

with the *same* $\beta = (1 - \alpha)(\delta + g + n)$ now appearing exactly. The log gap between the two countries is no longer exactly exponential, because the map from x to y is concave, so the answer depends on where the pair sits relative to the common steady state:

country 1 starts at $y_1(0)/y^*$	0.8	1.0	2.0
years until $y_1/y_2 = 1.1$	21.4	27.4	49.9

The log-linear figure of 37 years is of the right order but not accurate: when both countries are below the steady state the true convergence is faster (Exercise 3.3), and when both are above it is slower.

Either way the message is the one the book draws: the basic Solow model with a capital share of one third predicts that a twofold income gap between otherwise identical countries is largely gone within a few decades. That is far faster than anything in the data — the gaps documented in Chapter 1 have persisted for a century and more — which is one of the motivations for the augmented model of Section 3.3 (where a larger effective capital share slows convergence) and, ultimately, for models in which y^* itself differs permanently across countries.

Exercise 3.5

Index economies by i and let each be a Solow economy with its own saving rate s_i , depreciation rate δ_i , population growth rate n_i and technology level $A_i(t)$, with a common rate of technological progress g (so that a stable world income distribution is possible). Each economy in isolation obeys (3.7), so its steady-state effective capital-labor ratio k_i^* solves

$$s_i \frac{f(k_i^*)}{k_i^*} = \delta_i + g + n_i,$$

and its steady-state output per capita is $y_i^*(t) = A_i(t)f(k_i^*)$.

Repeating the derivation of (3.10) economy by economy — nothing in it used cross-country homogeneity — gives the country-specific convergence equation

$$\frac{\dot{y}_i(t)}{y_i(t)} \approx g - (1 - \varepsilon_k(k_i^*))(\delta_i + g + n_i)(\log y_i(t) - \log y_i^*(t)).$$

Writing $\beta_i \equiv (1 - \varepsilon_k(k_i^*))(\delta_i + g + n_i)$ and discretizing between $t - 1$ and t ,

$$g_{i,t,t-1} \approx \underbrace{[g + \beta_i \log y_i^*(t - 1)]}_{b_i^0} - \beta_i \log y_{i,t-1}. \quad (*)$$

This is (3.13), $g_{i,t,t-1} = b_i^0 + b_1 \log y_{i,t-1} + \varepsilon_{i,t}$, with $b_1 = -\beta_i$: the constant term is country specific, precisely because the steady state y_i^* is. Two observations complete the derivation.

The slope is common when the elasticity is. With a Cobb–Douglas technology $\varepsilon_k \equiv \alpha$ and $\beta_i = (1 - \alpha)(\delta_i + g + n_i)$, so the slope varies across countries only through $\delta_i + n_i$. Since $\delta + n$ moves within a narrow range across countries while y_i^* varies enormously, treating b_1 as common and b_i^0 as country specific — exactly what (3.13) does — is the natural first approximation. (That b_1 is in principle country specific is the caveat the book records after (3.13).)

The intercept is a function of observable country characteristics. With Cobb–Douglas technology $y_i^*(t) = A_i(t)[s_i/(\delta_i + g + n_i)]^{\alpha/(1-\alpha)}$, so

$$b_i^0 = g + (1 - \alpha)(\delta_i + g + n_i) \left[\log A_i(t-1) + \frac{\alpha}{1-\alpha} (\log s_i - \log(\delta_i + g + n_i)) \right].$$

Modeling b_i^0 as a linear function of $\log s_i$, $\log(\delta_i + g + n_i)$ and proxies for $\log A_i$ — schooling, institutions, openness, and the rest — delivers the empirical specification (3.14), $g_{i,t-1} = X_{i,t}^T \beta + b_1 \log y_{i,t-1} + \varepsilon_{i,t}$. Conditional convergence is then the statement that $b_1 < 0$ once these characteristics are held fixed, whereas unconditional convergence — the regression (3.12) with a common intercept — need not show $b_1 < 0$ at all: omitting b_i^0 puts it in the error term, and since economies with low y_i^* also tend to have low $y_{i,t-1}$, the resulting $\text{Cov}(b_i^0, \log y_{i,t-1}) > 0$ biases the estimate of b_1 upward.

Exercise 3.6

The system to be studied is

$$\dot{k} = G_k(k, h) \equiv s_k f(k, h) - (\delta_k + g + n)k, \quad \dot{h} = G_h(k, h) \equiv s_h f(k, h) - (\delta_h + g + n)h,$$

on $\mathbb{R}_{++}^2$, where $f(k, h) = F(k, h, 1)$ is strictly increasing and strictly concave in (k, h) by ASSUMPTION 1' and satisfies the Inada conditions of ASSUMPTION 2'. By PROPOSITION 3.1 there is exactly one rest point $(k^*, h^*) \in \mathbb{R}_{++}^2$. We must show that every solution starting in $\mathbb{R}_{++}^2$ converges to it.

Step 1: the system is cooperative, hence the flow is monotone. The off-diagonal entries of the Jacobian are

$$\frac{\partial G_k}{\partial h} = s_k f_h(k, h) > 0, \quad \frac{\partial G_h}{\partial k} = s_h f_k(k, h) > 0,$$

so the vector field is *cooperative*: each variable helps the other accumulate — more human capital raises output and therefore physical investment, and conversely. By Kamke's comparison theorem the flow preserves the componentwise order: if $(k_1(0), h_1(0)) \leq (k_2(0), h_2(0))$ then $(k_1(t), h_1(t)) \leq (k_2(t), h_2(t))$ for all $t \geq 0$. Two consequences will be used repeatedly. First, a solution that starts above (below) (k^*, h^*) stays above (below) it, since (k^*, h^*) is itself a solution. Second, if the vector field is nonpositive in both coordinates at the initial point, then the solution is nonincreasing in both coordinates for all t : the solution started at time $\tau > 0$ lies below the one started at time 0, and both solve the same autonomous equation.

Step 2: a supersolution and a subsolution. Recall from the proof of PROPOSITION 3.1 that the loci $\{G_k = 0\}$ and $\{G_h = 0\}$ are increasing curves in (k, h) space that cross exactly once, at (k^*, h^*) , with $\{G_k = 0\}$ steeper there; moreover $\{G_k = 0\}$ lies below $\{G_h = 0\}$ for small (k, h) and above it for large (k, h) . Since G_k is increasing in h and G_h is increasing in k ,

$$G_k < 0 \text{ to the right of } \{G_k = 0\}, \quad G_h < 0 \text{ above } \{G_h = 0\}.$$

For $k > k^*$ the region between the two curves is therefore a set on which $G_k < 0$ and $G_h < 0$; it is nonempty and, because both curves are increasing and unbounded (by ASSUMPTION 2', $f(k, h)/k \rightarrow 0$ as $k \rightarrow \infty$ forces $h \rightarrow \infty$ along $\{G_k = 0\}$), it contains points $(\bar{k}, \bar{h})$ with $\bar{k} \geq k(0)$ and $\bar{h} \geq h(0)$, as large as we please. Symmetrically, for $k < k^*$ the region between the two curves has $G_k > 0$ and $G_h > 0$ and contains points $(\underline{k}, \underline{h})$ with $\underline{k} \leq k(0)$, $\underline{h} \leq h(0)$, as small as we please.

Step 3: the sandwich. Consider the solution started at $(\bar{k}, \bar{h})$. By Step 1 it is nonincreasing in both coordinates, and it is bounded below by (k^*, h^*) — it started above the rest point, so it stays above. A monotone bounded trajectory converges, and its limit is a rest point of the system; by uniqueness, the limit is (k^*, h^*) . Exactly the same argument applied at $(\underline{k}, \underline{h})$ gives a trajectory that is nondecreasing, bounded above by (k^*, h^*) , and convergent to it. Finally, by monotonicity of the flow, the solution started at $(k(0), h(0))$ is trapped between these two for all $t \geq 0$:

$$(\underline{k}(t), \underline{h}(t)) \leq (k(t), h(t)) \leq (\bar{k}(t), \bar{h}(t)),$$

and both bounds converge to (k^*, h^*) . Hence $(k(t), h(t)) \rightarrow (k^*, h^*)$ from any initial condition in $\mathbb{R}_{++}^2$, which is the assertion of PROPOSITION 3.2.

Remark. The phase diagram in Figure 3.1 of the book is exactly this argument in pictures: the two loci cut the positive quadrant into four regions, in two of which both variables move toward the steady state, while in the other two the trajectory must cross into one of those. What the monotone-flow argument adds is the step the picture cannot supply — that no trajectory can cycle around the rest point instead of converging to it. Cooperativity is what rules that out; for planar cooperative systems there are no closed orbits.

Exercise 3.7

The steady state solves (3.17)–(3.18),

$$\Phi_1(k, h) \equiv s_k f(k, h) - (\delta_k + g + n)k = 0, \quad \Phi_2(k, h) \equiv s_h f(k, h) - (\delta_h + g + n)h = 0.$$

Write J for the Jacobian of (Φ_1, Φ_2) in (k, h) , evaluated at (k^*, h^*) :

$$J = \begin{pmatrix} s_k f_k - (\delta_k + g + n) & s_k f_h \\ s_h f_k & s_h f_h - (\delta_h + g + n) \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}.$$

Signs of the entries and of $\det J$. The off-diagonal entries are positive. For the diagonal ones, substitute $\delta_k + g + n = s_k f(k^*, h^*)/k^*$ from (3.17):

$$J_{11} = s_k \left[f_k(k^*, h^*) - \frac{f(k^*, h^*)}{k^*} \right] < 0,$$

because strict concavity of f in k together with $f(0, h^*) \geq 0$ gives $f > k f_k$; similarly $J_{22} < 0$. For the determinant, substituting both steady-state conditions,

$$\det J = (\delta_k + g + n)(\delta_h + g + n) - s_k f_k(\delta_h + g + n) - s_h f_h(\delta_k + g + n) = \frac{s_k s_h f}{k^* h^*} [f - k^* f_k - h^* f_h] > 0,$$

the bracket being positive by strict concavity of f (this is the same inequality that made the two loci cross only once in PROPOSITION 3.1).

The comparative statics. By the Implicit Function Theorem, $\partial(k^*, h^*)^T / \partial \theta = -J^{-1} \partial(\Phi_1, \Phi_2)^T / \partial \theta$ with

$$J^{-1} = \frac{1}{\det J} \begin{pmatrix} J_{22} & -J_{12} \\ -J_{21} & J_{11} \end{pmatrix}.$$

Since $\partial\Phi/\partial s_k = (f, 0)^T$, $\partial\Phi/\partial s_h = (0, f)^T$ and $\partial\Phi/\partial n = (-k^*, -h^*)^T$,

$$\begin{aligned} \frac{\partial k^*}{\partial s_k} &= \frac{-J_{22}f}{\det J} > 0, & \frac{\partial h^*}{\partial s_k} &= \frac{J_{21}f}{\det J} = \frac{s_h f_k f}{\det J} > 0, \\ \frac{\partial k^*}{\partial s_h} &= \frac{J_{12}f}{\det J} = \frac{s_k f_h f}{\det J} > 0, & \frac{\partial h^*}{\partial s_h} &= \frac{-J_{11}f}{\det J} > 0, \\ \frac{\partial k^*}{\partial n} &= \frac{J_{22}k^* - J_{12}h^*}{\det J} < 0, & \frac{\partial h^*}{\partial n} &= \frac{J_{11}h^* - J_{21}k^*}{\det J} < 0, \end{aligned}$$

the last two because $J_{11}, J_{22} < 0$ while $J_{12}, J_{21} > 0$. Since f is increasing in both arguments, output per effective unit of labor $\hat{y}^* = f(k^*, h^*)$ inherits the signs: it rises with either saving rate and falls with population growth.

Interpretation. The novelty relative to the basic model is the *cross* effect: a higher saving rate in physical capital raises the steady-state stock of *human* capital as well, and vice versa. The mechanism is the complementarity of the two factors in f — more physical capital raises output, and since a constant fraction s_h of output is invested in schooling, it raises human capital investment too. This is the “multiplier” that makes the augmented model fit cross-country income differences so much better than the basic one: the elasticity of steady-state income with respect to s_k is $\alpha/(1 - \alpha - \beta)$ rather than $\alpha/(1 - \alpha)$, as (3.22)–(3.23) show explicitly for the Cobb–Douglas case.

Exercise 3.8

The accounting equation. With $Y = F(K, H, AL)$, differentiate with respect to time and divide by Y :

$$\frac{\dot{Y}}{Y} = \frac{F_K K}{Y} \frac{\dot{K}}{K} + \frac{F_H H}{Y} \frac{\dot{H}}{H} + \frac{F_{AL} AL}{Y} \left(\frac{\dot{A}}{A} + \frac{\dot{L}}{L} \right),$$

where F_{AL} is the derivative with respect to the third argument. With competitive factor markets the elasticities are the observed income shares: $R = F_K$ gives $\alpha_K = F_K K/Y$; the rental rate of human capital $R_H = F_H$ gives $\alpha_H = F_H H/Y$; and the wage per worker is $w = AF_{AL}$, so $\alpha_L = wL/Y = F_{AL} AL/Y$. Constant returns in the three arguments make the three shares sum to one. Writing g_X for the growth rate of X ,

$$g_Y = \alpha_K g_K + \alpha_H g_H + \alpha_L (g_A + g_L)$$

so that the estimate of TFP growth — the part of output growth not accounted for by measured inputs — is

$$\hat{x} = g_Y - \alpha_K g_K - \alpha_H g_H - \alpha_L g_L, \quad \text{and} \quad g_A = \frac{\hat{x}}{\alpha_L}.$$

Two differences from the fundamental growth accounting equation (3.3) of the basic model deserve emphasis. First, the accumulation of human capital is now an *input*, so growth in schooling and skills is no longer part of the residual. Second, because technology is labor augmenting, the residual $\hat{x}$ measures $\alpha_L g_A$, not g_A : with a labor share of about 2/3, the implied rate of technological progress is one and a half times the measured residual.

Using data. One needs (i) output growth g_Y , from national accounts; (ii) capital stock growth g_K , from a perpetual inventory calculation on investment series; (iii) growth in efficiency units of labor g_H , constructed as in Section 3.5 from schooling data and Mincerian returns, and g_L from employment; and (iv) the three factor shares. The last item is the binding constraint, and it is where the exercise bites: national accounts report a split of value added into capital income and labor income, but labor income here is the sum of two conceptually different payments, $R_H H$ and wL . Some identifying assumption is unavoidable. The two used in practice are

- impose a functional form and calibrate: with $Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}$ as in (3.21), the shares are the constants α , β and $1 - \alpha - \beta$, and β is pinned down from microeconomic (Mincerian) estimates of the return to schooling; or
- avoid the split altogether by folding human capital into effective labor, $Y = F(K, AH)$, as in (3.29): then only two shares are needed, α_K and $1 - \alpha_K$, and the accounting equation collapses to $g_Y = \alpha_K g_K + (1 - \alpha_K)(g_A + g_H)$, at the price of imposing that raw labor and human capital are perfect substitutes in efficiency units.

Either way the residual absorbs everything the measurement misses — quality of schooling, capacity utilization, reallocation across sectors and firms, and pure measurement error — so it remains, in Abramovitz’s phrase, a measure of our ignorance; adding human capital narrows that ignorance rather than eliminating it.

Exercise 3.11

(a) *The accounting.* Work in per worker terms, $y_j \equiv Y_j/L_j$ and $\kappa_j \equiv K_j/L_j$. Constant returns make output per worker a function of the capital-labor ratio and technology alone, $y = \phi(\kappa; A)$, and competitive markets make the elasticity of that function equal to the *observed* capital share:

$$\frac{\partial \log y}{\partial \log \kappa} = \frac{F_K K}{Y} = \alpha^K.$$

This is all that is needed: no functional form, no assumption about how A enters. Order the countries by their capital-labor ratio, $\kappa_{(1)} \leq \kappa_{(2)} \leq \dots \leq \kappa_{(J)}$. If two adjacent countries had the same technology, the whole of their income difference would have to come from capital deepening,

$$\Delta \log y_{\text{capital}} = \bar{\alpha}^K \Delta \log \kappa, \quad \bar{\alpha}^K = \frac{1}{2}(\alpha_{(i)}^K + \alpha_{(i+1)}^K),$$

a Divisia (Törnqvist) index that uses the average of the two observed shares over the interval. The residual is the estimated technology gap,

$$\log \frac{A_{(i+1)}}{A_{(i)}} = (\log y_{(i+1)} - \log y_{(i)}) - \bar{\alpha}^K (\log \kappa_{(i+1)} - \log \kappa_{(i)}),$$

and cumulating these differences along the ordered cross section gives every country’s productivity relative to a numeraire country (the United States, say). Equivalently, and in one step for any pair,

$$\log \frac{A_j}{A_{\text{US}}} = \log \frac{y_j}{y_{\text{US}}} - \bar{\alpha}^K \log \frac{\kappa_j}{\kappa_{\text{US}}}.$$

The contribution of technology to cross-country income differences is then measured by the dispersion of $\log A_j$ relative to the dispersion of $\log y_j$, and the “predicted” income of country j

under a common technology is $\log \hat{y}_j = \log y_{\text{US}} + \bar{\alpha}^K \log(\kappa_j/\kappa_{\text{US}})$. (With a common Cobb–Douglas technology $\alpha_j^K \equiv \alpha$ and this is exactly the calibration of Section 3.5.) Only if the shares vary across countries does the chaining matter, and then the index is exact only up to the usual path dependence, which is second order when the share varies slowly.

(b) *What do large A differences mean?* Not necessarily that countries have access to different production possibility sets. The residual A_j is defined as whatever makes the accounting identity hold, so it absorbs every reason why a country produces less from the *measured* inputs than the numeraire country does:

- unmeasured or mismeasured inputs — human capital (part (c)), land and natural resources, capital quality and capacity utilization, hours per worker;
- allocative efficiency — misallocation of capital and labor across firms and sectors, distortions, credit constraints, monopoly power, informality;
- organization and institutions — the efficiency with which given inputs are combined;
- genuinely different technical knowledge or its adoption.

Only the last is technology in the engineering sense. Since a country can, in principle, adopt known technologies, the finding of very large A gaps is better read as a statement that *something other than the accumulation of measured factors* accounts for most of cross-country income differences — which is precisely what motivates Chapter 4’s search for fundamental causes.

(c) *What data are needed with $F(K, H, A)$?* Efficiency units of labor must be measured: the schooling distribution of the workforce (attainment by level), an estimate of how schooling maps into productivity (Mincerian returns, ideally allowing the return to vary by level and country), experience profiles, and — if one wants α_H separately from α_L — the split of labor income into payments to raw labor and to skills. Data on schooling quality (test scores, expenditure per pupil, teacher quality) are needed to avoid the bias in part (d), as are hours worked if labor input is to be measured consistently.

(d) *Quality of schooling shows up as productivity.* Suppose the truth is $Y_j = K_j^\alpha (AH_j)^{1-\alpha}$ with a *common* A , and that the true stock of efficiency units is $H_j = Q_j \hat{H}_j$, where $\hat{H}_j$ is the measure constructed from years of schooling as in Section 3.5 and $Q_j > 0$ captures the quality of a year of schooling. The calibration of Section 3.5 computes productivity as the residual from the measured stock $\hat{H}_j$:

$$\hat{A}_j \equiv \left[\frac{Y_j}{K_j^\alpha \hat{H}_j^{1-\alpha}} \right]^{\frac{1}{1-\alpha}} = \left[\frac{K_j^\alpha (AQ_j \hat{H}_j)^{1-\alpha}}{K_j^\alpha \hat{H}_j^{1-\alpha}} \right]^{\frac{1}{1-\alpha}} = AQ_j.$$

So the estimated productivity of country j relative to the United States is

$$\frac{\hat{A}_j}{\hat{A}_{\text{US}}} = \frac{Q_j}{Q_{\text{US}}},$$

even though all countries share the same technology: *every* estimated productivity difference is schooling quality in disguise. The bias goes the empirically awkward way, too. Poor countries are generally thought to have lower Q , so $\hat{A}_j$ is understated for them, and the calibration attributes to

“technology” what is really an unmeasured input — which is exactly the criticism of the Hall–Jones and Klenow–Rodríguez calibrations that the quality-adjustment literature (Caselli, 2005) takes up.

Chapter 4

Fundamental Determinants of Differences in Economic Performance

The two exercises of Chapter 4 left unsolved in the instructor's manual: Exercises 4.1 and 4.2, both on the Simon–Kremer model of Section 4.2. The manual solves 4.3. Throughout, world output is $Y(t) = L(t)^\alpha (A(t)Z)^{1-\alpha}$ with Z normalized to 1, so that $Y = L^\alpha A^{1-\alpha}$, and population responds to output through (4.2), $L(t) = \phi Y(t)$.

Exercise 4.1

Population as a function of the stock of ideas. Substituting the production function into (4.2),

$$L = \phi Y = \phi L^\alpha A^{1-\alpha} \quad \implies \quad L^{1-\alpha} = \phi A^{1-\alpha} \quad \implies \quad L(t) = \phi^{\frac{1}{1-\alpha}} A(t). \quad (*)$$

The Malthusian channel thus ties population to the stock of knowledge one for one: any improvement in technology is absorbed by a proportional increase in the number of people it has to feed.

Equation (4.3). Each individual discovers ideas at the rate λ , so by (4.1) $\dot{A} = \lambda L$. Substituting (*),

$$\dot{A}(t) = \lambda \phi^{\frac{1}{1-\alpha}} A(t),$$

which is (4.3): the *level* of the flow of new ideas is proportional to the existing stock, because the population that generates them is.

Equation (4.4). This is a linear differential equation with constant coefficient; separating variables, $d \log A = \lambda \phi^{1/(1-\alpha)} dt$, so with the initial condition $A(0) > 0$,

$$A(t) = \exp(\lambda \phi^{1/(1-\alpha)} t) A(0),$$

which is (4.4). Technology grows exponentially at the constant rate $g \equiv \lambda \phi^{1/(1-\alpha)}$, and so do population, by (*), and world output, since

$$Y = L^\alpha A^{1-\alpha} = (\phi^{\frac{1}{1-\alpha}} A)^\alpha A^{1-\alpha} = \phi^{\frac{\alpha}{1-\alpha}} A.$$

Remark. Income per capita, however, does not grow at all: $y = Y/L = 1/\phi$ at every date, directly from (4.2). The model produces a growing *world* — more people, more ideas, more output — with living standards pinned down by the Malthusian relation between income and population. That is the sense in which economies of scale in population are a theory of world growth rather than of prosperity, and it is the observation that Exercise 4.3(a) formalizes.

Exercise 4.2

Derivation. Simon and Kremer replace (4.1) by the assumption that population determines the *proportional* rate of improvement,

$$\frac{\dot{A}(t)}{A(t)} = \lambda L(t).$$

The static relations are unchanged, so (*) of Exercise 4.1 still gives $L(t) = \phi^{1/(1-\alpha)} A(t)$ and therefore

$$\dot{A}(t) = \lambda \phi^{\frac{1}{1-\alpha}} A(t)^2.$$

This is separable: $dA/A^2 = \lambda \phi^{1/(1-\alpha)} dt$, and integrating from 0 to t ,

$$\frac{1}{A(0)} - \frac{1}{A(t)} = \lambda \phi^{\frac{1}{1-\alpha}} t \quad \implies \quad A(t) = \frac{1}{A(0)^{-1} - \lambda \phi^{1/(1-\alpha)} t},$$

which is (4.5).

How the behavior differs. Under (4.4) the growth rate of technology is the constant $\lambda \phi^{1/(1-\alpha)}$. Under (4.5) it is

$$\frac{\dot{A}(t)}{A(t)} = \lambda \phi^{\frac{1}{1-\alpha}} A(t) = \frac{\lambda \phi^{1/(1-\alpha)}}{A(0)^{-1} - \lambda \phi^{1/(1-\alpha)} t},$$

which is increasing in t : growth accelerates. Indeed the acceleration is violent. The denominator of (4.5) vanishes at

$$T^* = \frac{1}{\lambda \phi^{1/(1-\alpha)} A(0)} < \infty,$$

so technology, population and output all become infinite in *finite* time; the solution does not merely grow fast, it blows up. Starting from a small $A(0)$, T^* is large and the early path is nearly flat, which is what gives the model its appeal as a story of millennia of stagnation followed by a takeoff.

Why. The difference is in how an idea is assumed to interact with the stock of ideas already available. Under (4.1) a discovery adds a fixed increment to the pool: a person is worth λ units of A per unit of time regardless of how much is already known. Under the Simon–Kremer assumption a discovery raises the stock *proportionally*: each person contributes λ percentage points of growth per unit of time, so a discovery is worth more the more is already known — the “standing on the shoulders of giants” externality. Since the Malthusian channel makes population itself proportional to A , the second assumption makes the growth rate proportional to A , and a differential equation whose growth rate grows linearly in the state explodes in finite time.

Which is more plausible? Neither is comfortable, and they fail in different places.

Taken as a description of the very long run, the Simon–Kremer version is the better one. It is essentially the only simple mechanism that delivers the qualitative shape of the historical record — hundreds of millennia of near-stagnation followed by an acceleration — and Kremer’s evidence that the growth rate of world population has risen with the *level* of world population over a million years is direct support for the proportional specification.

Taken literally as a model, however, it cannot be right. Finite-time blow-up is not a prediction about any observable world, so the specification must fail at some scale; and the mechanism is a pure scale effect, so it predicts that larger populations grow faster — a prediction rejected when the units are countries (the European population was below that of Asia for two millennia, yet modern growth began in Europe), and rescued only by insisting that the relevant unit is the world as a whole.

The assumption in (4.1) has the more defensible microfoundation — independent discoveries arriving at a constant rate per person, which is exactly the probabilistic story the section tells — but it can only generate steady exponential growth, never a takeoff.

The binding weakness, though, is shared and lies elsewhere: it is (4.2). Both models force income per capita to equal $1/\phi$ at every date, so neither can speak to living standards at all, and the Malthusian link between income and population on which they rest broke down precisely when modern growth began, with the demographic transition reversing the sign of the relation between income and fertility. A mechanism that is silent about income per capita and about *where* growth started is, as the chapter concludes, at best a theory of world growth — not a fundamental cause of cross-country differences.

Chapter 5

Foundations of Neoclassical Growth

The six exercises of Chapter 5 left unsolved in the instructor's manual: Exercises 5.3–5.8. The manual solves 5.1, 5.2 and 5.9–5.14. Notation follows the book: $\mathcal{H}$ is the set of households, $p = (p_1, \dots, p_N)$ the price vector, w^h household h 's income, $v^h(p, w^h)$ its indirect utility, ω^h its endowment and θ_f^h its share in firm f .

Exercise 5.3

Each household $h \in \mathcal{H}$ has indirect utility of the Gorman form (5.3),

$$v^h(p, w^h) = a^h(p) + b(p)w^h,$$

with a common $b(\cdot)$, and a positive demand for every commodity. We must show that aggregate demand is the demand of a single household with indirect utility $v(p, w) = a(p) + b(p)w$, where $a(p) \equiv \int_{h \in \mathcal{H}} a^h(p) dh$ and $w \equiv \int_{h \in \mathcal{H}} w^h dh$.

Step 1: what homogeneity forces. Any indirect utility function is homogeneous of degree 0 in (p, w) , so for every $\lambda > 0$ and every w^h ,

$$a^h(\lambda p) + b(\lambda p)\lambda w^h = a^h(p) + b(p)w^h.$$

Both sides are affine in w^h , and the household's income can be varied freely, so the constant terms and the slopes must agree separately:

$$a^h(\lambda p) = a^h(p) \quad \text{and} \quad b(\lambda p) = \lambda^{-1}b(p),$$

that is, a^h is homogeneous of degree 0 and b of degree -1 . Since v^h is strictly increasing in income, $b(p) > 0$.

Step 2: individual demands. Assume a^h and b differentiable. By Roy's Identity, household h 's demand for commodity j is

$$x_j^h(p, w^h) = -\frac{\partial v^h / \partial p_j}{\partial v^h / \partial w^h} = -\frac{1}{b(p)} \frac{\partial a^h(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w^h. \quad (*)$$

(The hypothesis that each household consumes a positive amount of every commodity is what makes the demands interior, so that Roy's Identity applies and no household sits at a corner where the envelope argument would fail.) Note the structure: the Engel curve of household h for commodity j is a straight line in w^h whose *slope*, $-b^{-1} \partial b / \partial p_j$, does not depend on h . All the household-specific information sits in the intercept.

Step 3: aggregation. Integrating $(*)$ over $\mathcal{H}$ and exchanging differentiation and integration,

$$\begin{aligned} x_j(p) &\equiv \int_{h \in \mathcal{H}} x_j^h(p, w^h) dh = -\frac{1}{b(p)} \frac{\partial}{\partial p_j} \int_{h \in \mathcal{H}} a^h(p) dh - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} \int_{h \in \mathcal{H}} w^h dh \\ &= -\frac{1}{b(p)} \frac{\partial a(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w. \end{aligned}$$

Now consider the hypothetical household with indirect utility $v(p, w) = a(p) + b(p)w$. It is a legitimate indirect utility function: a is homogeneous of degree 0 and b of degree -1 by Step 1, so v is homogeneous of degree 0; it is increasing in w and decreasing in each p_j wherever the individual ones are. Applying Roy's Identity to it gives demands

$$-\frac{1}{b(p)} \frac{\partial a(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w,$$

which is exactly $x_j(p)$. Hence aggregate demand is generated by the maximization problem of the single household with indirect utility $v(p, w)$: the economy admits a representative household, and THEOREM 5.2 is proved.

Step 4: the representative household is strong. Aggregate demand depends on the distribution of income only through the aggregate $w = \int w^h dh$, so any redistribution of income across households leaves it unchanged. The converse also holds, which is why the Gorman form is not merely sufficient but necessary: if two households h and h' had Engel curves of different slopes for commodity j , that is $b_h^{-1} \partial b_h / \partial p_j \neq b_{h'}^{-1} \partial b_{h'} / \partial p_j$, then transferring a unit of income from h' to h would change aggregate demand for j , and no single household's demand function could reproduce aggregate demand at all income distributions.

Remark. Nothing in the argument used $N < \infty$ beyond writing the derivatives coordinate by coordinate, and nothing used $\mathcal{H}$ finite: the integral is a Lebesgue integral and covers a continuum of households. What the theorem does *not* deliver is a *normative* representative household — the utility $v(p, w)$ reproduces aggregate demand but need not rank allocations the way any Pareto criterion would. That requires the additional structure of THEOREM 5.3, in particular $a^h(p) = a^h$ independent of p for the converse direction.

Exercise 5.4

Let the commodity space be indexed by $j \in J \equiv [0, 1]$ with the Lebesgue measure (nothing depends on the particular index set; $J = \mathbb{R}_+$ for a continuous-time horizon reads the same way). A consumption bundle is a measurable function $x : J \rightarrow \mathbb{R}$, and we take the commodity space to be $L^\infty(J)$ with prices in its predual $L^1(J)$, so that the value of a bundle is

$$p \cdot x = \int_J p(j)x(j) dj,$$

which is finite for $p \in L^1$, $x \in L^\infty$. (Any dual pair with a bilinear form of this kind will do; what matters is that values are given by integrals and that the dual space is rich enough to separate points.) Household h 's preferences have the Gorman representation $v^h(p, w^h) = a^h(p) + b(p)w^h$, each household demands a positive amount of (almost) every commodity, and the aggregate production possibilities set $Y \subset L^\infty(J)$ is convex. The claim is that THEOREM 5.3 survives verbatim:

- (1) any feasible allocation that maximizes $v(p, w) = \int_{\mathcal{H}} a^h(p) dh + b(p)w$, with $w = \int_{\mathcal{H}} w^h dh$, is Pareto optimal;
- (2) if in addition $a^h(p) = a^h$ for all p and all h , then every Pareto optimal allocation maximizes $v(p, w)$.

What carries over unchanged. The proof of THEOREM 5.3 is an argument about the *objective*

$$\max_{h \in \mathcal{H}} \sum \alpha^h v^h(p, w^h) = \max \left[\int_{\mathcal{H}} \alpha^h a^h(p) dh + b(p) \int_{\mathcal{H}} \alpha^h w^h dh \right]$$

subject to market clearing and to net supplies being profit maximizing at p , exactly as in (5.7). Every step of it is linear in the commodity index: sums $\sum_{j=1}^N p_j y_j$ become integrals $\int_J p(j) y(j) dj$, and Roy's Identity becomes the same formula with the functional (Gâteaux) derivatives $\delta a^h(p)/\delta p(j)$ and $\delta b(p)/\delta p(j)$ in place of partial derivatives. The linearity of the Gorman form in w^h — which is what makes the Pareto weights enter only through the weighted aggregate income and lets one substitute income transfers for commodity reallocations — is a statement about incomes, not about commodities, so it is untouched. With $b(p) > 0$, the objective is still increasing in aggregate income, and part 1 follows as before; with $a^h(p) = a^h$ constant, the argument by contradiction that produced (5.9)–(5.11) goes through with integrals.

What genuinely needs an extra assumption. Two things that are automatic in $\mathbb{R}^N$ are not automatic here.

- *Existence of supporting prices.* In $\mathbb{R}^N$ a convex set can always be separated from a disjoint convex set. In an infinite-dimensional space the separating hyperplane theorem requires a topological condition — for instance that Y (or the set of preferred allocations) have nonempty interior in the sup-norm topology, or that preferences satisfy Mas-Colell's properness condition. Without it a Pareto optimum need not be supported by any price system in the chosen dual space, and the phrase “at price vector p ” in the theorem would be empty. This is the same hypothesis the book adds in THEOREMS 5.6 and 5.7 when it proves the Second Welfare Theorem with infinitely many commodities; here it is needed for exactly the same reason.
- *Integrability.* The aggregates $a(p) = \int_{\mathcal{H}} a^h(p) dh$ and $w = \int_{\mathcal{H}} w^h dh$ must be finite, and the interchange of $\delta/\delta p(j)$ with $\int_{\mathcal{H}} dh$ must be legitimate — guaranteed, for instance, by a uniform integrable bound on the family $\{\delta a^h(p)/\delta p(j)\}$. With a continuum of commodities one must also require $p \cdot x^h < \infty$ for each household, which Exercise 5.5 delivers from local nonsatiation.

Conclusion. With these two provisos — a price system exists in the dual, and the household aggregates are well defined — THEOREM 5.3 holds word for word with $\sum_{j=1}^N$ replaced by $\int_J dj$. The economics of the result is therefore robust to the infinite-dimensional setting that growth models live in: it is the *linearity of Gorman preferences in income*, not the finiteness of the commodity space, that makes the representative household normative.

Exercise 5.5

“Household h chooses x^h at prices p ” means that x^h solves

$$\max_{x \in X^h} U^h(x) \quad \text{subject to} \quad p \cdot x \leq w^h, \quad w^h \equiv p \cdot \left(\omega^h + \sum_{f \in \mathcal{F}} \theta_f^h y^f \right),$$

and local nonsatiation is DEFINITION 5.3: at every $x \in X^h$, U^h is strictly increasing in at least one of its arguments, and $U^h(x) < \infty$.

Step 1: the income of a locally nonsatiated household that makes a choice is finite. Suppose $w^h = \infty$. Then the budget constraint does not bind anywhere and the problem reduces to maximizing U^h over X^h . But local nonsatiation says that at *every* point of X^h there is a coordinate j' in which U^h is strictly increasing, so for any candidate x the bundle $x(\varepsilon)$ that adds $\varepsilon > 0$ to coordinate j' — which is in X^h for small ε , this being exactly the construction used in the proof of THEOREM 5.5 — satisfies $U^h(x(\varepsilon)) > U^h(x)$ and is still affordable. No maximum exists, so no bundle is chosen, contradicting the hypothesis. Hence $w^h < \infty$.

Step 2: the chosen bundle is affordable, hence of finite value. Since x^h is in the budget set,

$$p \cdot x^h \leq w^h < \infty.$$

Step 3: in fact the budget holds with equality. Suppose $p \cdot x^h < w^h$. By local nonsatiation there is a coordinate j' in which U^h is strictly increasing at x^h ; for ε small enough the bundle $x^h(\varepsilon)$ obtained by adding ε to that coordinate still costs less than w^h and yields strictly higher utility, contradicting optimality. Therefore

$$p \cdot x^h = w^h < \infty.$$

Remark. In a finite-dimensional economy this is a triviality — a finite sum of finite terms. Its content is in the infinite-dimensional case relevant to growth theory, where $p \cdot x = \sum_{t=0}^{\infty} p_t x_t$ is an infinite series that may diverge. The result says that at any chosen bundle the series converges, and this is precisely what the proof of the First Welfare Theorem needs: (5.17)–(5.18) are summed over households, and the sums are meaningless unless each term is finite. It is also the reason DEFINITION 5.3 builds $U^h(x^h) < \infty$ into local nonsatiation: an infinite-utility plan would allow a household to be indifferent to budget constraints, and both the definition of a competitive equilibrium and the welfare theorems would lose their force.

Exercise 5.6

(a) *Demands and indirect utility.* Write $\hat{x}_j^h \equiv x_j^h - \xi_j^h$ and $\hat{w}^h \equiv w^h - \sum_j p_j \xi_j^h > 0$ (positive by the assumption $\sum_j p_j \bar{\xi} < w^h$). The problem

$$\max \left[\sum_{j=1}^N (\hat{x}_j^h)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad \text{s.t.} \quad \sum_{j=1}^N p_j \hat{x}_j^h \leq \hat{w}^h$$

is the textbook CES problem. The first-order conditions give $(\hat{x}_j^h)^{-1/\sigma} = \mu p_j$ for a multiplier $\mu > 0$, hence $\hat{x}_j^h \propto p_j^{-\sigma}$, and the budget constraint fixes the factor:

$$\hat{x}_j^h = \frac{\hat{w}^h p_j^{-\sigma}}{P}, \quad P \equiv \sum_{k=1}^N p_k^{1-\sigma}, \quad \text{that is} \quad x_j^h = \xi_j^h + \frac{(w^h - \sum_k p_k \xi_k^h) p_j^{-\sigma}}{\sum_k p_k^{1-\sigma}}.$$

Substituting back,

$$\sum_j (\hat{x}_j^h)^{\frac{\sigma-1}{\sigma}} = \left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} \sum_j p_j^{1-\sigma} = \left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} P,$$

so that

$$v^h(p, w^h) = \left[\left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} P \right]^{\frac{\sigma}{\sigma-1}} = \hat{w}^h P^{\frac{1}{\sigma-1}} = \frac{-\sum_j p_j \xi_j^h + w^h}{[\sum_j p_j^{1-\sigma}]^{\frac{1}{1-\sigma}}},$$

which is (5.5). It is of the Gorman form (5.3) with

$$b(p) = P^{\frac{1}{\sigma-1}} > 0 \quad \text{and} \quad a^h(p) = -\left(\sum_j p_j \xi_j^h\right) P^{\frac{1}{\sigma-1}},$$

and the household-specific term enters only through a^h , exactly as THEOREM 5.2 requires. It is homogeneous of degree 0: $\hat{w}^h$ is homogeneous of degree 1 in (p, w) and $P^{1/(\sigma-1)}$ of degree -1 .

(b) *The representative household.* The utility function (5.6) is the same CES form with $\xi_j \equiv \int_{\mathcal{H}} \xi_j^h dh$, so the computation in (a) applies verbatim and gives

$$v(p, w) = \frac{-\sum_j p_j \xi_j + w}{[\sum_j p_j^{1-\sigma}]^{\frac{1}{1-\sigma}}} = \int_{\mathcal{H}} a^h(p) dh + b(p)w,$$

using $\int_{\mathcal{H}} a^h(p) dh = -(\sum_j p_j \xi_j) P^{1/(\sigma-1)}$ and $w = \int_{\mathcal{H}} w^h dh$. This is precisely the indirect utility of the representative household constructed in THEOREM 5.2, so (5.6) is the utility function that rationalizes aggregate demand.

(c) *Dropping the outer exponent.* Let now $\tilde{U}^h = \sum_j (x_j^h - \xi_j^h)^{(\sigma-1)/\sigma}$. Take $\sigma > 1$, so that $(\sigma-1)/\sigma > 0$ and $\tilde{U}^h = (U^h)^{(\sigma-1)/\sigma}$ is a strictly increasing transformation of the utility in part (a). Demands are therefore unchanged, and the indirect utility function is the same transformation of (5.5):

$$\tilde{v}^h(p, w^h) = [\hat{w}^h P^{\frac{1}{\sigma-1}}]^{\frac{\sigma-1}{\sigma}} = (w^h - \sum_j p_j \xi_j^h)^{\frac{\sigma-1}{\sigma}} P^{\frac{1}{\sigma}}.$$

It is still homogeneous of degree 0 — the first factor has degree $(\sigma-1)/\sigma$ and the second $(1-\sigma)/\sigma$ — but it is *not* of the Gorman form: it is a strictly concave, not affine, function of w^h , so the Engel curves are no longer straight lines. The monotone transformation $z \mapsto z^{\sigma/(\sigma-1)}$ restores (5.5) and with it the Gorman form.

Is that enough? For the positive question — does the economy admit a (strong) representative household? — yes. Demand is an ordinal object: it is invariant to strictly increasing transformations of utility, so what matters is whether *some* representation of preferences takes the Gorman form, not whether the one we happened to write down does. This is the point the book makes right after THEOREM 5.2, and part (c) is the example that makes it concrete: aggregate demand here is exactly as in part (b).

Two caveats keep the answer from being an unqualified yes. First, under uncertainty utility is cardinal: the transformation $z \mapsto z^{\sigma/(\sigma-1)}$ changes the household's attitude to risk, so two representations that are equivalent for a deterministic exchange economy are *not* equivalent once expected utility is taken over lotteries — the aggregation result survives only for the representation that is also the Bernoulli utility index. Second, for a *normative* representative household the transformation matters as well: THEOREM 5.3(2) requires the Gorman representation to have $a^h(p) = a^h$ independent of prices, which is a restriction on a particular cardinalization, and Pareto weights attached to $\tilde{v}^h$ and to v^h define different social objectives. Aggregation of demand is ordinal; welfare analysis is not.

Exercise 5.7

Survival. Death arrives as the first event of a Poisson process with intensity ν , so the random date of death T is exponentially distributed:

$$\Pr\{T > t\} = e^{-\nu t}, \quad \mathbb{E}[T] = \frac{1}{\nu} < \infty.$$

The individual's expected lifetime is $1/\nu$, which is how one calibrates ν .

Expected utility. Normalize the flow utility after death to $u(0) = 0$, and let $[c(t)]_{t \geq 0}$ be the consumption plan conditional on being alive. Discounting the future at the pure rate ρ , expected lifetime utility at date 0 is

$$U_0 = \mathbb{E} \left[\int_0^T e^{-\rho t} u(c(t)) dt \right] = \mathbb{E} \left[\int_0^\infty e^{-\rho t} u(c(t)) \mathbf{1}\{T > t\} dt \right].$$

The integrand is nonnegative and measurable, so Fubini's theorem allows the expectation to be taken inside:

$$U_0 = \int_0^\infty e^{-\rho t} u(c(t)) \Pr\{T > t\} dt = \int_0^\infty e^{-\rho t} e^{-\nu t} u(c(t)) dt = \int_0^\infty e^{-(\rho+\nu)t} u(c(t)) dt.$$

This is the objective of an infinitely lived individual with discount rate

$\rho_{\text{eff}} = \rho + \nu$

Since the constraint set — the flow budget constraint and the no-Ponzi condition — is unaffected by the *form* of the objective, the individual's problem is identical to that of an infinitely lived agent who discounts at $\rho + \nu$. Her behavior, in particular her Euler equation $\dot{c}/c = (r(t) - \rho - \nu)/\theta$ for the CRRA case, is that of the infinite-horizon model with the larger discount rate.

Consistency with (5.12). In discrete time the book obtains $\beta = \hat{\beta}(1 - \nu)$; writing $\hat{\beta} = e^{-\rho\Delta}$ and $1 - \nu \approx e^{-\nu\Delta}$ for a period of length Δ gives $\beta \approx e^{-(\rho+\nu)\Delta}$, which is the same statement: the probability of not surviving acts exactly like impatience.

Remark. The result says that mortality raises the *effective discount rate*, not that it makes people poorer. Whether it also changes consumption growth depends on the asset market. With actuarially fair annuities — the Blanchard–Yaari setting of Chapter 9 — a surviving individual earns $r + \nu$ rather than r on her savings, because the assets of those who die are shared among the survivors, and the two effects cancel in the individual Euler equation: $\dot{c}/c = [(r + \nu) - (\rho + \nu)]/\theta = (r - \rho)/\theta$. What mortality then changes is the *aggregate*, through the arrival of newborns who hold no assets, which is exactly why the perpetual youth model breaks Ricardian equivalence while the individual problem looks infinite horizon.

Exercise 5.8

(a) *Different utility functions across generations.* Yes — the dynastic recursion still collapses into an infinite-horizon maximization, but the problem loses stationarity. Let generation t value its own consumption through u_t and the continuation value of its descendant with factor β :

$$V_t(y_t) = \max_{c_t + b_t \leq y_t} \{u_t(c_t) + \beta V_{t+1}(b_t + w)\}.$$

Iterating forward,

$$V_0 = \max \sum_{t=0}^{\infty} \beta^t u_t(c_t),$$

provided the sum converges (e.g. if the u_t are uniformly bounded). So a dynasty of one-period-lived altruistic individuals again behaves like a single infinitely lived agent.

Two things should be said about what does and does not survive. The objective remains *time-consistent*: viewed from date t , the dynasty maximizes $\sum_{s \geq t} \beta^{s-t} u_s(c_s)$, which ranks continuation plans exactly as $\sum_{s \geq 0} \beta^s u_s(c_s)$ does, because the two differ by the positive factor β^t and by terms that are already sunk. Geometric discounting, not the identity of the period utility functions, is what delivers consistency (compare EXERCISE 5.1). What is lost is *stationarity*: the maximization problem is no longer autonomous, so the stationary dynamic programming of Chapter 6 does not apply directly and one needs its nonstationary version (Section 6.7), with value functions indexed by time. Balanced growth also imposes a restriction: if $u_t(c) = \gamma^t u(c)$ for a constant $\gamma > 0$ — taste change at a constant rate — the objective is $\sum (\beta\gamma)^t u(c_t)$ and stationarity is restored with the modified discount factor $\beta\gamma$, which must satisfy $\beta\gamma < 1$. For general sequences $\{u_t\}$ neither a steady state nor a balanced growth path need exist.

(b) *Caring about grandchildren directly.* Now

$$V_t = u(c_t) + \beta V_{t+1} + \delta V_{t+2}, \quad \beta, \delta > 0.$$

Expanding, the weight a_s that generation t attaches to $u(c_{t+s})$ obeys the linear recursion

$$a_0 = 1, \quad a_1 = \beta, \quad a_s = \beta a_{s-1} + \delta a_{s-2} \quad (s \geq 2),$$

whose solution is a combination of the powers of the roots of $x^2 = \beta x + \delta$, the dominant one being $\varphi = (\beta + \sqrt{\beta^2 + 4\delta})/2$. Three consequences follow.

First, the objective is still an infinite discounted sum, $V_t = \sum_{s \geq 0} a_s u(c_{t+s})$, and it converges if and only if $\varphi < 1$, which is equivalent to

$$\beta + \delta < 1.$$

So the extra altruism must be paid for out of the total weight on the future.

Second — and this is the substantive difference — the discounting is *not* geometric. The ratios of successive weights are

$$\frac{a_1}{a_0} = \beta, \quad \frac{a_2}{a_1} = \beta + \frac{\delta}{\beta} > \beta,$$

and they keep changing (converging to φ only in the limit). A sequence of weights that falls at a varying rate is exactly the structure of the hyperbolic discounting of EXERCISE 5.1(b), and it has the same consequence: the plan that generation t would like its descendants to follow is not the plan that generation $t+1$ chooses, because generation t discounts dates $t+1$ and $t+2$ in the ratio β to $\beta^2 + \delta$, while generation $t+1$ discounts the same two dates in the ratio 1 to β . Preferences are *time-inconsistent*, and the dynasty can no longer be represented as a single maximizing agent.

Third, the modeling consequence: the right solution concept is not a maximization problem but an intergenerational *game*, in which each generation takes the decision rules of its descendants as given; the natural equilibrium notion is a Markov-perfect (or “consistent-planning”) equilibrium,

and saving behavior in it generally differs from the commitment solution — the dynasty saves less than generation t would wish, since it cannot bind its descendant to treat the grandchild as generously as it would like. The economics is the same as in the individual hyperbolic-discounting model: an extra, non-geometric concern for a more distant date creates a conflict between successive decision makers.

Chapter 6

Infinite-Horizon Optimization and Dynamic Programming

The fourteen exercises of Chapter 6 left unsolved in the instructor's manual: Exercises 6.1, 6.4, 6.5, 6.6, 6.10, 6.11, 6.13, 6.14, 6.15, 6.16, 6.17, 6.19, 6.20 and 6.21. The manual solves 6.2, 6.3, 6.7, 6.8, 6.9, 6.12 and 6.18. Notation follows the book: PROBLEM 6.2 is the sequence problem, PROBLEM 6.3 the Bellman equation, T the operator (6.18), $\Phi(x(0))$ the set of feasible plans, and π the policy function. Assumptions numbered 6.1–6.5 are the dynamic programming assumptions; Assumptions 1, 2 and 3' are those on technology and preferences. The numerical part is computed by `code/ch6.py`.

Exercise 6.1

Following EXAMPLE 6.1, the state is $x = k$, the control is next period's capital $y = k'$, and

$$U(k, k') = u(g(k) - k'), \quad g(k) \equiv f(k) + (1 - \delta)k, \quad G(k) = [0, g(k)],$$

the constraint $k' \leq g(k)$ being $c \geq 0$. Assumptions 1 and 2 of Chapter 2 restrict f only, so throughout we also use the standard assumptions on the utility function collected in Assumption 3': u is continuous, strictly increasing, strictly concave, and (for ASSUMPTION 6.5) continuously differentiable.

A compact state space. First note that the state can be confined to a compact interval. By Assumption 2, $\lim_{k \rightarrow \infty} f(k)/k = 0$, so there is a unique $\bar{k} > 0$ with $f(\bar{k}) = \delta\bar{k}$: at $\bar{k}$ the whole of output is needed to replace depreciated capital, and no feasible plan starting below $\bar{k}$ can ever exceed it. Hence, with $\vec{k} \equiv g(\max\{k(0), \bar{k}\})$, every feasible plan from $k(0)$ remains in $X \equiv [0, \vec{k}]$, a compact set.

ASSUMPTION 6.1. $G(k) \ni 0$ is nonempty for every $k \in X$. Since X is compact, g is continuous and u is continuous, U is bounded on X_G (a continuous function on a compact set), say $|U| \leq \bar{U} < \infty$. Then for any feasible plan $\sum_{t=0}^n \beta^t U(k(t), k(t+1))$ is a Cauchy sequence, since the tail is bounded by $\bar{U}\beta^{n+1}/(1 - \beta)$; the limit exists and is finite.

ASSUMPTION 6.2. X is a compact subset of $\mathbb{R}$. $G(k) = [0, g(k)]$ is nonempty- and compact-valued, and it is continuous: both endpoints are continuous functions of k , which gives upper and lower hemicontinuity. U is continuous on $X_G = \{(k, k') : 0 \leq k' \leq g(k)\}$ as the composition of the continuous functions u and $(k, k') \mapsto g(k) - k'$.

ASSUMPTION 6.3. Take $(k, k'), (\tilde{k}, \tilde{k}') \in X_G$ and $\alpha \in (0, 1)$. Since f is concave (Assumption 1), so is g , and since u is increasing and concave,

$$\begin{aligned} U(\alpha k + (1 - \alpha)\tilde{k}, \alpha k' + (1 - \alpha)\tilde{k}') &= u(g(\alpha k + (1 - \alpha)\tilde{k}) - \alpha k' - (1 - \alpha)\tilde{k}') \\ &\geq u(\alpha[g(k) - k'] + (1 - \alpha)[g(\tilde{k}) - \tilde{k}']) \\ &\geq \alpha U(k, k') + (1 - \alpha)U(\tilde{k}, \tilde{k}'), \end{aligned}$$

the first inequality by concavity of g and monotonicity of u , the second by concavity of u . If $k \neq \tilde{k}$ the first inequality is strict, because f is strictly concave (Assumption 1 has $f'' < 0$) and u is strictly increasing — which is exactly the strict inequality required by ASSUMPTION 6.3. Convexity of G is the same computation: if $k' \leq g(k)$ and $\tilde{k}' \leq g(\tilde{k})$ then $\alpha k' + (1 - \alpha)\tilde{k}' \leq \alpha g(k) + (1 - \alpha)g(\tilde{k}) \leq g(\alpha k + (1 - \alpha)\tilde{k})$.

ASSUMPTION 6.4. $U(\cdot, k')$ is strictly increasing because $\partial U / \partial k = u'(c)g'(k) = u'(c)[f'(k) + 1 - \delta] > 0$, using $u' > 0$ and $f' > 0$ (Assumption 1). And g is increasing, so $k \leq \tilde{k}$ implies $G(k) = [0, g(k)] \subset [0, g(\tilde{k})] = G(\tilde{k})$: a larger capital stock relaxes the constraint set.

ASSUMPTION 6.5. On the interior of X_G — where $c = g(k) - k' > 0$ — U is continuously differentiable, since u is continuously differentiable and f is twice differentiable by Assumption 1.

All five assumptions therefore hold, and THEOREMS 6.1–6.6 apply to the optimal growth model, which is what licenses the recursive treatment in Sections 6.6 and 6.8.

Remark. Only two properties of the technology did real work: concavity of f (for ASSUMPTION 6.3) and the Inada condition $f(k)/k \rightarrow 0$ (for compactness of X). The second is indispensable in this form of the argument: an AK technology, which violates Assumption 2, has an unbounded state space, and the contraction-based theory of this chapter does not apply directly — which is why the book turns to continuous-time methods for endogenous growth models.

Exercise 6.4

Recall the setting: $\dot{x}(t) = f(x(t))$ with $x(0) = c$, where f is Lipschitz with constant M , and for $g \in C[0, s]$,

$$Tg(z) = c + \int_0^z f(g(x)) \, dx, \quad z \in [0, s].$$

(a) T maps $C[0, s]$ into itself, and its fixed points are the solutions. If g is continuous then $f \circ g$ is continuous, so the integral is well defined and Tg is continuously differentiable, hence continuous: $T : C[0, s] \rightarrow C[0, s]$. If $Tg = g$ then $g(z) = c + \int_0^z f(g(x)) \, dx$, so $g(0) = c$ and, by the Fundamental Theorem of Calculus, g is differentiable with $\dot{g}(z) = f(g(z))$: g solves (6.7). Conversely, integrating (6.7) shows every continuous solution is a fixed point. For the contraction property, Lipschitz continuity gives, for every $z \in [0, s]$,

$$|Tg(z) - T\tilde{g}(z)| \leq \int_0^z |f(g(x)) - f(\tilde{g}(x))| \, dx \leq M \int_0^z |g(x) - \tilde{g}(x)| \, dx \leq Mz\|g - \tilde{g}\|, \quad (*)$$

so $\|Tg - T\tilde{g}\| \leq Ms\|g - \tilde{g}\|$. For $s < 1/M$ this is a contraction, and since $C[0, s]$ with the sup norm is complete, THEOREM 6.7 gives a unique fixed point: (6.7) has exactly one continuous solution on $[0, s]$ when $s < 1/M$.

(b) *The second iterate.* The useful form of (*) is the *pointwise* bound $|Tg(z) - T\tilde{g}(z)| \leq Mz\|g - \tilde{g}\|$. Applying T again and using it inside the integral,

$$|T^2g(z) - T^2\tilde{g}(z)| \leq M \int_0^z |Tg(x) - T\tilde{g}(x)| dx \leq M \int_0^z Mx\|g - \tilde{g}\| dx = \frac{M^2 z^2}{2} \|g - \tilde{g}\|,$$

and taking the supremum over $z \in [0, s]$ gives $\|T^2g - T^2\tilde{g}\| \leq M^2 \frac{s^2}{2} \|g - \tilde{g}\|$.

(c) *Induction.* Suppose $|T^n g(z) - T^n \tilde{g}(z)| \leq M^n \frac{z^n}{n!} \|g - \tilde{g}\|$ for all $z \in [0, s]$, which is (*) for $n = 1$. Then

$$|T^{n+1}g(z) - T^{n+1}\tilde{g}(z)| \leq M \int_0^z \frac{M^n x^n}{n!} \|g - \tilde{g}\| dx = \frac{M^{n+1} z^{n+1}}{(n+1)!} \|g - \tilde{g}\|,$$

completing the induction; the sup over $[0, s]$ gives the stated bound.

(d) *Any interval.* Fix $s \in \mathbb{R}_+$ arbitrary. Since $M^n s^n / n! \rightarrow 0$ as $n \rightarrow \infty$ — the terms of the convergent series for e^{Ms} — there is n with $M^n s^n / n! < 1$, so T^n is a contraction on the complete metric space $C[0, s]$. By EXERCISE 6.2, if some iterate T^n is a contraction then T itself has a unique fixed point. Hence (6.7) has a unique continuous solution on $[0, s]$, for *every* s , not merely for $s < 1/M$.

Remark. This is the Picard–Lindelöf theorem, proved by contraction. The restriction $s < 1/M$ in EXAMPLE 6.3 is an artifact of using T rather than its iterates: the factorial in part (c) beats the geometric growth of $M^n s^n$, so the solution never breaks down at a finite horizon. The same device — show that some iterate is a contraction even when the map itself is not — is what makes the contraction method usable in dynamic programming problems where the one-step operator is not a contraction, for instance with $\beta = 1$ but a bounded number of periods of “effective” discounting.

Exercise 6.5

The hypotheses are that $\varphi : \mathbb{R} \times [a, b] \rightarrow \mathbb{R}$ is continuously differentiable with $0 < m \leq \partial\varphi(y, x)/\partial y \leq M < \infty$ for all (y, x) .

(a) *T is a contraction.* For $y, \tilde{y}$ and any $x \in [a, b]$, the Mean Value Theorem gives a ξ between $y(x)$ and $\tilde{y}(x)$ with $\varphi(y(x), x) - \varphi(\tilde{y}(x), x) = \varphi_y(\xi, x)[y(x) - \tilde{y}(x)]$. Therefore

$$Ty(x) - T\tilde{y}(x) = [y(x) - \tilde{y}(x)] \left[1 - \frac{\varphi_y(\xi, x)}{M} \right],$$

and since $m \leq \varphi_y \leq M$ we have $0 \leq 1 - \varphi_y/M \leq 1 - m/M$. Hence

$$\|Ty - T\tilde{y}\| \leq \left(1 - \frac{m}{M}\right) \|y - \tilde{y}\|, \quad \beta \equiv 1 - \frac{m}{M} \in [0, 1),$$

so T is a contraction with modulus β . That T maps continuous functions to continuous functions is immediate from continuity of φ .

(b) *The theorem.* $C([a, b])$ with the sup norm is a complete metric space, so by THEOREM 6.7 T has a unique fixed point y^* . By construction

$$Ty^* = y^* \iff \frac{\varphi(y^*(x), x)}{M} = 0 \iff \varphi(y^*(x), x) = 0 \quad \text{for all } x \in [a, b],$$

which is the existence and uniqueness assertion of the Implicit Function Theorem. Differentiability follows once existence is in hand: since φ is continuously differentiable and $\varphi_y \geq m > 0$, the identity

$\varphi(y^*(x), x) = 0$ can be differentiated to give

$$\frac{dy^*(x)}{dx} = -\frac{\varphi_x(y^*(x), x)}{\varphi_y(y^*(x), x)},$$

which is continuous, so $y^* \in C^1([a, b])$.

Remark. The exercise suggests working directly in $C^1([a, b])$. That is the natural guess but the wrong space: with the C^1 norm $\|y\| + \|\dot{y}\|$, T need not be a contraction, because differentiating Ty brings in φ_x and the derivative $\dot{y}$ with a coefficient that is not bounded by β . The clean route is the one above — contract on $C([a, b])$, where the sup-norm estimate is exactly the one-dimensional bound $|1 - \varphi_y/M| \leq 1 - m/M$, and recover continuous differentiability afterwards from the implicit relation. The uniform bound $\varphi_y \geq m > 0$ is what makes the modulus strictly less than one, and it is the same hypothesis that makes the implicit function well defined globally on $[a, b]$ rather than only locally.

Exercise 6.6

Recall

$$TV(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad V \in C(X).$$

Under ASSUMPTION 6.2, X is compact, G is nonempty- and compact-valued and continuous, and U is continuous; so the maximum is attained (Weierstrass) and $TV \in C(X)$ by Berge's Maximum Theorem. Since X is compact, every $V \in C(X)$ is bounded, and $C(X)$ with the sup norm is a complete metric space. We verify Blackwell's two sufficient conditions (THEOREM 6.9).

Monotonicity. Let $V, W \in C(X)$ with $V(y) \leq W(y)$ for all y . For every x and every $y \in G(x)$,

$$U(x, y) + \beta V(y) \leq U(x, y) + \beta W(y) \leq TW(x),$$

the last step because $TW(x)$ is the maximum over $y \in G(x)$. Taking the maximum of the left-hand side over $y \in G(x)$ gives $TV(x) \leq TW(x)$ for all x , that is $TV \leq TW$.

Discounting. For a constant $c \geq 0$,

$$T(V + c)(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y) + \beta c\} = \left[\max_{y \in G(x)} \{U(x, y) + \beta V(y)\} \right] + \beta c = TV(x) + \beta c,$$

because the additive constant βc does not affect which y attains the maximum. So $T(V + c) \leq TV + \beta c$, with $\beta \in [0, 1)$ the discount factor.

Both of Blackwell's conditions hold, so by THEOREM 6.9 T is a contraction on $C(X)$ with modulus β , and THEOREM 6.7 then delivers the unique fixed point $V \in C(X)$ that solves PROBLEM 6.3.

Remark. The proof shows that discounting does all the work: the modulus of the contraction *is* the discount factor. This is why $\beta < 1$ is not a technical convenience but the substantive hypothesis of the whole theory — with $\beta = 1$ the operator satisfies monotonicity but not discounting, the fixed point need not be unique or even exist, and problems like EXERCISE 6.15 have no solution at all.

Exercise 6.10

It is enough to exhibit one. Take the log/Cobb–Douglas optimal growth model of EXAMPLE 6.4 with full depreciation:

$$\max \sum_{t=0}^{\infty} \beta^t \log c(t) \quad \text{s.t.} \quad k(t+1) = k(t)^\alpha - c(t), \quad k(0) > 0.$$

The Euler equation (6.50) here is

$$\frac{1}{c(t)} = \beta \alpha k(t+1)^{\alpha-1} \frac{1}{c(t+1)}. \quad (*)$$

The Euler equations alone leave one degree of freedom. Let $u(t) \equiv c(t)/k(t)^\alpha$ be the consumption share, so that $k(t+1) = (1 - u(t))k(t)^\alpha$. Substituting into (*) and simplifying, the Euler equation is equivalent to the first-order difference equation

$$u(t+1) = \frac{\alpha\beta u(t)}{1 - u(t)},$$

with fixed points $u^* = 1 - \alpha\beta$ — the familiar optimal policy, $c = (1 - \alpha\beta)k^\alpha$ — and $u = 0$. The derivative of the map at u^* is $\alpha\beta/(1 - u^*)^2 = 1/(\alpha\beta) > 1$, so u^* is *unstable* going forward, while $u = 0$ is stable (derivative $\alpha\beta < 1$). Consequently, for any $u(0) \neq u^*$ the resulting path satisfies *every* Euler equation, and:

- if $u(0) > u^*$ the consumption share rises until it exceeds 1 and capital would have to be negative: the plan is infeasible after finitely many periods;
- if $u(0) < u^*$ the path is feasible forever, but $u(t) \rightarrow 0$ geometrically — the economy saves an ever larger fraction of output and consumption vanishes relative to output.

The second family is the answer: these plans are feasible, satisfy all the Euler equations, and are strictly suboptimal.

They fail transversality. The transversality condition (6.51) here is

$$\lim_{t \rightarrow \infty} \beta^t f'(k(t)) u'(c(t)) k(t) = \lim_{t \rightarrow \infty} \beta^t \frac{\alpha k(t)^\alpha}{c(t)} = \lim_{t \rightarrow \infty} \frac{\alpha\beta^t}{u(t)} = 0.$$

On the optimal path $u(t) \equiv u^*$ and the expression is $\alpha\beta^t/u^* \rightarrow 0$. On an over-saving path $u(t+1) \approx \alpha\beta u(t)$ once u is small, so $u(t) \sim C(\alpha\beta)^t$ and

$$\frac{\alpha\beta^t}{u(t)} \sim \frac{\alpha}{C} \left(\frac{1}{\alpha}\right)^t \rightarrow \infty,$$

so transversality fails as badly as possible.

And they are strictly worse. With $\alpha = 1/3$, $\beta = 0.9$ and $k(0) = 1$, so that $u^* = 0.7$:

	optimal, $u(0) = 0.70$	$u(0) = 0.60$	$u(0) = 0.50$
value $\sum \beta^t \log c(t)$	−8.7266	−98.1473	−105.2498
loss relative to the optimum	—	89.42	96.52
$\beta^t \alpha / u(t)$ at $t = 100$	4.3×10^{-7}	4.1×10^{46}	9.8×10^{46}

Figure 4 plots the consumption share and the transversality expression along one such path.

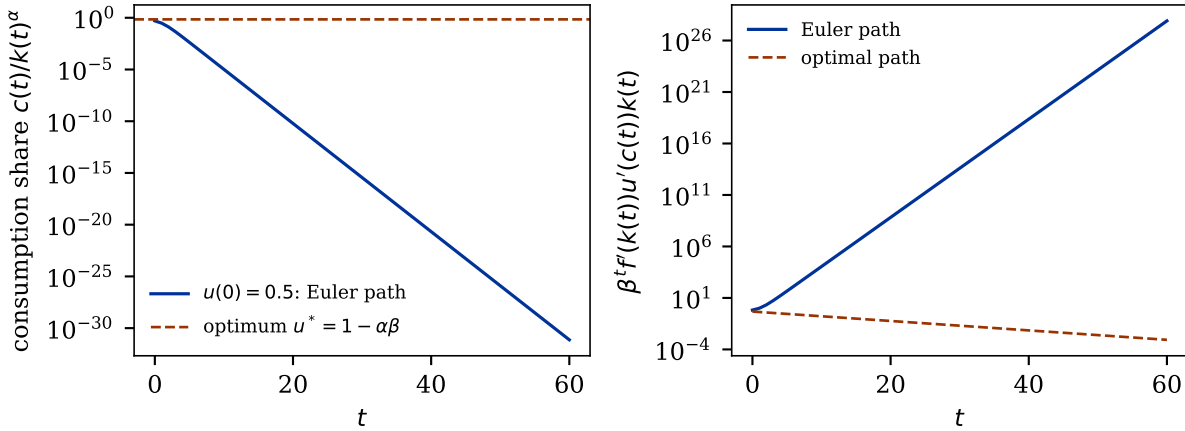


FIGURE 4. An Euler path that is not optimal. Left: the consumption share $u(t) = c(t)/k(t)^\alpha$ drifts away from the optimal $u^* = 1 - \alpha\beta$ and vanishes. Right: the transversality expression $\beta^t f'(k(t)) u'(c(t)) k(t)$, which tends to zero on the optimal path and diverges on the Euler path.

The economics is transparent: the household keeps accumulating capital it never consumes. Each Euler equation is a statement about a *local, one-period* deviation — shifting a little consumption between t and $t+1$ cannot improve matters — and every such deviation is indeed unprofitable along these paths. What they miss is the *global* deviation of consuming a fraction of the terminal capital stock, which is what the transversality condition prices: when $\lim \beta^t D_x U(x^*(t), x^*(t+1)) \cdot x^*(t+1) > 0$, the plan leaves asymptotically valuable capital unconsumed, and shrinking the whole capital path by a factor $1 - \varepsilon$ raises utility. That is exactly the perturbation used in the proof of THEOREM 6.10.

Remark. The instability of the Euler map is not a defect of the example but the general situation: the Euler equations form a second-order difference equation in the state, whose steady state is a saddle, so the initial condition $k(0)$ pins down only one of the two degrees of freedom and transversality selects the saddle path. It also means that one cannot compute the optimum by iterating the Euler equation forward — even starting exactly at u^* , rounding error grows by a factor $1/(\alpha\beta)$ per period. The accompanying program therefore imposes the optimal path exactly rather than iterating it.

Exercise 6.11

Shifting consumption up is always affordable. The only constraint is the flow budget constraint

$$a(t+1) = (1 + r(t))a(t) + w(t) - c(t), \quad a(0) \text{ given.}$$

Given a path $\{c(t)\}$ with assets $\{a(t)\}$, define $c'(t) \equiv c(t) + \gamma$ and let $\{a'(t)\}$ be generated by the same flow constraint from the same $a(0)$. The difference $d(t) \equiv a'(t) - a(t)$ satisfies $d(0) = 0$ and $d(t+1) = (1 + r(t))d(t) - \gamma$, so

$$d(t) = -\gamma \sum_{s=0}^{t-1} \prod_{j=s+1}^{t-1} (1 + r(j)) < 0.$$

Every equation of the flow constraint holds by construction, so $\{c'(t)\}$ is feasible: the extra consumption is simply borrowed, and the debt is rolled over forever. With $r(t) \geq 0$ the debt grows at least linearly, and $a'(t) \rightarrow -\infty$.

Hence no optimum exists. Since u is strictly increasing,

$$\sum_{t=0}^{\infty} \beta^t u(c(t) + \gamma) > \sum_{t=0}^{\infty} \beta^t u(c(t)) \quad \text{for every } \gamma > 0,$$

so no feasible path can be optimal: whatever is proposed, adding γ to consumption at every date is also feasible and strictly better. If in addition u is unbounded above, letting $\gamma \rightarrow \infty$ makes the supremum $+\infty$; if u is bounded above by $\bar{u}$, the supremum is $\bar{u}/(1 - \beta)$ and is still not attained. Either way the maximization problem is not well posed — ASSUMPTION 6.1 fails, and with it the entire dynamic programming apparatus.

Why market economies exclude these paths. The path finances consumption by borrowing and never repaying: the present value of the consumer's debt does not go to zero, so in present-value terms he is consuming resources he never earns. Nobody would take the other side of that trade. Formally, the lender's own optimization imposes that the value of the loan be recovered, so equilibrium requires a *no-Ponzi-game* condition

$$\lim_{t \rightarrow \infty} \frac{a(t)}{\prod_{j=0}^{t-1} (1 + r(j))} \geq 0,$$

or, equivalently and more primitively, the *natural debt limit* (6.44): the consumer can borrow at most the present value of his future labor income, since that is all he can ever repay with nonnegative consumption. Either restriction restores a compact feasible set and a well-posed problem, and in general equilibrium the restriction is not an extra assumption at all — aggregate assets must equal the aggregate capital stock, which is finite, so debts must be backed by someone's assets.

Remark. This is the discrete-time counterpart of the transversality condition of EXERCISE 6.10, seen from the household's side. There the danger was accumulating capital forever and never consuming it; here it is borrowing forever and never repaying. Both are ruled out by the same kind of terminal condition, and both show that the Euler equations alone do not characterize optimal behavior in infinite horizon.

Exercise 6.13

Write $q(t) \equiv \prod_{j=0}^{t-1} (1 + r(j))^{-1}$ for the price at date 0 of a unit of the good at date t (with $q(0) = 1$), and let

$$W(t) \equiv \sum_{s=0}^{\infty} \left(\prod_{j=0}^s \frac{1}{1 + r(t+j)} \right) w(t+s)$$

be the present value at t of current and future labor income, so that $\bar{W} = \max_t W(t)$.

(a) *The natural debt limit.* Iterating the flow budget constraint forward from t and dividing by the accumulated interest factors,

$$\sum_{s=0}^T \left(\prod_{j=0}^s \frac{1}{1 + r(t+j)} \right) c(t+s) = a(t) + \sum_{s=0}^T \left(\prod_{j=0}^s \frac{1}{1 + r(t+j)} \right) w(t+s) - \left(\prod_{j=0}^T \frac{1}{1 + r(t+j)} \right) a(t+T+1).$$

The debt is repayable only if the household can make the left-hand side nonnegative for every T while letting the last term vanish. Since consumption cannot be negative, the most the household can ever dedicate to repayment is its entire future labor income; taking $T \rightarrow \infty$ and imposing $c \geq 0$ gives

$$a(t) \geq -\sum_{s=0}^{\infty} \left(\prod_{j=0}^s \frac{1}{1+r(t+j)} \right) w(t+s) = -W(t),$$

which is (6.44). It binds exactly when the household plans zero consumption forever, which is why it is called *natural*: it is implied by feasibility alone, not by any extra market restriction.

(b) *Bounds on assets.* The lower bound is immediate: $a(t) \geq -W(t) \geq -\bar{W}$. For the upper bound, work with total wealth $\Omega(t) \equiv a(t) + W(t) \geq 0$, which evolves as $\Omega(t+1) = (1+r(t))(\Omega(t) - c(t))$. So Ω is nonincreasing as soon as the household consumes at least the interest on its wealth,

$$c(t) \geq \frac{r(t)}{1+r(t)} \Omega(t),$$

and a sufficient condition for that at the optimum is $\beta(1+r(t)) \leq 1$ for all t : with the Euler equation (6.45), $\beta(1+r(t+1)) \leq 1$ makes consumption nonincreasing, which (given the intertemporal budget constraint) requires consuming at least the annuity value of wealth. Under that condition $\Omega(t) \leq \Omega(0) = a(0) + W(0) \leq a(0) + \bar{W}$, and since $W(t) \geq 0$,

$$a(t) = \Omega(t) - W(t) \leq a(0) + \bar{W}.$$

So the conditions are: $\bar{W} < \infty$ (finite lifetime resources) and $\beta(1+r(t)) \leq 1$ for all t (the household is at least as impatient as the market). Both are needed — EXERCISE 6.12(b) shows that with $r > \beta^{-1} - 1$ assets grow without bound and no such $\bar{a}$ exists.

(c) *The level of initial consumption.* The Euler equation (6.45) determines the whole path from its first element: define recursively $c(t) = h_t(c(0))$ by $u'(c(t)) = \beta(1+r(t+1))u'(c(t+1))$, so each h_t is strictly increasing in $c(0)$ because u' is strictly decreasing. The transversality condition — equivalently the no-Ponzi condition holding with equality, $\lim_{t \rightarrow \infty} q(t)a(t) = 0$ — turns the flow constraints into the single intertemporal budget constraint

$$\sum_{t=0}^{\infty} q(t) h_t(c(0)) = a(0) + \sum_{t=0}^{\infty} q(t) w(t) \equiv \Omega(0), \quad (*)$$

which determines $c(0)$ implicitly. The left-hand side is continuous and strictly increasing in $c(0)$ (from 0 to ∞), so the solution exists and is unique. Differentiating (*),

$$\frac{\partial c(0)}{\partial a(0)} = \left[\sum_{t=0}^{\infty} q(t) h'_t(c(0)) \right]^{-1} > 0 :$$

an increase in initial assets raises initial consumption. With CRRA utility $h_t(c(0)) = c(0) \prod_{j=1}^t [\beta(1+r(j))]^{1/\theta}$, so $c(0) = \Omega(0) / \sum_t q(t) \prod_{j \leq t} [\beta(1+r(j))]^{1/\theta}$ and the derivative is the marginal propensity to consume out of wealth, a number in $(0, 1]$.

(d) *Retiming earnings.* Let $\{\tilde{w}(t)\}$ satisfy $\tilde{w}(t) > w(t)$ for $t < T$, $\tilde{w}(t) \leq w(t)$ for $t \geq T$, and $\sum_t (1+r)^{-t} w(t) = \sum_t (1+r)^{-t} \tilde{w}(t)$. Then $\Omega(0)$ is unchanged, so by (*) the whole consumption path $\{c(t)\}$ — not merely $c(0)$ — is unchanged. What changes is the asset path: with income

arriving earlier, the household saves the difference, so

$$\tilde{a}(t) - a(t) = \sum_{s < t} (1+r)^{t-s} [\tilde{w}(s) - w(s)] > 0 \quad \text{for } 0 < t \leq T,$$

and the extra assets are run down thereafter, reaching zero in present value again.

The intuition is the permanent income hypothesis in its purest form. With perfect capital markets, a complete set of borrowing and lending opportunities at the market interest rate, and no binding constraint, the household cares only about the *present value* of its earnings; the timing affects only the portfolio decision, not consumption. Two qualifications are worth noting, because they are where the result breaks in applications: if the natural debt limit (6.44) or a tighter borrowing constraint binds, a shift of income toward the present relaxes it and does raise consumption; and if the interest rate at which the household can borrow exceeds that at which it can lend, the equivalence fails as well. Here neither applies — the change makes the household unambiguously less constrained early on.

Exercise 6.14

Conditions. The natural hypotheses are the stationary ones imposed *uniformly in t*:

- (1) $X \subset \mathbb{R}^K$ is compact, and for every t , $G(t, \cdot)$ is nonempty-valued, compact-valued and continuous, with the graph $X_G(t) = \{(x, y) : y \in G(t, x)\}$ closed;
- (2) $U(t, \cdot, \cdot)$ is continuous on $X_G(t)$ for each t , and *uniformly bounded*, in the sense that $\bar{U} \equiv \sup_t \sup_{(x, y) \in X_G(t)} |U(t, x, y)|$ is finite;
- (3) $\beta \in [0, 1)$.

Let $C(\mathbb{Z}_+ \times X)$ be the space of functions $V(t, x)$, continuous in x for each t and bounded, with the sup norm $\|V\| = \sup_{t, x} |V(t, x)|$. This is a complete metric space.

T is a well-defined self-map. For each (t, x) the maximand is continuous in y on the compact set $G(t, x)$, so the maximum is attained (Weierstrass); by Berge's Maximum Theorem $TV(t, \cdot)$ is continuous; and $\|TV\| \leq \bar{U} + \beta\|V\| < \infty$, so $TV \in C(\mathbb{Z}_+ \times X)$. Uniform boundedness of U is exactly what is needed here — without it the sup norm over all t is infinite and the space is not the right one.

T is a contraction. Blackwell's conditions hold verbatim as in Exercise 6.6, because t is carried along as a parameter: if $V \leq W$ pointwise then $TV \leq TW$; and $T(V + c)(t, x) = TV(t, x) + \beta c$ for any constant c . By THEOREM 6.9, T is a contraction of modulus β , so by THEOREM 6.7 it has a unique fixed point $V \in C(\mathbb{Z}_+ \times X)$, the value function of the nonstationary problem.

The equivalents of THEOREMS 6.2–6.6. Write $\Phi(t, x)$ for the set of feasible plans from state x at date t , and $V^*(t, x)$ for the supremum of $\sum_{s \geq t} \beta^{s-t} U(s, x(s), x(s+1))$.

- *Principle of optimality (6.2).* $V^*(t, x^*(t)) = U(t, x^*(t), x^*(t+1)) + \beta V^*(t+1, x^*(t+1))$ for an optimal plan, and any feasible plan satisfying this relation at every t attains $V^*(t, x)$. The proof is the stationary one with the date carried along: truncate the plan at n , use uniform boundedness to control the tail by $\bar{U}\beta^{n+1}/(1-\beta)$, and pass to the limit.

- *Existence (6.3)*. The fixed point V exists and is unique by the contraction argument above, and $V = V^*$; the maximum in the Bellman equation is attained, which defines a (possibly set-valued) optimal policy $\pi(t, x)$. Equivalently one can argue directly on the sequence problem: $\Phi(t, x)$ is compact in the product topology (Tychonoff), the objective is continuous, and Weierstrass applies.
- *Concavity (6.4)*. If in addition $U(t, \cdot, \cdot)$ is concave and $G(t, \cdot)$ has convex graph for each t , then $V(t, \cdot)$ is concave for each t . Proof: the set of V that are concave in x for every t is a closed subset of $C(\mathbb{Z}_+ \times X)$, and T maps it into the (smaller) set of functions strictly concave in x when U is strictly concave; apply THEOREM 6.8(2).
- *Monotonicity (6.5)*. If $U(t, \cdot, y)$ is strictly increasing and $G(t, \cdot)$ is monotone for each t , the same closed-subset argument gives $V(t, \cdot)$ strictly increasing.
- *Differentiability (6.6)*. If $U(t, \cdot, \cdot)$ is continuously differentiable and the optimal policy is interior, $x \in \text{Int } X$ and $\pi(t, x) \in \text{Int } G(t, x)$, then $V(t, \cdot)$ is differentiable at x with $D_x V(t, x) = D_x U(t, x, \pi(t, x))$. The Benveniste–Scheinkman argument is unchanged: $W(z) \equiv U(t, z, \pi(t, x)) + \beta V(t+1, \pi(t, x))$ is differentiable, lies below $V(t, \cdot)$, and touches it at x .

Remark. The one substantive difference from the stationary theory is uniformity. Continuity and boundedness must hold uniformly in t , since otherwise $C(\mathbb{Z}_+ \times X)$ with the sup norm is not the relevant space and the contraction modulus is not β . Economically this rules out problems in which the environment deteriorates or improves without bound — for instance a wage sequence that diverges, which is precisely why EXAMPLE 6.5 needs $\bar{W} < \infty$. The book’s own treatment in Section 6.7 proves the analogous results by working directly with the sequence problem; the contraction route above is shorter but buys that brevity with the uniform bounds.

Exercise 6.15

Note two features: there is no discounting ($\beta = 1$) and no production, so the problem is to divide a fixed cake $k(0)$ among infinitely many dates. Feasibility requires $c(t) \geq 0$ and $\sum_{t=0}^{\infty} c(t) \leq k(0)$. Normalize $u(0) = 0$; this is without loss, since if $u(0) > 0$ then every plan that consumes nothing in infinitely many periods already has value $+\infty$ and the problem is ill posed for a still cruder reason, while if $u(0) < 0$ the same is true with $-\infty$.

The supremum is not attained. Consider the plans

$$c_n(t) = \begin{cases} k(0)/n, & t = 0, \dots, n-1, \\ 0, & t \geq n, \end{cases}$$

which are feasible and give value $V_n = n u(k(0)/n)$. Because u is strictly concave with $u(0) = 0$, the ratio $u(c)/c$ is strictly decreasing in c , so

$$V_n = k(0) \cdot \frac{u(k(0)/n)}{k(0)/n}$$

is strictly increasing in n , with

$$\lim_{n \rightarrow \infty} V_n = k(0) \cdot \lim_{c \downarrow 0} \frac{u(c)}{c} = k(0) u'(0^+) \in (0, \infty].$$

On the other hand every feasible plan satisfies, again by strict concavity and $u(0) = 0$,

$$\sum_t u(c(t)) < u'(0^+) \sum_t c(t) \leq u'(0^+)k(0)$$

whenever $u'(0^+) < \infty$ and some $c(t) > 0$. So the supremum equals $u'(0^+)k(0)$ and is approached by the sequence $\{c_n\}$ but attained by no feasible plan: any candidate can be improved by splitting its consumption into more, smaller pieces. If $u'(0^+) = \infty$ the same computation shows the supremum is $+\infty$, and again no plan attains it. In both cases there is no solution.

Why. Without discounting there is no cost to postponing consumption, while strict concavity means that the *average* utility per unit consumed, $u(c)/c$, rises as the portions shrink. Spreading the cake ever more thinly over ever more dates therefore always raises total utility, and the “optimum” would be to consume infinitesimal amounts forever — which is not a feasible plan. In the language of the chapter, ASSUMPTION 6.1 fails: with $\beta = 1$ the limit $\lim_n \sum_{t=0}^n U(x(t), x(t+1))$ need not behave well, the objective is not continuous in the product topology (the tail of a plan matters in the limit), and the Bellman operator satisfies Blackwell’s monotonicity but not the discounting condition of Exercise 6.6, so it is not a contraction. Boundedness of u does not rescue matters — it makes the supremum finite when $u'(0^+) < \infty$, but attainment is what fails.

Remark. The example isolates discounting as the substantive assumption. Economists sometimes treat $\beta < 1$ as a mere technical device; here it is what makes the problem have a solution at all. When one does want to study undiscounted problems — the golden rule, or the Ramsey problem with $\beta = 1$ — the optimality criterion itself has to be weakened, to overtaking or catching-up criteria, as the footnote to ASSUMPTION 6.1 notes.

Exercise 6.16

(a) *Recursive formulation.* With k the state and next period’s capital k' the control,

$$V(k) = \max_{k' \in G(k)} \{u(f(k) - k') + \beta V(k')\}, \quad G(k) = [0, f(k)].$$

As in Exercise 6.1, the state can be confined to the compact interval $[0, \bar{k}]$, where $\bar{k}$ solves $f(\bar{k}) = \bar{k}$ (full depreciation makes the maximum sustainable stock the fixed point of f) and $\bar{k} = f(\max\{k(0), \bar{k}\})$.

(b) *Existence, uniqueness and shape.* The problem satisfies ASSUMPTIONS 6.1–6.4 by the argument of Exercise 6.1, with $g(k) = f(k)$. Hence:

- THEOREM 6.3 gives a unique continuous bounded V solving the Bellman equation, and an optimal plan exists;
- THEOREM 6.4 gives strict concavity of V , since $U(k, k') = u(f(k) - k')$ satisfies ASSUMPTION 6.3;
- COROLLARY 6.1 then gives a *unique* optimal plan, generated by a continuous policy function $k' = \pi(k)$, equivalently $c = \pi_c(k) = f(k) - \pi(k)$;
- THEOREM 6.5 gives that V is strictly increasing.

That π is nondecreasing follows from the revealed-preference argument used for PROPOSITION 6.1: if $k' > k$ but $\pi(k) > \pi(k')$, adding the two optimality inequalities gives

$$u(z - \pi(k')) - u(z - \pi(k)) \leq u(z' - \pi(k')) - u(z' - \pi(k)), \quad z \equiv f(k) < z' \equiv f(k'),$$

which contradicts strict concavity of u , since the two consumption differences are equal in size but taken at strictly higher levels on the right. Monotonicity of the consumption function follows from the Euler equation as in Exercise 6.17.

(c) *Differentiability.* THEOREM 6.6 gives differentiability of V at any k such that u and f are continuously differentiable near the relevant point (ASSUMPTION 6.5) and the optimal policy is interior, $\pi(k) \in \text{Int } G(k)$, that is $0 < \pi(k) < f(k)$ — equivalently both consumption and investment are strictly positive. Interiority is guaranteed if u and f satisfy Inada conditions ($u'(0) = \infty$ rules out $c = 0$; $f'(0) = \infty$ rules out $k' = 0$). Then

$$V'(k) = \frac{\partial U(k, \pi(k))}{\partial k} = f'(k) u'(f(k) - \pi(k)).$$

Without differentiability of u (or at corner solutions) V can have kinks, though it always has left and right derivatives by concavity.

(d) *The Euler equation.* The first-order condition of the Bellman problem is $u'(c) = \beta V'(k')$, and the envelope condition just derived is $V'(k) = f'(k) u'(c)$. Combining, with $c(t) = f(k(t)) - k(t+1)$,

$$\boxed{u'(c(t)) = \beta f'(k(t+1)) u'(c(t+1))}$$

the familiar condition that the marginal utility given up today equals the discounted marginal utility gained tomorrow, multiplied by the marginal product of the capital carried forward.

(e) *Is it enough?* No. Substituting $c(t) = f(k(t)) - k(t+1)$, the Euler equation is a *second order* difference equation in $\{k(t)\}$,

$$u'(f(k(t)) - k(t+1)) = \beta f'(k(t+1)) u'(f(k(t+1)) - k(t+2)),$$

so the initial condition $k(0)$ leaves a one-parameter family of solutions, as EXERCISE 6.10 shows explicitly. The missing condition is the transversality condition

$$\lim_{t \rightarrow \infty} \beta^t f'(k(t)) u'(c(t)) k(t) = 0,$$

which by THEOREM 6.10 is necessary and sufficient, together with the Euler equations, for optimality of an interior plan. Intuitively it says that the present value of the capital stock left over at the end of time must be zero: capital is worth holding only for the consumption it eventually delivers, so a plan that keeps accumulating capital of positive present value forever is wasting resources. It is the infinite-horizon analogue of the terminal condition $k(T+1) = 0$ in a finite-horizon problem — one would never end a finite life with capital left in the ground.

Exercise 6.17

The consumption function is nondecreasing. By PROPOSITION 6.1 the policy function $s(k)$ — the capital carried into the next period — is nondecreasing, and by PROPOSITION 6.2 the value function V is differentiable; it is strictly concave by THEOREM 6.4, so V' is strictly decreasing. The

first-order condition of the recursive problem (6.48) is

$$u'(c(k)) = \beta V'(s(k)).$$

Take $k' > k$. Then $s(k') \geq s(k)$, so $V'(s(k')) \leq V'(s(k))$, and therefore

$$u'(c(k')) = \beta V'(s(k')) \leq \beta V'(s(k)) = u'(c(k)).$$

Since u is strictly concave, u' is strictly decreasing, and the inequality inverts: $c(k') \geq c(k)$. Hence $c(\cdot)$ is nondecreasing, as claimed.

Monotone convergence of consumption. Suppose $k(0) < k^*$. By PROPOSITION 6.3 the capital stock increases monotonically to the steady state, $k(t) \uparrow k^*$. Composing with the nondecreasing and continuous function $c(\cdot)$,

$$c(t) = c(k(t)) \text{ is nondecreasing, and } c(t) \rightarrow c(k^*)$$

by continuity of $c(\cdot)$ (COROLLARY 6.1). The limit is the steady-state consumption level: at k^* the resource constraint with $k(t+1) = k(t) = k^*$ gives

$$c^* = f(k^*) + (1 - \delta)k^* - k^* = f(k^*) - \delta k^*.$$

Thus $c(t) \uparrow c^*$. If instead $k(0) > k^*$, PROPOSITION 6.3 gives $k(t) \downarrow k^*$ and the same argument gives $c(t) \downarrow c^*$.

Remark. The result is worth contrasting with the Solow model. There the saving rate is constant, so consumption is a fixed fraction of output and inherits its monotonicity mechanically. Here the saving rate is chosen, and could in principle move in either direction; what pins the sign down is the combination of two facts — the economy accumulates capital monotonically, and the marginal value of capital falls as capital rises. Note also what is *not* claimed: the saving *rate* need not be monotone along the transition, and generally is not.

Exercise 6.19

Suppose, to get a contradiction, that there are arbitrarily small $\varepsilon > 0$ with $s(\varepsilon) \leq \varepsilon$. Along the optimal path starting from $k = \varepsilon$, the Euler equation (6.50) holds:

$$u'(c(\varepsilon)) = \beta \left[f'(s(\varepsilon)) + 1 - \delta \right] u'(c(s(\varepsilon))).$$

By PROPOSITION 6.4 (Exercise 6.17) the consumption function is nondecreasing, so $s(\varepsilon) \leq \varepsilon$ implies $c(s(\varepsilon)) \leq c(\varepsilon)$ and hence $u'(c(s(\varepsilon))) \geq u'(c(\varepsilon)) > 0$, because u' is decreasing. Dividing the Euler equation by $u'(c(\varepsilon))$,

$$1 = \beta \left[f'(s(\varepsilon)) + 1 - \delta \right] \frac{u'(c(s(\varepsilon)))}{u'(c(\varepsilon))} \geq \beta \left[f'(s(\varepsilon)) + 1 - \delta \right],$$

that is

$$f'(s(\varepsilon)) \leq \frac{1}{\beta} - 1 + \delta < \infty.$$

But $0 < s(\varepsilon) \leq \varepsilon$, so letting $\varepsilon \rightarrow 0$ forces $s(\varepsilon) \rightarrow 0$, and Assumption 2 (the Inada condition $\lim_{k \rightarrow 0} f'(k) = \infty$) gives $f'(s(\varepsilon)) \rightarrow \infty$ — contradicting the bound just derived. Hence for all sufficiently small ε we must have $s(\varepsilon) > \varepsilon$.

Intuition. Near zero the marginal product of capital is arbitrarily large, so an extra unit of saving returns arbitrarily much tomorrow. A household with a finite rate of time preference — the right-hand side $\beta^{-1} - 1 + \delta$ is a fixed, finite number, the effective required return — will always take that trade: at very low capital the return to accumulation exceeds the required return by an unbounded margin. Thus the economy cannot optimally run its capital stock down toward zero, and $k = 0$, though a rest point of the resource constraint, is never approached from a positive initial stock. This is precisely what the proof of PROPOSITION 6.3 needs to conclude that the monotone sequence $\{k(t)\}$ converges to the *positive* steady state k^* rather than to zero.

Exercise 6.20

The claims of PROPOSITION 6.1 are: there exist a unique value function V and consumption function $c(k)$, with $s(k) = f(k) + (1 - \delta)k - c(k)$; V is strictly increasing and strictly concave; and $s(k)$ is nondecreasing. Assume u is continuous, strictly increasing and strictly concave, but *not* necessarily differentiable, and let f satisfy Assumptions 1 and 2.

Step 1: the dynamic programming assumptions still hold. Exercise 6.1 verified ASSUMPTIONS 6.1–6.4 using only continuity, monotonicity and strict concavity of u together with concavity and monotonicity of f ; differentiability entered only in ASSUMPTION 6.5. So ASSUMPTIONS 6.1–6.4 hold as before, while ASSUMPTION 6.5 is dropped.

Step 2: existence, uniqueness, concavity, monotonicity. THEOREMS 6.3, 6.4 and 6.5 and COROLLARY 6.1 require only ASSUMPTIONS 6.1–6.4. They give, in order: a unique continuous bounded V solving the Bellman equation (6.48), with an optimal plan existing; V strictly concave; V strictly increasing; and a unique optimal plan generated by a continuous policy function. None of these arguments differentiates anything: the contraction argument uses Blackwell's conditions (Exercise 6.6), and strict concavity is obtained from THEOREM 6.8(2), by showing that T maps the closed set of weakly concave functions into the set of strictly concave ones.

Step 3: $s(k)$ is nondecreasing. This is the only part of PROPOSITION 6.1 whose proof might seem to need calculus, and it does not: the book's argument is a revealed-preference argument. Suppose $k' > k$ but $s(k) > s(k')$. Both $s(k)$ and $s(k')$ are feasible at both states, since $f(k) + (1 - \delta)k < f(k') + (1 - \delta)k'$ and $s(k') < s(k)$. Optimality at k and at k' gives

$$\begin{aligned} u(z - s(k)) + \beta V(s(k)) &\geq u(z - s(k')) + \beta V(s(k')), \\ u(z' - s(k')) + \beta V(s(k')) &\geq u(z' - s(k)) + \beta V(s(k)), \end{aligned}$$

with $z \equiv f(k) + (1 - \delta)k < z' \equiv f(k') + (1 - \delta)k'$. Adding and rearranging eliminates V :

$$u(z - s(k')) - u(z - s(k)) \leq u(z' - s(k')) - u(z' - s(k)).$$

But the two consumption increments are equal, $(z - s(k')) - (z - s(k)) = (z' - s(k')) - (z' - s(k)) = s(k) - s(k') > 0$, while the right-hand pair is evaluated at strictly higher consumption levels ($z' > z$). Strict concavity of u makes equal increments worth strictly less at higher levels, so the right-hand side is *strictly smaller* than the left — a contradiction. Hence $s(k)$ is nondecreasing.

This completes the proof of PROPOSITION 6.1 using only continuity, monotonicity and strict concavity.

Remark. What is genuinely lost without differentiability is PROPOSITION 6.2 — that V is differentiable — and with it the Euler equation (6.50) in its classical form. Concavity still gives V left and right derivatives everywhere, and the first-order condition becomes the inclusion $u'_-(c) \geq \beta V'_+(s)$ and $u'_+(c) \leq \beta V'_-(s)$, or in the language of convex analysis, $0 \in \partial[u(z - s) + \beta V(s)]$ with ∂ the superdifferential. All the qualitative conclusions of Sections 6.8 — uniqueness of the steady state, monotone convergence — survive, because they rest on the monotonicity of $s(\cdot)$ established above rather than on calculus.

Exercise 6.21

The two problems generate the same difference equations. In the competitive equilibrium, firms hire capital and labor competitively, so

$$R(t) = f'(k(t)), \quad 1 + r(t+1) = f'(k(t)) + 1 - \delta, \quad w(t) = f(k(t)) - k(t)f'(k(t)),$$

and asset market clearing requires $a(t) = k(t)$. Substituting the factor prices into the household's flow budget constraint (6.54),

$$a(t+1) = (1 + r(t))a(t) + w(t) - c(t) = k(t) + [f'(k(t)) - \delta]k(t) + f(k(t)) - k(t)f'(k(t)) - c(t),$$

that is

$$k(t+1) = f(k(t)) + (1 - \delta)k(t) - c(t),$$

which is exactly the resource constraint (6.47) of the optimal growth problem. Substituting the same factor prices into the household's Euler equation (6.55) gives (6.56),

$$u'(c(t)) = \beta [f'(k(t+1)) + 1 - \delta] u'(c(t+1)),$$

which is identical to the planner's Euler equation (6.50). The two problems also share the initial condition $k(0)$, and the household's no-Ponzi condition with $a(t) = k(t)$, $\lim_{t \rightarrow \infty} \beta^t u'(c(t))k(t+1) = 0$, is the planner's transversality condition (6.51).

Uniqueness closes the argument. By THEOREM 6.10, for this problem the Euler equations together with the transversality condition are necessary *and* sufficient for optimality, and by COROLLARY 6.1 — the objective being strictly concave and the constraint set convex — the optimal plan is unique. So there is exactly one sequence $\{k(t), c(t)\}$ satisfying (i) the resource constraint, (ii) the Euler equation, (iii) $k(0)$ given, and (iv) transversality. The competitive equilibrium path satisfies all four, and so does the optimal growth path; therefore they coincide.

Both directions. The argument above takes an equilibrium and shows it solves the planner's problem. The converse is equally immediate: given the optimal path $\{k(t), c(t)\}$, define prices by the marginal product conditions above and set $a(t) = k(t)$. Then firms maximize profits by construction (they take prices as given and earn zero profits, by Euler's Theorem and constant returns), the household's budget constraint holds at every date by the computation above, its Euler equation is the planner's, and its transversality condition holds. Hence the optimal allocation, supported by these prices, is a competitive equilibrium.

Remark. This is the First and Second Welfare Theorems of Chapter 5 in action, made concrete: the equivalence holds because there are no externalities, no distortions and no missing markets, and because the natural debt limit rules out the Ponzi schemes of EXERCISE 6.11. The practical payoff is large — it is why one

may compute equilibria of the neoclassical growth model by solving a single planner's dynamic programming problem, a device used throughout the rest of the book. It is also worth noting where it stops: as soon as Chapter 11 onward introduces externalities or monopolistic competition, the planner's and the equilibrium paths separate, and this shortcut is no longer available.

Chapter 7

An Introduction to the Theory of Optimal Control

The sixteen exercises of Chapter 7 left unsolved in the instructor's manual: Exercises 7.3, 7.4, 7.6–7.9, 7.11–7.16, 7.20, 7.22, 7.27 and 7.29. The manual solves 7.1, 7.2, 7.5, 7.10, 7.17–7.19, 7.21, 7.23–7.26 and 7.28. Notation follows the book: x is the state, y the control, λ the costate, $H(t, x, y, \lambda) = f + \lambda g$ the Hamiltonian (7.16), $M(t, x, \lambda) = \max_y H$ the maximized Hamiltonian (7.20), and $\hat{H} = e^{\rho t} H$ the current-value Hamiltonian with costate $\mu = e^{\rho t} \lambda$. The numerical part is computed by `code/ch7.py`.

Exercise 7.3

Put the problem in control form by treating the derivative as the control: set

$$y(t) \equiv \dot{x}(t), \quad f(t, x, y) \equiv F(t, x, y), \quad g(t, x, y) \equiv y,$$

so the state equation $\dot{x} = g(t, x, y) = y$ holds by definition and the problem is exactly the one covered by THEOREM 7.2. The Hamiltonian (7.16) is

$$H(t, x, y, \lambda) = F(t, x, y) + \lambda(t)y.$$

THEOREM 7.2 gives a continuously differentiable costate λ with, for all $t \in [0, t_1]$,

$$H_y = 0 : \quad \frac{\partial F(t, x, \dot{x})}{\partial \dot{x}} + \lambda(t) = 0, \tag{*}$$

$$\dot{\lambda} = -H_x : \quad \dot{\lambda}(t) = -\frac{\partial F(t, x, \dot{x})}{\partial x}. \tag{**}$$

From (*), $\lambda(t) = -F_{\dot{x}}(t, x(t), \dot{x}(t))$, which is differentiable in t along the (continuously differentiable) solution. Differentiating and substituting into (**),

$$-\frac{d}{dt} \left[\frac{\partial F(t, x(t), \dot{x}(t))}{\partial \dot{x}} \right] = -\frac{\partial F(t, x(t), \dot{x}(t))}{\partial x},$$

that is,

$$\boxed{\frac{\partial F(t, x(t), \dot{x}(t))}{\partial x} - \frac{d}{dt} \left[\frac{\partial F(t, x(t), \dot{x}(t))}{\partial \dot{x}} \right] = 0}$$

which is the Euler–Lagrange equation. The costate has a clean reading here: $\lambda(t) = -F_{\dot{x}}$ is (minus) the marginal value of the rate of change of the state, and the Euler–Lagrange equation is the statement that its rate of change must equal the marginal value of the state itself.

Remark. The book writes the last term as $\partial^2 F / \partial \dot{x} \partial t$. That notation is correct only if the symbol is read as the *total* derivative along the solution path,

$$\frac{d}{dt} F_{\dot{x}} = F_{\dot{x}t} + F_{\dot{x}x} \dot{x}(t) + F_{\dot{x}\dot{x}} \ddot{x}(t),$$

and not as the partial derivative of $F_{\dot{x}}$ with respect to its explicit time argument, which would drop the last two terms. It is the total derivative that the variational argument produces, as (*) and (**) show.

Exercise 7.4

Here $F(t, x, \dot{x}) = -\sqrt{1 + \dot{x}^2}$, which does not depend on x or on t explicitly, so

$$F_x = 0, \quad F_{\dot{x}} = -\frac{\dot{x}}{\sqrt{1 + \dot{x}^2}}.$$

The Euler–Lagrange equation of Exercise 7.3, $F_x - \frac{d}{dt}F_{\dot{x}} = 0$, therefore reduces to

$$\frac{d}{dt} \left[\frac{x'(t)}{\sqrt{1 + (x'(t))^2}} \right] = 0.$$

(The same conclusion follows from the general fact that when F does not depend on x , the costate is constant: $\dot{\lambda} = -F_x = 0$ and $\lambda = -F_{\dot{x}}.$)

Carrying out the differentiation,

$$\frac{d}{dt} \left[x'(1 + (x')^2)^{-1/2} \right] = x''(1 + (x')^2)^{-1/2} - x' \cdot x'x''(1 + (x')^2)^{-3/2} = \frac{x''(t)}{(1 + (x'(t))^2)^{3/2}}.$$

Since $(1 + (x')^2)^{3/2} > 0$ always, the Euler–Lagrange equation holds if and only if

$$x''(t) = 0 \quad \text{for all } t,$$

so $x(t) = a + bt$: the extremal is a straight line, and the constants are pinned down by the endpoints, $b = (u_1 - u_0)/(z_1 - z_0)$ and $a = u_0$ after the normalization $(z_0, u_0) = (0, 0)$.

Finally, the extremal really is the minimizer of length rather than merely a stationary point: F does not depend on x at all, so the maximized Hamiltonian $M(t, x, \lambda)$ is constant — in particular concave — in x , and THEOREM 7.6 applies, giving a global maximum of $-\int \sqrt{1 + (x')^2}$, that is a global minimum of arc length. Equivalently, $p \mapsto -\sqrt{1 + p^2}$ is strictly concave, so the stationary path is the unique minimizer.

Remark. The expression printed in the book, $d[x'(t)(1 + (x'(t))^2)]/dt = 0$, has the bracket raised to the power +1; the derivative of $-\sqrt{1 + \dot{x}^2}$ gives the power $-1/2$, as above. The conclusion $x'' = 0$ is unaffected — indeed the misprinted expression also implies $x'' = 0$, since $\frac{d}{dt}[x'(1 + (x')^2)] = x''(1 + 3(x')^2)$ and $1 + 3(x')^2 > 0$ — but only the form with the power $-1/2$ is the Euler–Lagrange equation of this problem.

Exercise 7.6

Setting. The problem is $\max \int_0^1 e^{-\rho t} u(c) dt$ subject to $\dot{a} = ra + w - c$, $a(t) \geq 0$, $a(0) > 0$ given, and — as the book argues — $a(1) = 0$. THEOREM 7.2 gives the costate equation and the first-order condition

$$e^{-\rho t} u'(c(t)) = \lambda(t), \quad \dot{\lambda}(t) = -r\lambda(t) \implies \lambda(t) = \lambda(0)e^{-rt},$$

so that consumption satisfies

$$u'(c(t)) = u'(c(0))e^{(\rho-r)t}. \quad (*)$$

Integrating the flow constraint with $a(1) = 0$ gives the intertemporal budget constraint

$$\int_0^1 e^{-rt} c(t) dt = a(0) + \int_0^1 e^{-rt} w dt = a(0) + \frac{w(1 - e^{-r})}{r} \equiv \Omega. \quad (**)$$

(a) *Initial consumption.* Equation (*) writes every $c(t)$ as an increasing function of $c(0)$: $c(t) = h_t(c(0))$ with $h_t \equiv (u')^{-1}[u'(c(0))e^{(\rho-r)t}]$, increasing because u' is strictly decreasing. Substituting into (**),

$$\int_0^1 e^{-rt} h_t(c(0)) dt = \Omega,$$

whose left-hand side is continuous and strictly increasing in $c(0)$, from 0 to ∞ : $c(0)$ is uniquely determined. With the CRRA specification $u(c) = c^{1-\theta}/(1-\theta)$ this can be solved in closed form. Then (*) gives $c(t) = c(0)e^{(r-\rho)t/\theta}$, and (**) gives

$$c(0) \int_0^1 e^{-\zeta t} dt = \Omega, \quad \zeta \equiv r - \frac{r-\rho}{\theta},$$

so that

$$c(0) = \frac{\zeta}{1-e^{-\zeta}} \left[a(0) + \frac{w(1-e^{-r})}{r} \right]$$

(with $c(0) = \Omega$ in the knife-edge case $\zeta = 0$). For log utility ($\theta = 1$), $\zeta = \rho$ and $c(0) = \rho\Omega/(1-e^{-\rho})$.

(b) *A rise in initial assets.* Differentiating,

$$\frac{\partial c(0)}{\partial a(0)} = \frac{\zeta}{1-e^{-\zeta}} > 0,$$

and since $c(t) = c(0)e^{(r-\rho)t/\theta}$ with the growth rate independent of wealth, the *entire* consumption path shifts up by the same proportional factor: preferences are homothetic, so wealth effects are neutral across dates. The slope of the path, $(r-\rho)/\theta$, is unchanged — consumption grows if $r > \rho$ and falls if $r < \rho$, whatever the level of assets. (For general u the same argument gives $\partial c(0)/\partial a(0) = [\int_0^1 e^{-rt} h'_t dt]^{-1} > 0$, though the path need no longer scale proportionally unless u is CRRA.)

(c) *A time-varying earnings profile.* Replacing w by $[w(t)]_{t=0}^1$ changes *only* the constant Ω , which becomes $a(0) + \int_0^1 e^{-rt} w(t) dt$: the Euler equation (*) is untouched, because labor income does not enter the first-order conditions. Hence

- if the present value of earnings is unchanged, the consumption path is exactly unchanged — only the asset path moves, the household saving when income arrives early and borrowing when it arrives late;
- if the present value rises, the whole path shifts up proportionally, as in part (b).

This is the permanent income logic: with a perfect capital market the timing of income is irrelevant, only its present value matters.

The qualification matters here, though, because EXAMPLE 7.1 imposes $a(t) \geq 0$: the household cannot borrow at all. If the earnings profile is back-loaded enough that the unconstrained solution would require $a(t) < 0$ on some interval, the constraint binds there. On such an interval the household simply consumes its income, $c(t) = w(t)$ and $a(t) = 0$, and the Euler equation holds with an inequality ($e^{-\rho t} u'(c(t)) \geq \lambda(t)$, the multiplier on $a \geq 0$ being positive). In that case the timing of income *does* affect consumption, and a shift of income toward the present raises welfare even holding its present value fixed.

Exercise 7.7

Let x_1, x_2 be in the (convex) domain, $\alpha \in (0, 1)$, and let y_1, y_2 attain the maxima defining $M(x_1)$ and $M(x_2)$. Then $(\alpha x_1 + (1 - \alpha)x_2, \alpha y_1 + (1 - \alpha)y_2)$ is a feasible pair, so

$$M(\alpha x_1 + (1 - \alpha)x_2) \geq \varphi(\alpha x_1 + (1 - \alpha)x_2, \alpha y_1 + (1 - \alpha)y_2) \geq \alpha \varphi(x_1, y_1) + (1 - \alpha)\varphi(x_2, y_2),$$

and the right-hand side is $\alpha M(x_1) + (1 - \alpha)M(x_2)$: the first inequality holds because M is a maximum and the second by joint concavity of φ . Hence M is concave.

If φ is *strictly* jointly concave and $x_1 \neq x_2$, the second inequality is strict — strict concavity applies because the two points (x_1, y_1) and (x_2, y_2) differ (their first coordinates do) — and therefore M is strictly concave.

If the maximum is not attained, the same argument runs with ε -maximizers: choose y_i with $\varphi(x_i, y_i) \geq M(x_i) - \varepsilon$, conclude $M(\alpha x_1 + (1 - \alpha)x_2) \geq \alpha M(x_1) + (1 - \alpha)M(x_2) - \varepsilon$, and let $\varepsilon \rightarrow 0$.

Remark. This little lemma is what makes Arrow's sufficiency conditions usable. THEOREMS 7.6 and 7.8 require concavity of the *maximized* Hamiltonian $M(t, x, \lambda)$ in x , which is not an assumption one can check directly; the lemma reduces it to joint concavity of $H(t, x, y, \lambda)$ in (x, y) , which is usually immediate from concavity of f and linearity or concavity of g — provided $\lambda(t) \geq 0$, since $H = f + \lambda g$ inherits concavity from g only when the multiplier is nonnegative.

Exercise 7.8

Claim. Consider $\max \int_0^{t_1} f(t, x, y) dt$ subject to $\dot{x} = g(t, x, y)$, $x(0) = x_0$ and $x(t_1) = x_1$, with f, g continuously differentiable. Suppose the admissible pair $(\hat{x}, \hat{y})$ satisfies the necessary conditions of THEOREM 7.2: $H_y(t, \hat{x}, \hat{y}, \lambda) = 0$, $\dot{\lambda} = -H_x(t, \hat{x}, \hat{y}, \lambda)$ and $\hat{\dot{x}} = g$. If X is convex and $M(t, x, \lambda) \equiv \max_y H(t, x, y, \lambda)$ is concave in x for every $t \in [0, t_1]$, then $(\hat{x}, \hat{y})$ attains the global maximum; if M is strictly concave in x , it is the unique solution.

Proof. Let (x, y) be any admissible pair, so that $x(0) = x_0$ and — this is the only change from THEOREM 7.6 — $x(t_1) = x_1 = \hat{x}(t_1)$. By definition of M and of the Hamiltonian,

$$f(t, x, y) = H(t, x, y, \lambda) - \lambda g(t, x, y) \leq M(t, x, \lambda) - \lambda(t)\dot{x}(t),$$

with equality along $(\hat{x}, \hat{y})$ because $\hat{y}$ maximizes H (which is what $H_y = 0$ plus concavity gives). Integrating and subtracting,

$$W(x, y) - W(\hat{x}, \hat{y}) \leq \int_0^{t_1} [M(t, x, \lambda) - M(t, \hat{x}, \lambda)] dt - \int_0^{t_1} \lambda(t) [\dot{x}(t) - \dot{\hat{x}}(t)] dt.$$

Concavity of M in x gives $M(t, x, \lambda) - M(t, \hat{x}, \lambda) \leq M_x(t, \hat{x}, \lambda) [x(t) - \hat{x}(t)]$, and by the envelope theorem (EXERCISE 7.2) $M_x(t, \hat{x}, \lambda) = H_x(t, \hat{x}, \hat{y}, \lambda) = -\dot{\lambda}(t)$. Hence

$$W(x, y) - W(\hat{x}, \hat{y}) \leq - \int_0^{t_1} \dot{\lambda} [x - \hat{x}] dt - \int_0^{t_1} \lambda [\dot{x} - \dot{\hat{x}}] dt = - \left[\lambda(t) (x(t) - \hat{x}(t)) \right]_0^{t_1},$$

integrating by parts. At $t = 0$ both paths start at x_0 , so the lower limit vanishes. At $t = t_1$ the *terminal constraint* gives $x(t_1) = \hat{x}(t_1) = x_1$, so the upper limit vanishes as well. Therefore

$$W(x, y) \leq W(\hat{x}, \hat{y}),$$

which is the assertion. If M is strictly concave in x , the concavity inequality is strict whenever $x(t) \neq \hat{x}(t)$ on a set of positive measure, so any other admissible pair is strictly worse and the solution is unique.

Remark. The comparison with THEOREM 7.6 is instructive: there the boundary term is killed by the transversality condition $\lambda(t_1) = 0$, which says that leftover state is worthless; here it is killed by the constraint $x(t_1) = x_1$, which says that every admissible path ends at the same place, so the value of leftover state is irrelevant. Exactly one of the two is needed, which is why THEOREMS 7.2 and 7.1 differ only in this boundary condition.

Exercise 7.9

The problem is that of THEOREM 7.2 with the terminal condition weakened to $x(t_1) \geq x_1$. Let $(\hat{x}, \hat{y})$ be an interior continuous solution and consider, as in the proofs of THEOREMS 7.1 and 7.2, the variations $y(t, \varepsilon) = \hat{y}(t) + \varepsilon \eta(t)$ with the induced state path $x(t, \varepsilon)$, $x(0, \varepsilon) = x_0$. Writing $W(\varepsilon)$ for the value of the objective and introducing a continuously differentiable costate λ , the same integration by parts as in the proof of THEOREM 7.1 gives

$$\left. \frac{dW}{d\varepsilon} \right|_{\varepsilon=0} = \int_0^{t_1} \left[H_x(t, \hat{x}, \hat{y}, \lambda) + \dot{\lambda}(t) \right] x_\varepsilon(t, 0) dt + \int_0^{t_1} H_y(t, \hat{x}, \hat{y}, \lambda) \eta(t) dt - \lambda(t_1) x_\varepsilon(t_1, 0). \quad (*)$$

The interior conditions. Restrict attention first to variations with $x_\varepsilon(t_1, 0) = 0$ — feasible whatever the terminal constraint, since they do not move the endpoint. Choosing λ to solve the linear terminal-value problem $\dot{\lambda} = -H_x(t, \hat{x}, \hat{y}, \lambda)$ kills the first integral, and optimality then requires $\int H_y \eta dt = 0$ for all such η , whence $H_y = 0$ for all t , that is (7.11); and $\dot{\lambda} = -H_x$ is (7.12). These are exactly the conditions of THEOREM 7.2.

The terminal condition. With (7.11) and (7.12) in hand, (*) reduces to

$$\left. \frac{dW}{d\varepsilon} \right|_{\varepsilon=0} = -\lambda(t_1) x_\varepsilon(t_1, 0).$$

Two cases arise.

- $\hat{x}(t_1) > x_1$ (*the constraint is slack*). For small ε the terminal constraint continues to hold whichever sign $x_\varepsilon(t_1, 0)$ takes, so both signs are feasible; optimality requires $dW/d\varepsilon = 0$ for both, hence $\lambda(t_1) = 0$.
- $\hat{x}(t_1) = x_1$ (*the constraint binds*). Only variations with $x_\varepsilon(t_1, 0) \geq 0$ are feasible, so optimality requires $-\lambda(t_1) x_\varepsilon(t_1, 0) \leq 0$ for all such variations, that is $\lambda(t_1) \geq 0$; nothing forces it to vanish.

In both cases

$$\lambda(t_1) \geq 0 \quad \text{and} \quad \lambda(t_1)(\hat{x}(t_1) - x_1) = 0,$$

which is the complementary slackness condition of THEOREM 7.3.

Interpretation. $\lambda(t_1)$ is the shadow value of the state at the terminal date. If the planner ends with more of the state than required, the marginal unit must be worthless — otherwise it would have been converted into payoff earlier; if the terminal requirement binds, the marginal unit has positive value, exactly as the multiplier on a binding constraint in static optimization.

Exercise 7.11

The problem is

$$\max_{x,y} \int_0^{t_1} f(t, x(t), y(t)) dt \quad \text{s.t.} \quad \dot{x}(t) = G(t, x(t), y(t)), \quad x(0) = x_0,$$

with $x(t) \in X \subset \mathbb{R}^{K_x}$, $y(t) \in Y \subset \mathbb{R}^{K_y}$, and an interior continuous solution $(\hat{x}, \hat{y})$. The argument is the one-dimensional proof of THEOREM 7.1 written with inner products.

Variations. Take any piecewise continuous $\eta : [0, t_1] \rightarrow \mathbb{R}^{K_y}$ and set $y(t, \varepsilon) = \hat{y}(t) + \varepsilon \eta(t)$. Since $\hat{y}(t) \in \text{Int } Y$ and $[0, t_1]$ is compact, $y(t, \varepsilon) \in Y$ for $|\varepsilon|$ small. Let $x(t, \varepsilon)$ be the corresponding state path, which exists, is unique and is differentiable in ε because G is continuously differentiable; write $x_\varepsilon(t) \equiv \partial x(t, \varepsilon) / \partial \varepsilon|_{\varepsilon=0}$, which satisfies the linearized state equation

$$\dot{x}_\varepsilon(t) = D_x G(t, \hat{x}, \hat{y}) x_\varepsilon(t) + D_y G(t, \hat{x}, \hat{y}) \eta(t), \quad x_\varepsilon(0) = 0.$$

The derivative of the objective. For any continuously differentiable $\lambda : [0, t_1] \rightarrow \mathbb{R}^{K_x}$ the state equation lets us write, with H as in (7.28),

$$W(\varepsilon) = \int_0^{t_1} \left[H(t, x(t, \varepsilon), y(t, \varepsilon), \lambda) - \lambda(t) \cdot \dot{x}(t, \varepsilon) \right] dt,$$

and integrating the last term by parts,

$$W(\varepsilon) = \int_0^{t_1} \left[H + \dot{\lambda}(t) \cdot x(t, \varepsilon) \right] dt - \lambda(t_1) \cdot x(t_1, \varepsilon) + \lambda(0) \cdot x_0.$$

Differentiating at $\varepsilon = 0$,

$$\left. \frac{dW}{d\varepsilon} \right|_0 = \int_0^{t_1} \left[D_x H(t, \hat{x}, \hat{y}, \lambda) + \dot{\lambda}(t) \right] \cdot x_\varepsilon(t) dt + \int_0^{t_1} D_y H(t, \hat{x}, \hat{y}, \lambda) \cdot \eta(t) dt - \lambda(t_1) \cdot x_\varepsilon(t_1).$$

Choosing the costate. Let λ solve the linear terminal-value problem

$$\dot{\lambda}(t) = -D_x H(t, \hat{x}(t), \hat{y}(t), \lambda(t)), \quad \lambda(t_1) = 0,$$

which has a unique continuously differentiable solution on $[0, t_1]$ because the right-hand side is linear in λ with continuous coefficients. This is (7.30) and the terminal condition in (7.31), and it annihilates the first and third terms above, leaving

$$\left. \frac{dW}{d\varepsilon} \right|_0 = \int_0^{t_1} D_y H(t, \hat{x}, \hat{y}, \lambda) \cdot \eta(t) dt.$$

Conclusion. Since $(\hat{x}, \hat{y})$ is optimal and the variations are two-sided, this derivative must vanish for every admissible η . Taking η supported on a small interval around any t and equal to each coordinate direction in turn, and using continuity of $D_y H$ along the solution, gives

$$D_y H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0 \quad \text{for all } t \in [0, t_1],$$

which is (7.29). Finally $\dot{x} = D_\lambda H = G$ is the state equation with $x(0) = x_0$, completing (7.31). This proves THEOREM 7.7.

Exercise 7.12

Let $(\hat{x}, \hat{y})$ satisfy (7.29)–(7.31) and let X be convex. Write

$$M(t, x, \lambda) \equiv \max_{y \in Y} H(t, x, y, \lambda)$$

and suppose it is concave in x . Take any admissible pair (x, y) . As in Exercise 7.8, for every t ,

$$f(t, x, y) = H(t, x, y, \lambda) - \lambda(t) \cdot G(t, x, y) \leq M(t, x, \lambda) - \lambda(t) \cdot \dot{x}(t),$$

with equality along $(\hat{x}, \hat{y})$ since $\hat{y}$ maximizes H pointwise. Subtracting and using concavity of M in x together with the envelope relation $D_x M(t, \hat{x}, \lambda) = D_x H(t, \hat{x}, \hat{y}, \lambda) = -\dot{\lambda}(t)$,

$$\begin{aligned} W(x, y) - W(\hat{x}, \hat{y}) &\leq \int_0^{t_1} \left[D_x M(t, \hat{x}, \lambda) \cdot (x - \hat{x}) - \lambda \cdot (\dot{x} - \dot{\hat{x}}) \right] dt \\ &= \int_0^{t_1} \left[-\dot{\lambda} \cdot (x - \hat{x}) - \lambda \cdot (\dot{x} - \dot{\hat{x}}) \right] dt = - \left[\lambda(t) \cdot (x(t) - \hat{x}(t)) \right]_0^{t_1}. \end{aligned}$$

The lower limit vanishes because both paths start at x_0 , and the upper limit vanishes because $\lambda(t_1) = 0$ by (7.31). Hence $W(x, y) \leq W(\hat{x}, \hat{y})$ for every admissible pair: $(\hat{x}, \hat{y})$ is a global maximum.

If $M(t, \cdot, \lambda)$ is strictly concave, the first inequality is strict unless $x(t) = \hat{x}(t)$ almost everywhere, so the maximizer is unique (and then $y = \hat{y}$ wherever H has a unique maximizer).

Remark. Exercise 7.7 is what makes the hypothesis checkable: joint concavity of H in (x, y) — which follows from concavity of f , linearity of G in (x, y) , and $\lambda \geq 0$ — implies concavity of M in x . In the growth applications of the next chapters G is linear (capital accumulation) and f is concave (utility), so the conditions hold automatically.

Exercise 7.13

Let $(\hat{x}, \hat{y})$ be admissible and satisfy (7.39)–(7.41), let X be convex, let $M(t, x, \lambda)$ be concave in x for all t , and suppose

$$\lim_{t \rightarrow \infty} \lambda(t)(\hat{x}(t) - \tilde{x}(t)) \leq 0 \quad \text{for every admissible } \tilde{x}. \quad (*)$$

Take any admissible pair $(\tilde{x}, \tilde{y})$. Exactly as in Exercise 7.8, for each finite T ,

$$\int_0^T f(t, \tilde{x}, \tilde{y}) dt - \int_0^T f(t, \hat{x}, \hat{y}) dt \leq - \left[\lambda(t)(\tilde{x}(t) - \hat{x}(t)) \right]_0^T = \lambda(T)(\hat{x}(T) - \tilde{x}(T)),$$

using $\tilde{x}(0) = \hat{x}(0) = x_0$ for the lower limit. This is the “additional term” the hint refers to: in finite horizon it was eliminated by $\lambda(t_1) = 0$ or by a fixed endpoint, and here it survives.

Now let $T \rightarrow \infty$. The left-hand side converges to $W(\tilde{x}, \tilde{y}) - W(\hat{x}, \hat{y})$ whenever the two improper integrals exist (which admissibility guarantees), and the right-hand side is bounded above by zero by hypothesis (*). Hence

$$W(\tilde{x}, \tilde{y}) - W(\hat{x}, \hat{y}) \leq \limsup_{T \rightarrow \infty} \lambda(T)(\hat{x}(T) - \tilde{x}(T)) \leq 0,$$

so $(\hat{x}, \hat{y})$ achieves the global maximum of (7.32). If $M(t, \cdot, \lambda)$ is strictly concave, the inequality leading to the display is strict unless $\tilde{x} = \hat{x}$ almost everywhere, so the solution is unique.

Remark. Condition (*) is the honest price of infinite horizon, and the book rightly calls it hard to check: it involves *all* admissible comparison paths. Two special cases make it usable. If the state is nonnegative and $\lambda(t) \geq 0$ with $\lim_{t \rightarrow \infty} \lambda(t)\hat{x}(t) = 0$ — the transversality condition (7.69) — then $\lambda(T)(\hat{x}(T) - \tilde{x}(T)) \leq \lambda(T)\hat{x}(T) \rightarrow 0$ and (*) holds automatically. This is exactly how THEOREM 7.14 is used in the applications: check the transversality condition, and concavity does the rest.

Exercise 7.14

The discounted problem has $f(t, x, y) = e^{-\rho t} f(x, y)$ and $g(t, x, y) = g(x, y)$. Write the present-value Hamiltonian and its current-value counterpart,

$$H(t, x, y, \lambda) = e^{-\rho t} f(x, y) + \lambda(t)g(x, y), \quad \hat{H}(t, x, y, \mu) \equiv e^{\rho t} H(t, x, y, \lambda) = f(x, y) + \mu(t)g(x, y),$$

where the current-value costate is

$$\mu(t) \equiv e^{\rho t} \lambda(t) \quad \Longleftrightarrow \quad \lambda(t) = e^{-\rho t} \mu(t).$$

THEOREM 7.9 applies to the problem and gives, at points of continuity of $\hat{y}$, the necessary conditions (7.39)–(7.41). Translate each in turn.

(7.65). $H_y = 0$ reads $e^{-\rho t} f_y + \lambda g_y = 0$. Multiplying by $e^{\rho t} > 0$,

$$f_y(x, y) + \mu(t)g_y(x, y) = \hat{H}_y(t, \hat{x}, \hat{y}, \mu) = 0,$$

which is (7.65).

(7.66). Differentiating $\lambda = e^{-\rho t} \mu$,

$$\dot{\lambda}(t) = e^{-\rho t} [\dot{\mu}(t) - \rho \mu(t)],$$

while $H_x = e^{-\rho t} f_x + \lambda g_x = e^{-\rho t} [f_x + \mu g_x] = e^{-\rho t} \hat{H}_x$. Substituting into $\dot{\lambda} = -H_x$ and cancelling $e^{-\rho t} > 0$,

$$\dot{\mu}(t) - \rho \mu(t) = -\hat{H}_x(t, \hat{x}, \hat{y}, \mu) \quad \Longleftrightarrow \quad \rho \mu(t) - \dot{\mu}(t) = \hat{H}_x(t, \hat{x}, \hat{y}, \mu),$$

which is (7.66).

(7.67). The state equation is unchanged: $\dot{x} = H_\lambda = g(x, y) = \hat{H}_\mu$, with $x(0) = x_0$ and the limiting constraint $\lim_{t \rightarrow \infty} b(t)x(t) \geq x_1$ carried over from (7.41).

Since THEOREM 7.9 gives (7.39)–(7.41) as necessary conditions for an interior piecewise continuous solution, and the three translations above are equivalences, (7.65)–(7.67) are necessary as well.

Remark. The point of the current-value formulation is visible in (7.66): the term $\rho \mu$ appears because μ is measured in date- t rather than date-0 units, so its required rate of return has to cover the discount rate. Written as $\dot{\mu}/\mu = \rho - \hat{H}_x/\mu$, the condition is an asset-pricing equation — the shadow price of the state must appreciate at the discount rate less its “dividend”.

Exercise 7.15

(a) *Rewriting the objective.* For any continuously differentiable λ , the constraint $\dot{x} = g$ gives $f = H - \lambda\dot{x}$ with $H(t, x, y) \equiv f(t, x, y) + \lambda(t)g(t, x, y)$. Hence

$$W = \int_0^{t_1} f \, dt = \int_0^{t_1} [H - \lambda(t)\dot{x}(t)] \, dt,$$

and integrating the last term by parts, $\int_0^{t_1} \lambda\dot{x} \, dt = \lambda(t_1)x(t_1) - \lambda(0)x_0 - \int_0^{t_1} \dot{\lambda}x \, dt$, so

$$W(x(t), y(t)) = \int_0^{t_1} [H(t, x(t), y(t)) + \dot{\lambda}(t)x(t)] \, dt - \lambda(t_1)x(t_1) + \lambda(0)x_0,$$

as required.

(b) *Feasible variations.* Let $(\hat{x}, \hat{y}, \hat{t}_1)$ be optimal. Perturb the control on the original interval and extend the horizon:

$$y(t, \varepsilon) = \hat{y}(t) + \varepsilon\eta(t) \quad \text{on } [0, \hat{t}_1], \quad y(t, \varepsilon) = \hat{y}(\hat{t}_1) + \varepsilon\eta(t) \quad \text{on } [\hat{t}_1, \hat{t}_1 + \varepsilon\Delta t],$$

with $t_1 = \hat{t}_1 + \varepsilon\Delta t$. For $|\varepsilon|$ small this is feasible: $\hat{y}(t)$ is interior to Y and $\hat{x}(t)$ interior to X on a compact interval, so small perturbations stay inside, and the state path $x(t, \varepsilon)$ generated by $\dot{x} = g$ is well defined and differentiable in ε (the extension is over an interval of length $O(\varepsilon)$, on which the state moves by $O(\varepsilon)$).

(c) *The derivative.* Differentiating the expression in (a) at $\varepsilon = 0$, three groups of terms appear. The interior variation contributes $\int_0^{\hat{t}_1} [H_x + \dot{\lambda}]x_\varepsilon \, dt + \int_0^{\hat{t}_1} H_y\eta \, dt$. The moving upper limit contributes $H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1))\Delta t$ from the integral and $-\lambda(\hat{t}_1)\dot{x}(\hat{t}_1)\Delta t$ from $-\lambda(t_1)x(t_1)$; the endpoint term also contributes $-\lambda(\hat{t}_1)x_\varepsilon(\hat{t}_1)$. Since the second of these enters $-\lambda(t_1)x(t_1)$ with a minus sign and the total derivative of $x(t_1, \varepsilon)$ with respect to ε is $x_\varepsilon(\hat{t}_1) + \dot{x}(\hat{t}_1)\Delta t$, collecting terms gives exactly

$$\left. \frac{dW}{d\varepsilon} \right|_0 = \int_0^{\hat{t}_1} [H_x + \dot{\lambda}]x_\varepsilon \, dt + \int_0^{\hat{t}_1} H_y\eta \, dt + H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1))\Delta t - \lambda(\hat{t}_1)x_\varepsilon(\hat{t}_1) + \lambda(\hat{t}_1)\dot{x}(\hat{t}_1)\Delta t,$$

the expression in the exercise.

(d) *Why it vanishes.* $(\hat{x}, \hat{y}, \hat{t}_1)$ maximizes W over a set that contains the whole family $\{(x(\cdot, \varepsilon), y(\cdot, \varepsilon), \hat{t}_1 + \varepsilon\Delta t)\}$ for ε in a neighborhood of 0 — both signs are feasible, since the horizon may be lengthened or shortened and the control perturbed in either direction. A differentiable function of ε attaining an interior maximum at $\varepsilon = 0$ has zero derivative there.

The conditions of the Maximum Principle follow. Taking $\Delta t = 0$ and η arbitrary, and choosing λ to satisfy $\dot{\lambda} = -H_x$ with $\lambda(\hat{t}_1) = 0$, gives $H_y = 0$ as usual. Then, with $\lambda(\hat{t}_1) = 0$, the remaining terms leave $H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1))\Delta t = 0$ for every Δt , so

$$\boxed{H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1), \lambda(\hat{t}_1)) = 0}$$

— the extra transversality condition associated with a free terminal date. Its meaning is an optimal stopping rule: continue as long as the flow value of continuing, measured by the Hamiltonian, is positive, and stop when it reaches zero.

(e) Letting $\hat{t}_1 \rightarrow \infty$. The condition becomes

$$\lim_{t \rightarrow \infty} H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0,$$

which is precisely the infinite-horizon transversality condition (7.56) of THEOREM 7.12. So the “vanishing Hamiltonian” that is usually presented as a separate infinite-horizon result is the same condition as the free-terminal-time condition of the finite-horizon problem: an infinite horizon is a horizon that is optimally never reached, and the value of stopping at the margin must be zero at every sufficiently distant date.

Exercise 7.16

The discounted problem has $f(t, x, y) = e^{-\rho t} f(x, y)$ and $g(t, x, y) = g(x, y)$: neither the payoff apart from the discount factor nor the constraint depends on calendar time.

Admissibility. The continuation satisfies $\dot{\hat{x}}(t) = g(\hat{x}(t), \hat{y}(t))$ for all $t \geq s$ and $\hat{x}(s)$ is its own initial condition, with $(\hat{x}(t), \hat{y}(t)) \in X \times Y$; so it is admissible for the problem started at s with initial state $\hat{x}(s)$.

Optimality. The objective of the date- s problem is

$$W_s(x, y) \equiv \int_s^\infty e^{-\rho(t-s)} f(x(t), y(t)) dt = e^{\rho s} \int_s^\infty e^{-\rho t} f(x(t), y(t)) dt,$$

so it ranks continuation plans exactly as the date-0 objective does: the two differ by the positive constant $e^{\rho s}$. Suppose the continuation were not optimal. Then there is an admissible $(\tilde{x}, \tilde{y})$ on $[s, \infty)$ with $\tilde{x}(s) = \hat{x}(s)$ and

$$\int_s^\infty e^{-\rho t} f(\tilde{x}, \tilde{y}) dt > \int_s^\infty e^{-\rho t} f(\hat{x}, \hat{y}) dt.$$

Splice the plans:

$$(\vec{x}(t), \vec{y}(t)) = \begin{cases} (\hat{x}(t), \hat{y}(t)), & t \in [0, s), \\ (\tilde{x}(t), \tilde{y}(t)), & t \geq s. \end{cases}$$

This is admissible from x_0 : it satisfies the state equation on each piece, and the two pieces agree at $t = s$ because $\tilde{x}(s) = \hat{x}(s)$, so $\vec{x}$ is continuous and satisfies $\dot{\vec{x}} = g(\vec{x}, \vec{y})$ throughout. Its value at date 0 is

$$\int_0^s e^{-\rho t} f(\hat{x}, \hat{y}) dt + \int_s^\infty e^{-\rho t} f(\tilde{x}, \tilde{y}) dt > \int_0^\infty e^{-\rho t} f(\hat{x}, \hat{y}) dt,$$

contradicting the optimality of $(\hat{x}, \hat{y})$ from date 0. Hence the continuation is optimal for the date- s problem.

Remark. Two features do the work, and both are needed: the objective is *exponentially* discounted, so that re-dating multiplies it by a constant and leaves the ranking of continuations unchanged; and g is autonomous, so that the constraint faced from s onward is the same correspondence, only with a different initial state. Drop the first and time consistency fails — this is exactly EXERCISE 7.22(b), where $\beta(t) = e^{-\rho t}$ turns out to be necessary as well as sufficient. The statement in the exercise writes the date- s initial condition as $x(s) = x_0$; what is meant, and what is proved above, is that the continuation is optimal for the problem started at the state the optimal path has reached, $x(s) = \hat{x}(s)$.

Exercise 7.20

Recall the problem: maximize $\int_0^\infty e^{-\rho t} u(y(t)) dt$ subject to $\dot{x} = -y$, $x(0) = 1$, $x(t) \geq 0$, with $u : [0, 1] \rightarrow \mathbb{R}$ strictly increasing, continuously differentiable and strictly concave. The Hamiltonian is $H(t, x, y, \lambda) = e^{-\rho t} u(y) - \lambda y$, and THEOREM 7.9 gives (7.58) and (7.59):

$$u'(\hat{y}(t)) = e^{\rho t} \lambda(t), \quad \dot{\lambda}(t) = 0 \implies \lambda(t) \equiv \lambda(0) > 0,$$

so that the Hotelling rule $\dot{\mu}/\mu = \rho$ holds for $\mu = e^{\rho t} \lambda$ and

$$\hat{y}(t) = (u')^{-1}[\lambda(0)e^{\rho t}].$$

Since u' is strictly decreasing and $\lambda(0)e^{\rho t} \rightarrow \infty$, extraction falls monotonically with $\hat{y}(t) \rightarrow 0$; the constant $\lambda(0)$ is fixed by exhaustion, $\int_0^\infty \hat{y}(t) dt = x(0)$.

Checking (7.56). The condition is $\lim_{t \rightarrow \infty} H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0$. Here

$$H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = e^{-\rho t} u(\hat{y}(t)) - \lambda(0) \hat{y}(t).$$

Both terms vanish in the limit:

- $\hat{y}(t) \rightarrow 0$ and u is continuous on $[0, 1]$, so $u(\hat{y}(t)) \rightarrow u(0)$, a finite number, and $e^{-\rho t} u(\hat{y}(t)) \rightarrow 0$;
- $\lambda(0) \hat{y}(t) \rightarrow 0$ because $\hat{y}(t) \rightarrow 0$ and $\lambda(0)$ is a constant.

Hence $\lim_{t \rightarrow \infty} H = 0$: the transversality condition (7.56) holds.

An explicit case. With $u(y) = \log y$ (unbounded below at 0, so strictly outside the stated domain, but the standard illustration), $\hat{y}(t) = e^{-\rho t} / \lambda(0)$, and exhaustion $\int_0^\infty \hat{y} dt = 1/(\rho \lambda(0)) = x(0)$ gives $\lambda(0) = 1/(\rho x(0))$ and $\hat{y}(t) = \rho x(0) e^{-\rho t}$: the resource is depleted at the constant proportional rate ρ . Then

$$H(t) = e^{-\rho t} \log(\rho x(0) e^{-\rho t}) - \frac{1}{\rho x(0)} \rho x(0) e^{-\rho t} = e^{-\rho t} [\log(\rho x(0)) - \rho t - 1] \longrightarrow 0,$$

since $t e^{-\rho t} \rightarrow 0$.

Remark. Note which transversality condition is being verified. The *stronger* condition (7.69) requires $\lim_{t \rightarrow \infty} e^{-\rho t} \mu(t) \hat{x}(t)$ to vanish; since $e^{-\rho t} \mu(t) = \lambda(0)$ here, it reduces to $\lambda(0) \lim \hat{x}(t) = 0$, which holds precisely because the resource is fully exhausted, $\hat{x}(t) \rightarrow 0$. EXERCISE 7.18 modifies the problem so that a positive stock x_1 must be left behind. The stronger condition then fails, while the weaker one still holds — which is exactly why the book states (7.56) as the general result.

Exercise 7.22

(a) *Necessary conditions.* The Hamiltonian is

$$H(t, x, y, \lambda) = \beta(t) f(x, y) + \lambda(t) g(x, y),$$

and THEOREM 7.9 gives, along an interior solution,

$$H_y = 0 : \quad \beta(t) f_y(x, y) + \lambda(t) g_y(x, y) = 0, \quad \dot{\lambda} = -H_x : \quad \dot{\lambda}(t) = -[\beta(t) f_x + \lambda(t) g_x],$$

together with $\dot{x} = g(x, y)$, $x(0) = x_0$ and the transversality condition (7.56). Assume $g_y \neq 0$, so the first condition can be solved for the costate:

$$\lambda(t) = -\beta(t) \frac{f_y(x(t), y(t))}{g_y(x(t), y(t))} \equiv \beta(t) \Lambda(t), \quad (*)$$

where Λ depends on the path only, not on the discount function.

(b) *Time consistency forces exponential discounting.* Consider re-solving the problem at date $s > 0$ with the same structure — that is, with discount function $\beta(t - s)$ applied to returns $t - s$ away — starting from the state $\hat{x}(s)$ that the original plan has reached. Its necessary conditions are the same with $\beta(t)$ replaced by $\beta(t - s)$ and costate λ_s . Time consistency means that the original path $(\hat{x}, \hat{y})$ continues to solve this problem, so it must satisfy both sets of conditions. From (*) applied to each,

$$\lambda(t) = \beta(t)\Lambda(t), \quad \lambda_s(t) = \beta(t - s)\Lambda(t) \implies \lambda_s(t) = k(t)\lambda(t), \quad k(t) \equiv \frac{\beta(t - s)}{\beta(t)}.$$

Both costates satisfy their own costate equations along the *same* path:

$$\dot{\lambda} = -[\beta(t)f_x + \lambda g_x], \quad \dot{\lambda}_s = -[\beta(t - s)f_x + \lambda_s g_x] = -[k(t)\beta(t)f_x + k(t)\lambda g_x] = k(t)\dot{\lambda}(t).$$

But differentiating $\lambda_s = k\lambda$ directly gives $\dot{\lambda}_s = \dot{k}\lambda + k\dot{\lambda}$. Comparing, $\dot{k}(t)\lambda(t) = 0$, and since $\lambda(t) \neq 0$ along an interior solution, $\dot{k}(t) = 0$: the ratio $\beta(t - s)/\beta(t)$ is independent of t . Evaluating it at $t = s$ gives $k = \beta(0)/\beta(s)$, so for all $t \geq s \geq 0$,

$$\beta(t - s)\beta(s) = \beta(t)\beta(0).$$

Normalizing $\beta(0) = 1$, this is the Cauchy functional equation $\beta(t - s)\beta(s) = \beta(t)$; its only positive continuous solutions are

$$\beta(t) = e^{-\rho t} \quad \text{for some } \rho,$$

with $\rho \geq 0$ if the discount function is nonincreasing (and $\rho > 0$ needed for the objective to converge). Conversely, if $\beta(t) = e^{-\rho t}$, then $\beta(t - s) = e^{\rho s}\beta(t)$, the two systems differ by the positive constant $e^{\rho s}$, and the original path satisfies both — time consistency, which is Exercise 7.16.

(c) *Interpretation, and the contrast with LEMMA 7.1.* The result is the continuous-time counterpart of the discrete-time statement in EXERCISE 5.1: only exponential discounting makes the trade-off between two future dates independent of the date from which it is viewed. With any other β — hyperbolic discounting, say — the planner's relative valuation of dates t and t' changes as time passes, so at date s she would like to revise the plan made at date 0 even though nothing has been learned and no state has been surprised.

The contrast with LEMMA 7.1 is worth stating carefully, since the lemma is also called a principle of optimality and holds for general $f(t, x, y)$. LEMMA 7.1 says that the value function computed *from the date-0 perspective* is recursive: $V(t_0, x) = \int_{t_0}^{t_1} f dt + V(t_1, \hat{x}(t_1))$. That is a statement about one fixed objective, and it is true whatever the discount function, because $V(t_1, \cdot)$ is itself defined with the date-0 weights. Time consistency in the sense of part (b) is a different claim: that the plan optimal under the date-0 objective is still optimal under the *date-s* objective, which applies $\beta(t - s)$ rather than $\beta(t)$. Only exponential discounting makes the two objectives agree, up to scale, on the

ranking of continuations. In short, LEMMA 7.1 is about a consistent bookkeeping of one plan; part (b) is about whether a later self would keep it.

Exercise 7.27

The problem of Section 7.8 is

$$\max \int_0^\infty e^{-rt} [f(K) - I - \phi(I)] dt \quad \text{s.t.} \quad \dot{K} = I - \delta K, \quad K(0) > 0,$$

with f satisfying Assumptions 1 and 2 of Chapter 2 and ϕ strictly increasing, strictly convex, continuously differentiable, with $\phi(0) = \phi'(0) = 0$.

(a) ASSUMPTION 7.1 on a compact state space. Restrict $K \in [\varepsilon, \bar{k}]$ with $0 < \varepsilon < \bar{k} < \infty$; investment can then also be restricted to a compact interval, since $\dot{K} = I - \delta K$ and K is bounded, and since paying $I + \phi(I)$ for arbitrarily large I is never optimal. Now check the three parts:

- (1) *Monotonicity.* The payoff $f(K) - I - \phi(I)$ is increasing in K (as $f' > 0$) and decreasing in I (as $1 + \phi' > 0$), so it is weakly monotone in each argument; $g(K, I) = I - \delta K$ is increasing in I , decreasing in K , and independent of t .
- (2) $|g_I| \geq m$. Here $g_I \equiv 1$, so the condition holds with $m = 1$ — the control moves the state one for one.
- (3) $|f_I| \leq M$. Here $|f_I| = 1 + \phi'(I)$, which is bounded on the compact interval to which I has been restricted, so the condition holds with $M = 1 + \max \phi'$.

It is the third condition that fails on the unbounded state space, since ϕ' is unbounded; this is exactly why the compactification is needed before THEOREM 7.13 can be invoked.

(b) *From the modified problem to the original one.* Suppose the modified problem has a solution $(\hat{K}, \hat{I})$ with $\hat{K}(t) > \varepsilon$ for all t , so the artificial constraint never binds. Then $(\hat{K}, \hat{I})$ satisfies the necessary conditions (7.85) of the unconstrained problem, including the transversality condition $\lim_{t \rightarrow \infty} e^{-rt} q(t) \hat{K}(t) = 0$. Now appeal to sufficiency rather than necessity: the current-value Hamiltonian

$$\hat{H}(K, I, q) = f(K) - I - \phi(I) + q[I - \delta K]$$

is strictly concave in (K, I) — f is strictly concave, $-\phi$ is strictly concave, and the remaining terms are linear — so by Exercise 7.7 the maximized Hamiltonian is strictly concave in K , and THEOREM 7.14 applies on the *whole* feasible set $K \in \mathbb{R}_+$: any path satisfying the necessary conditions together with the transversality condition is the unique global maximum among all feasible paths, not merely among those confined to $[\varepsilon, \bar{k}]$. Hence $(\hat{K}, \hat{I})$ solves the original problem. (Note that sufficiency does not use ASSUMPTION 7.1 at all; the assumption was needed only to obtain the strong transversality condition as a *necessary* condition.)

(c) *Such an ε exists.* The steady state of the system (7.84), (7.88) satisfies

$$I^* = \delta K^*, \quad f'(K^*) = (r + \delta)(1 + \phi'(I^*)),$$

which has a unique positive solution because the left side is strictly decreasing in K from ∞ to 0 (Assumptions 1 and 2) while the right side is strictly increasing in K . The optimal path is the stable arm of a saddle point — the linearized system has one positive and one negative eigenvalue

— and $K(t)$ moves monotonically from $K(0)$ to K^* . Therefore

$$\inf_{t \geq 0} K(t) = \min\{K(0), K^*\} > 0,$$

and any $\varepsilon < \min\{K(0), K^*\}$ works: the modified problem's solution never touches the artificial lower bound, and part (b) applies.

With $f(K) = K^{1/3}$, $\phi(I) = I^2$, $\delta = r = 0.05$, the numbers are

$$K^* = 3.767391, \quad I^* = \delta K^* = 0.188370, \quad q^* = 1 + \phi'(I^*) = 1.376739,$$

and the eigenvalues of the linearized system are -0.108440 and $+0.158440$ — one of each sign, as the saddle-path structure requires. Figure 5 shows the phase diagram and the stable arm, which stays well away from $K = 0$.

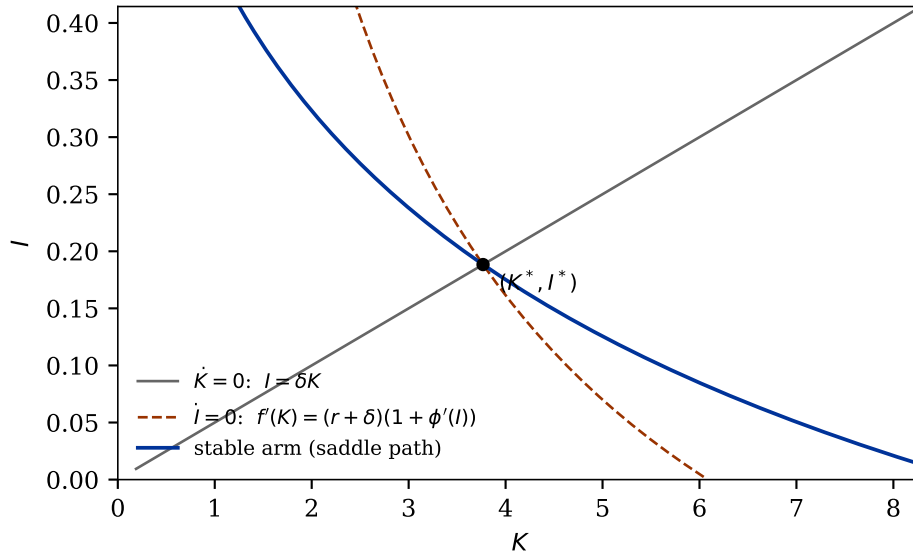


FIGURE 5. The q -theory phase diagram: the loci $\dot{K} = 0$ and $\dot{I} = 0$, their intersection at (K^*, I^*) , and the stable arm along which the firm approaches the steady state.

Remark. Adjustment costs lower the steady-state capital stock: with $\phi \equiv 0$ the steady state would solve $f'(K) = r + \delta$, giving $K = 6.085806$ in this parameterization, against 3.767391 with adjustment costs. The reason is visible in the steady-state condition — the required marginal product is scaled up by $q^* = 1 + \phi'(I^*) > 1$, the shadow price of installed capital, because replacing depreciated capital is itself costly.

Exercise 7.29

Suppose the candidate path has $I(t) = 0$ for all $t \geq t'$. Then from (7.84) its capital stock decays exponentially,

$$K(t) = K(t')e^{-\delta(t-t')} \longrightarrow 0 \quad \text{as } t \rightarrow \infty.$$

The deviation. Fix $T > t'$ and $\epsilon > 0$ small, and consider investing ϵ at every date $t \geq T$ instead of nothing. Two things make the accounting simple. First, the extra *cost* at each date is $\epsilon + \phi(\epsilon)$, and since $\phi(0) = \phi'(0) = 0$ we have $\phi(\epsilon) = o(\epsilon)$: to first order the adjustment cost of a small investment is zero, so the cost is ϵ per unit time. Second, the extra *capital* raises output at the margin by

$f'(K(t))$ per unit. To first order in ϵ the change in the present value of profits is therefore

$$\Delta = \epsilon \int_T^\infty e^{-rt} [f'(K(t)) - 1] dt + o(\epsilon).$$

The sign. By Assumption 2 of Chapter 2, $\lim_{K \rightarrow 0} f'(K) = \infty$. Since $K(t) \rightarrow 0$ along the candidate path, there is a $\tilde{K} > 0$ with $f'(K) > 1$ for all $K < \tilde{K}$, and a date T with $K(t) < \tilde{K}$ — hence $f'(K(t)) > 1$ — for all $t \geq T$. For that T the integrand is strictly positive throughout, so $\Delta > 0$ for ϵ small: the deviation strictly raises the present value of profits, and the candidate path is not optimal.

Quantitatively, with $f(K) = AK^\alpha$ the threshold is explicit: $f'(K) > 1$ if and only if $K < (A\alpha)^{1/(1-\alpha)}$, which equals 0.192450 for $A = 1$, $\alpha = 1/3$. Starting from the steady state $K^* = 3.767391$ and switching investment off, capital falls below that threshold after $\log(K^*/0.19245)/\delta \approx 59.5$ periods, and $f'(K(t)) \rightarrow \infty$ thereafter.

Conclusion. No optimal path sets investment to zero permanently after a finite date: gross investment is strictly positive in the long run, and indeed the optimal path converges to the steady state with $I^* = \delta K^* > 0$. The economics is the Inada condition doing its usual work — capital that has become scarce is arbitrarily productive at the margin, while the marginal cost of a small amount of investment is only its purchase price, adjustment costs being second order at $I = 0$.

Remark. The same argument shows why the constraint $I \geq 0$ studied in EXERCISE 7.28(f) — irreversible investment — binds only on the way *down*, never permanently: a firm may wish to disinvest faster than depreciation allows, but never to abstain from investment forever.